

**ELECTRICAL RESISTANCE WELDING  
WELDING TECHNIQUES  
GENERAL SPECIFICATIONS**

Page 1/69

**Restrictions described in the norm**

This standard is published by the PSA Standards Department (DRD/DAPF/RHN/NCF)  
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This standard is available in the Intranet [NORMES](#) and on the [B2B gate](#).

In the event of litigation, only the French version of this standard is taken.

**WARNING**

*This PSA norm has been translated by the BIWE DCHM into English.*

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**FOREWORD**

At the date of publication of this document, there is not international, European or French norm with the same object.

This present norm is the complement related to the welding process of the norms [B13 1220](#) and [B13 1226](#) which deals with the product.

It defines:

- *welding parameters to apply for each resistance welding equipment,*
- *welding equipments to use according to the products requirements (quality and appearance of welding).*

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| 10/02/2015                                | -         | 10/02/2015                                                      | -         | 10/02/2015                              | -         |

|                                                          |              |      |
|----------------------------------------------------------|--------------|------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 2/69 |
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## RECORDS

| Index | Date       | Nature of modifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OR    | 01/03/2001 | Recovery under GEODE addition of the paragraphs 1.4.5, 1.5.2, 1.7.3, 1.7.5, 2.1, 2.4 – Transition to an internal use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| A     | 04/10/2001 | ADDITION OF PARAGRAPHS 1.4.5, 1.5.2, 1.7.3, 1.7.5, 2.1 and 2.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| B     | 01/09/2011 | <ul style="list-style-type: none"> <li>SUPPRESSION OF THESE PARAGRAPHS : <ul style="list-style-type: none"> <li>PARAMETERS CLASSES (§ 2.1 IN THE INDEX A),</li> <li>DETERMINATION OF THE WELDING EQUIPMENTS - PHASE PREPARATORY STAGE (§ 5 IN THE INDEX A),</li> <li>APPENDIX B.</li> </ul> </li> <li>CREATION OF THE PARAGRAPHS : <ul style="list-style-type: none"> <li>4.7.1.2 WELDING ASSEMBLY THICKNESS OF REFERENCE <math>\leq 1,27</math> MM WITH <math>\varnothing 8</math> MM ELECTRODES OF ACTIVE FACE,</li> <li>5.5 THE WELDING OF THE ASSEMBLIES 4 THICKNESSES IN 50HZ,</li> <li>5.8 WELDING ON AVERAGE FREQUENCY AND SELF-ADAPTING INTERMEDIATE FREQUENCY,</li> <li>5.9.4.2 EMBOSSINGS FORMS "D" FOR REDUCED MILLSTEPS,</li> <li>7.7.2 WELDING POINT IN FUNCTIONAL AREA,</li> <li>7.9.7 DRESSING INTERVALS FOR ROBOT ACCORDING TO THICKNESSES OF REF. AND TYPICAL CASES.</li> </ul> </li> </ul> |
| D     | 26/03/2014 | <p>Norm modification.</p> <ul style="list-style-type: none"> <li>INDEX C; WELDING RULES – STEEL SHEETS THICKNESS AND GATHERING – WELDING MODES – RECOMMENDATION OF USE – WELDING SCREWS INTRODUCTION – WELDING NUT INTRODUCTION – WELDING WAYS – WELDING TECHNIQUES – MIXED RANGE – STEEL SHEETS SHADES CLASSIFICATION – ALUMINIUM WELDING – WELDING PARAMETERS – WELDING ASSEMBLIES 4 EP – SCREWS AND NUTS WELDING PARAMETERS – DEAD GUN – ASSEMBLY RANGE – PROCESS DEFINITION – WELDING IN MFAA – PRIMARY CONNECTION OF EQUIPMENTS – PRIMARY THERMAL INTENSITY – COOLING WATER – ELECTRICAL AND MAGNETIC INSULATION – ATTRIBUTION OF THE FUNCTIONAL PTSE AREA – PTSE ASPECT – ROBOT INCREMENTATION DIAGRAM VERY TIDY PRESENTATION – MANUAL WELDING GUN INCREMENTATION DIAGRAM VERY TIDY WELDING SPOT– MSA INCREMENTATION DIAGRAM VERY TIDY PRESENTATION – DRESSING INTERVALS</li> </ul>                    |
| E     | 10/02/2015 | MODIFICATIONS OF THE NORM FOLLOWING EXPERT RESISTANCE WELDING REQUESTS, UPDATIND OF THE LIST OF PARTICIPANTS, ADDITION OF HYPertext LINK TOWARDS MENTIONED DOCUMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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## CONTENTS

|                                                                                                                                                   |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1. OBJECT AND FIELD OF APPLICATION</b>                                                                                                         | <b>6</b>  |
| <b>2. REFERENTIAL DOCUMENTARY</b>                                                                                                                 | <b>6</b>  |
| <b>2.1. NORMS</b>                                                                                                                                 | <b>6</b>  |
| <b>2.2. REGULATIONS</b>                                                                                                                           | <b>6</b>  |
| <b>2.3. OTHER DOCUMENTS</b>                                                                                                                       | <b>7</b>  |
| <b>2.4. EXPRESSION ON DOCUMENTS</b>                                                                                                               | <b>7</b>  |
| <b>3. TERMINOLOGY</b>                                                                                                                             | <b>7</b>  |
| <b>3.1. DEFINITIONS</b>                                                                                                                           | <b>7</b>  |
| <b>3.2. ACRONYMS</b>                                                                                                                              | <b>7</b>  |
| <b>4. RULES OF WELDING</b>                                                                                                                        | <b>9</b>  |
| <b>4.1. DETERMINATION OF REFERENCE THICKNESS</b>                                                                                                  | <b>9</b>  |
| <b>4.2. THICKNESSES OF SHEETS AND REGROUPING FOR THE CALCULATION OF THE THICKNESS OF REFERENCE AND THE INCREMENTING OF THE WELDING PARAMETERS</b> | <b>10</b> |
| <b>4.3. WELDING MODES</b>                                                                                                                         | <b>10</b> |
| 4.3.1. Field of application                                                                                                                       | 10        |
| 4.3.2. Specific instructions for screws and nuts                                                                                                  | 11        |
| <b>4.4. TYPE OF WELDING</b>                                                                                                                       | <b>17</b> |
| <b>4.5. WELDING TECHNIQUE</b>                                                                                                                     | <b>18</b> |
| <b>4.6. OUTSIDE THE NORMS OF PRODUCT &amp; PROCESS WELDING</b>                                                                                    | <b>19</b> |
| 4.6.1. Spot welding                                                                                                                               | 19        |
| 4.6.2. WELDING screws and nuts                                                                                                                    | 19        |
| 4.6.3. Embossment welding                                                                                                                         | 20        |
| <b>4.7. MIXED RANGE</b>                                                                                                                           | <b>20</b> |
| 4.7.1. Restrictions of use                                                                                                                        | 20        |
| <b>4.8. WELDING ON DEAD MECHANICAL WELDING GUN</b>                                                                                                | <b>21</b> |
| 4.8.1. Impedance                                                                                                                                  | 22        |
| 4.8.2. Transformer of the robot welding gun or manual                                                                                             | 22        |
| 4.8.3. Supply of current                                                                                                                          | 23        |
| <b>4.9. SHEET WELDED ON TUBE</b>                                                                                                                  | <b>24</b> |
| 4.9.1. Principle                                                                                                                                  | 24        |
| 4.9.2. Thickness ratio → See norm B13 1220                                                                                                        | 24        |
| 4.9.3. Electrodes                                                                                                                                 | 24        |
| <b>4.10. CHARACTERISTICS FOR WELDING SHEETS</b>                                                                                                   | <b>25</b> |
| 4.10.1. Classification of the sheet SHades                                                                                                        | 25        |
| 4.10.2. Dissymmetrical assemblies                                                                                                                 | 27        |
| 4.10.3. Welding of aluminum                                                                                                                       | 27        |
| <b>5. WELDING PARAMETERS</b>                                                                                                                      | <b>28</b> |
| <b>5.1. WELDING PARAMETERS CONTINUOUS ROD MILL FOR THE 50HZ APPLICATIONS - SE PTS OF Ø 3 MM</b>                                                   | <b>28</b> |
| <b>5.2. WELDING PARAMETERS CONTINUOUS ROD MILL FOR APPLICATIONS 50 HZ - PTS SE Ø 4 MM</b>                                                         | <b>29</b> |
| 5.2.1. Coatings ≤ 10 µm                                                                                                                           | 29        |
| 5.2.2. Coatings > 10 µm                                                                                                                           | 30        |
| <b>5.3. PULSATION WELDING PARAMETERS FOR APPLICATIONS 50 HZ</b>                                                                                   | <b>31</b> |
| <b>5.4. WELDING OF THE ASSEMBLIES 4 THICKNESSES IN 50HZ</b>                                                                                       | <b>33</b> |
| <b>5.5. WELDING TECHNIQUE 60 HZ</b>                                                                                                               | <b>33</b> |
| <b>5.6. LEVEL OF REGROUPING FOR WELDING PROGRAMS</b>                                                                                              | <b>34</b> |
| <b>5.7. WELDING PARAMETERS FOR SCREWS AND NUTS</b>                                                                                                | <b>34</b> |
| 5.7.1. Screw of definition conform to norms PSA Peugeot Citroën STNG SCREW 9691205699 (01266_12_00682)                                            | 35        |
| 5.7.2. Nut of definition conform to norm PSA Peugeot Citroën STNG NUT 9691172999 (01266_12_00912)                                                 | 36        |
| 5.7.3. Nut of mass                                                                                                                                | 36        |
| 5.7.4. Welding of embossment (ref. norm PSA Peugeot Citroën B13 1230)                                                                             | 36        |
| <b>6. MEDIUM FREQUENCY WELDING (ROBOTIZED PROCESS) AND AUTO ADAPTIVE MEDIUM FREQUENCY (MANUAL PROCESS)</b>                                        | <b>37</b> |
| <b>6.1. PREAMBLE MF WELDING</b>                                                                                                                   | <b>37</b> |

| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS                                                       |  | E34.03.180.G | 5/69      |
|----------------------------------------------------------------------------------------------------------------|--|--------------|-----------|
| 6.1.1. Robotized process                                                                                       |  |              | 37        |
| 6.1.2. Manual process                                                                                          |  |              | 37        |
| <b>6.2. RULES TO RESPECT</b>                                                                                   |  |              | <b>38</b> |
| <b>6.3. WELDABILITY FIELD</b>                                                                                  |  |              | <b>38</b> |
| <b>6.4. WELDING SETTING</b>                                                                                    |  |              | <b>41</b> |
| 6.4.1. Medium frequency (robot or manual)                                                                      |  |              | 41        |
| 6.4.2. adaptive medium frequency (manual)                                                                      |  |              | 41        |
| <b>7. ASSEMBLIES RANGES</b>                                                                                    |  |              | <b>43</b> |
| <b>7.1. RESORE</b>                                                                                             |  |              | <b>43</b> |
| 7.1.1. Installation AND maintenance                                                                            |  |              | 43        |
| 7.1.2. RESORE file                                                                                             |  |              | 43        |
| 7.1.3. Making of range                                                                                         |  |              | 43        |
| <b>8. DEFINITION OF THE PROCESS</b>                                                                            |  |              | <b>43</b> |
| <b>8.1. EQUIPMENTS</b>                                                                                         |  |              | <b>44</b> |
| 8.1.1. Initial adjustments and conditions of implementation                                                    |  |              | 44        |
| 8.1.2. Use in mass production                                                                                  |  |              | 44        |
| <b>8.2. PRIMARY CONNECTION OF THE EQUIPMENTS</b>                                                               |  |              | <b>45</b> |
| 8.2.1. Connection IG2 / Welding cabinet                                                                        |  |              | 45        |
| 8.2.2. Connection welding cabinet / Welding equipments                                                         |  |              | 47        |
| <b>8.3. THERMAL INTENSITY</b>                                                                                  |  |              | <b>47</b> |
| 8.3.1. Primary thermal intensity                                                                               |  |              | 47        |
| 8.3.2. Secondary thermal intensity                                                                             |  |              | 48        |
| <b>8.4. WORKING FACTOR</b>                                                                                     |  |              | <b>48</b> |
| <b>8.5. COOLING WATER OF THE WELDING EQUIPMENTS</b>                                                            |  |              | <b>48</b> |
| <b>8.6. MAGNETIC AND ELECTRICAL INSULATION</b>                                                                 |  |              | <b>48</b> |
| <b>8.7. USE OF CONNECTION BY BARS</b>                                                                          |  |              | <b>50</b> |
| 8.7.1. U Fastening for the bar hold up                                                                         |  |              | 51        |
| 8.7.2. Determination of the bar section                                                                        |  |              | 51        |
| <b>8.8. ATTRIBUTE OF SPOTS WELD - FUNCTIONAL AREAS</b>                                                         |  |              | <b>52</b> |
| 8.8.1. Regulation of welding spots                                                                             |  |              | 52        |
| 8.8.2. welding spot in functional area                                                                         |  |              | 52        |
| <b>8.9. DIAMETERS OF SPOTS WELD</b>                                                                            |  |              | <b>52</b> |
| <b>8.10. WELDING SPOTS ASPECT</b>                                                                              |  |              | <b>53</b> |
| <b>8.11. SPOTS MADE IN 2 THEN 3 THICKNESSES</b>                                                                |  |              | <b>54</b> |
| <b>8.12. INCREMENTAL LAW FOR WELDING CURRENTS</b>                                                              |  |              | <b>55</b> |
| 8.12.1. For robot and coating thicknesses $\leq 10 \mu\text{m}$                                                |  |              | 56        |
| 8.12.2. For M.S.A and coating thicknesses $\leq 10 \mu\text{m}$                                                |  |              | 58        |
| 8.12.3. For P.A.P. and coating thicknesses $\leq 10 \mu\text{m}$                                               |  |              | 60        |
| 8.12.4. For robot and coating thicknesses $> 10 \mu\text{m}$                                                   |  |              | 62        |
| 8.12.5. For M.S.A. and for coating thicknesses $> 10 \mu\text{m}$                                              |  |              | 63        |
| 8.12.6. For P A P. and coating thicknesses $> 10 \mu\text{m}$                                                  |  |              | 64        |
| 8.12.7. Intervals of dressing for robot according to thicknesses of reference and typical cases (50Hz and MF)  |  |              | 65        |
| <b>APPENDIX 1 METHODS OF WELDING</b>                                                                           |  |              | <b>66</b> |
| <b>APPENDIX 2 TABLE OF DEFINITION OF THE SECTIONS OF CABLE (50HZ) VALID FOR CABLES TYPE H07RNF AND A07RN-F</b> |  |              | <b>69</b> |

## 1.OBJECT AND FIELD OF APPLICATION

This document joins together the necessary information to the study and the realization of the welding equipments used by the PSA Peugeot Citroën group.

It is used by the integrators of PSA Peugeot Citroën group, the door-fitting methods department and the fabrication department.

## 2.REFERENTIAL DOCUMENTARY

### 2.1.NORMS

#### 2.1.1 INTERNAL PSA NORMS

|                              |                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------|
| <a href="#">B13 1220</a>     | RESISTANCE SPOT WELDING OF STEEL SHEETS ASSEMBLY DEFINITION                                         |
| <a href="#">B13 1226</a>     | RESISTANCE SPOT WELDING OF STEEL SHEETS - ASSEMBLY QUALITY                                          |
| <a href="#">B13 1230</a>     | RESISTANCE EMBOSMENT WELDING ASSEMBLY DEFINITION                                                    |
| <a href="#">C00 1000</a>     | WATERTIGHTNESS, AIRTIGHTNESS AND ACOUSTIC SEALING OF REPETITIVE FIXATIONS                           |
| <a href="#">E03.65.015.G</a> | MACHINES AND INDUSTRIAL INSTALLATION OF AUTOMATIC FUNCTIONING: TECHNICAL AND TRANSVERSE REFERENTIAL |
| <a href="#">GE34-007G</a>    | TECHNICAL GUIDE FOR WORKING PARTS FOR ELECTRICAL RESISTANCE WELDING                                 |
| <a href="#">GE34-009G</a>    | ELECTRICAL RESISTANCE WELDING TECHNICAL GUIDE FOR THE DESIGN OF AUTOMATIC WELDING EQUIPMENT         |

#### 2.1.1.EXTERNAL NORMS

|                |                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------|
| ISO 5826       | RESISTANCE WELDING EQUIPMENT - TRANSFORMERS - GENERAL SPECIFICATIONS APPLICABLE TO ALL TRANSFORMERS |
| NF EN ISO 5182 | RESISTANCE WELDING - MATERIALS FOR ELECTRODES AND ADDITIONAL EQUIPMENT                              |

### 2.2.REGULATIONS

Not applicable.

## 2.3.OTHER DOCUMENTS

|                                 |                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">01342_09_00020</a>  | Collection of bodyshop tests procedure (=PSA_FER_182)                                                                                           |
| <a href="#">01342_11_00009</a>  | Management methodology of Functional Areas.                                                                                                     |
| <a href="#">01342_11_00036</a>  | Guide and milestone of the assemblers' activities.                                                                                              |
| <a href="#">01266_12_00682</a>  | Norm STNG welding screw with cylindrical end– metric threading ISO (=STNG ECROUS 9691172999)                                                    |
| <a href="#">01266_12_00912</a>  | Norm STNG hexagonal nuts to weld by resistance (=STNG VIS 9691205699)                                                                           |
| <a href="#">01342_12_00210</a>  | Terms and conditions of pedestal welding machine (=PSA_FER_122).                                                                                |
| <a href="#">01353_09_00111</a>  | Recommendations of design favorable to a welding PtSE 4 thicknesses                                                                             |
| 9691172999                      | STNG Screws<br>(Accessible under DEVIN)                                                                                                         |
| <a href="#">MXP_CEB00_0022</a>  | Procedure - Overcoding of designations sheet - 30/01/2008<br>(Managed under GRCP)                                                               |
| <a href="#">MXP_CEB07_0644</a>  | Cracking of SE Pt. of the B58 cradle<br>Technical report available at DRD/DCHM/BIWE/ADEE/ISES                                                   |
| Note n° 4277                    | DMEI/CEI/EVC Technical note available at DRD/DCHM/BIWE/ADEE/ISES                                                                                |
| <a href="#">PMXP_CEB09_0014</a> | Procedure of definition of welding parameters and control fixings welded by embossing and doesn't satisfy standards PSA<br>(Managed under GRCP) |
| PSA.VEH.182                     | Collection of the bodyshop procedures of tests<br>(Managed under GRCP)                                                                          |

## 2.4.EXPRESSION ON DOCUMENTS

Not applicable.

## 3.TERMINOLOGY

A dictionary (glossary) of the main terms and their definitions used within the activities of the R&D can be consulted in-house via the [DESP](#) or by clicking on the following link: [SCPO\\_MGPJ07\\_0116](#).

### 3.1.DEFINITIONS

For the requirements of this norm, the following term is defined:

**SAFIR / SAPHIR :** Software of sheets overcoding.

### 3.2.ACRONYMS

For the requirements of this norm, the following acronyms are defined:

|              |                                   |
|--------------|-----------------------------------|
| <b>AC:</b>   | Alternative Current               |
| <b>aM:</b>   | Engine accompaniment.             |
| <b>BI:</b>   | Low Impedance.                    |
| <b>CdC:</b>  | Specifications.                   |
| <b>CE:</b>   | Counter Electrode.                |
| <b>CPS:</b>  | Programmable controls of Welding. |
| <b>DPPP:</b> | Double Spot Push-Pull.            |
| <b>ep:</b>   | thickness (of reference)          |
| <b>FDM:</b>  | Duty ratio.                       |
| <b>HLE:</b>  | High Yield stress.                |
| <b>I2p:</b>  | Permanent secondary intensity.    |

|                                                                 |                     |             |
|-----------------------------------------------------------------|---------------------|-------------|
| <b>ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS</b> | <b>E34.03.180.G</b> | <b>8/69</b> |
|-----------------------------------------------------------------|---------------------|-------------|

|                |                                            |
|----------------|--------------------------------------------|
| <b>Iq:</b>     | Quadratic intensity.                       |
| <b>Is:</b>     | Welding Current                            |
| <b>Ith:</b>    | Thermal intensity.                         |
| <b>LAF:</b>    | Cold rolling.                              |
| <b>MAS:</b>    | Pedestal welding Machine.                  |
| <b>MF:</b>     | Medium frequency.                          |
| <b>MSA:</b>    | Automatic machine of Welding.              |
| <b>PaP:</b>    | Manual welding gun equipment.              |
| <b>PSE:</b>    | File "Electric Welding spots".             |
| <b>Pso:</b>    | Tidy presentation.                         |
| <b>Pst:</b>    | Standard presentation.                     |
| <b>Pts:</b>    | Very tidyTidy presentation.                |
| <b>PtSE</b>    | Electric spot Welding.                     |
| <b>Puls:</b>   | Pulsation.                                 |
| <b>RESORE:</b> | Referential Welding Resistance.(software)  |
| <b>RO:</b>     | Tools realization.                         |
| <b>ROB:</b>    | Robot.                                     |
| <b>RPF:</b>    | Welding fixed Gun for robot                |
| <b>RSE:</b>    | Electric robot of Welding.                 |
| <b>SP:</b>     | Direct Simple spot.                        |
| <b>SPI:</b>    | Indirect Simple spot.                      |
| <b>STNG:</b>   | Generic Normative Technical Specification. |
| <b>TC:</b>     | Continuous rod mill.                       |
| <b>TM:</b>     | Holding time.                              |
| <b>TS:</b>     | Welding time.                              |
| <b>VI:</b>     | Industrial Validation                      |
| <b>ZF:</b>     | Functional Area                            |
| <b>HZF:</b>    | Outside Functional Area                    |
| <b>RAZ:</b>    | Reset                                      |
| <b>R:</b>      | Resistance                                 |
| <b>daN:</b>    | Deca-newton                                |

## 4. RULES OF WELDING

### 4.1. DETERMINATION OF REFERENCE THICKNESS

Welding parameters (intensity, force, time) and the choice of welding mode to be used for a determined assembly depend on a thickness of reference "E" defined in table 1.

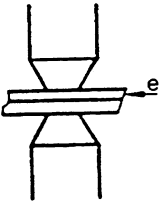
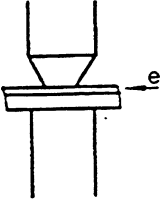
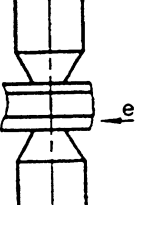
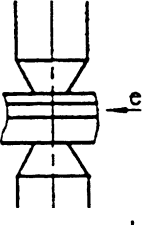
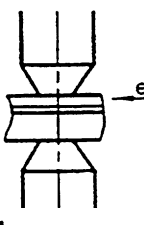
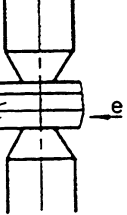
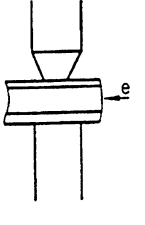
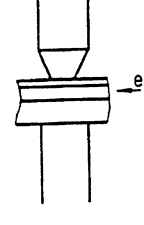
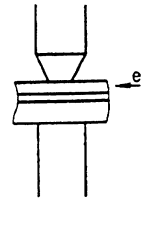
|          | Between electrodes                                                                                                                                                                                                                                                                                                                                                             | On counter electrode (for appearance)                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 sheets | Valid for simple spot and double push-pull spot                                                                                                                                                                                                                                                                                                                                | Valid for simple spot, double pspot range and double push-pull spot                                                                                                                                                                                                                                                                                                                                                                                              |
|          | <p>e = the thinnest sheet</p>                                                                                                                                                                                                                                                                 | <p>e = sheet on the side of active electrode</p>  <p>Max. thickness of the strongest sheet:<br/>E max. <math>\leq 1,3</math> mm</p>                                                                                                                                                                                                                                            |
| 3 sheets | e = average sheet                                                                                                                                                                                                                                                                                                                                                              | Valid for simple spot and double spot push-pull *                                                                                                                                                                                                                                                                                                                                                                                                                |
|          |     <p>Sheets of<br/>thickness</p> | <p>e = thickest of the two sheets which is not in contact with opposite electrode</p>    <p>E max. <math>\leq 1,3</math> mm</p> <p><b>Welding pulsation on opposite electrode is forbidden</b></p> |

Table 1

**Note:** For other configurations of assembly, need for carrying out preliminary tests of weldability with the PSA Peugeot Citroën qualified department.

Definition of opposite electrode: flat electrode, electrode with radius 40 and 80, having an active face higher of 2 mm than the traditional active faces of  $\varnothing 6$  mm and  $\varnothing 8$  mm.

**\* The 6 coatings assemblies are forbidden.**

For the assemblies from 1 to 5 coatings, tests of weldability are carried out by the Body-in-white Department.

The validations are defined individually according to certain criteria:

- positions, thicknesses, sheets shapes...
- numbers and positions of the coatings...

## 4.2.THICKNESSES OF SHEETS AND REGROUPING FOR THE CALCULATION OF THE THICKNESS OF REFERENCE AND THE INCREMENTING OF THE WELDING PARAMETERS

| Real thickness<br>mm | Thickness retained for calculations of the welding<br>parameters (thicknesses of reference +<br>incrementing)<br>mm |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| From 0,55 to 0,64    | 0,6                                                                                                                 |
| From 0,65 to 0,74    | 0,7                                                                                                                 |
| From 0,75 to 0,84    | 0,8                                                                                                                 |
| From 0,85 to 0,94    | 0,9                                                                                                                 |
| From 0,95 to 1,09    | 1                                                                                                                   |
| From 1,10 to 1,30    | 1,2                                                                                                                 |
| From 1,31 to 1,69    | 1,5                                                                                                                 |
| From 1,7 to 2,29     | 2                                                                                                                   |
| From 2,3 to 2,79     | 2,5                                                                                                                 |
| From 2,8 to 3,00     | 3                                                                                                                   |

Table 2

## 4.3.WELDING MODES

The welding modes define the architecture and the secondary circuit connections mode and are:

- Simple direct spot.
- Simple indirect spot.
- Double push-pull spot.
- Double spot range.

See the representation of the welding modes in Appendix 1.

### 4.3.1.FIELD OF APPLICATION

The field of application of the modes of welding modes is as follows:

| Mode of welding              | Type of equipments | Welding technique               | Types of point                  | Type of welding                             |
|------------------------------|--------------------|---------------------------------|---------------------------------|---------------------------------------------|
| SP (simple direct spot)      | ROB, PAP           | 50 Hz - 60 Hz *-<br>(AC) MF(DC) | PtSE                            | Continuous rod<br>mill (TC) or<br>Pulsation |
|                              | MSA, MAS           | 50 Hz - 60 Hz *-<br>(AC) MF(DC) | PtSE, Screw, Nut,<br>Embossment |                                             |
| SPI (simple indirect spot)   | MSA, MAS           | 50 Hz - 60 Hz* -<br>(AC) MF(DC) | PtSE, Screw, Nut,<br>Embossment | TC                                          |
| DPPP (double push-pull spot) | MSA                | 50 Hz - 60 Hz*<br>(AC)          | PtSE                            | TC or Pulsation                             |

\*: 50 or 60Hz (AC) according to countries. (AC = Alternative Current)  
(MF: Intermediate frequency) (DC: Direct Current)

### Recommendations of uses

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 11/69 |
|----------------------------------------------------------|--------------|-------|

#### 4.3.1.1.Simple direct spot

No specific recommendation.

#### 4.3.1.2.Simple indirect spot

As far as possible, the studs of indirect contacts must be left closed.

In the event of impossibility, faces of contact (up and down) must be imperatively protected from projections of welding.

#### 4.3.1.3.Double push-pull spot

In accordance with guide [GE34-009G](#), by convention, the elements of welding carrying out the same Pt SE are connected to the same polarity of the transformers in opposite. The inversion of phase is obligatorily carried out on the primary of the lower transformer.

#### 4.3.1.4.Double serial spot

This welding mode is **forbidden** for welding:

- of PtSE,
- of screw and nut with sealing constraint.

Case of screws and nuts:

Respect the in-between axes electrodes between screws or nuts which are in the table of chapters [4.3.2.1](#) and 4.3.2.2.

In case of problem with sealing, and welding on a same steel sheet, ask the bodyshop Technician Referential to define the necessary in-between axle.

The following instructions, as a reminder, come to complete the STNG screw 9691205699 ([01266 12 00682](#)) and the STNG nuts 9691172999 ([01266 12 00912](#)).

A request must be done to the PSA Fixation Pole department (POFI) for the missing torques data and for the special screwing.

### 4.3.2. SPECIFIC INSTRUCTIONS FOR SCREWS AND NUTS

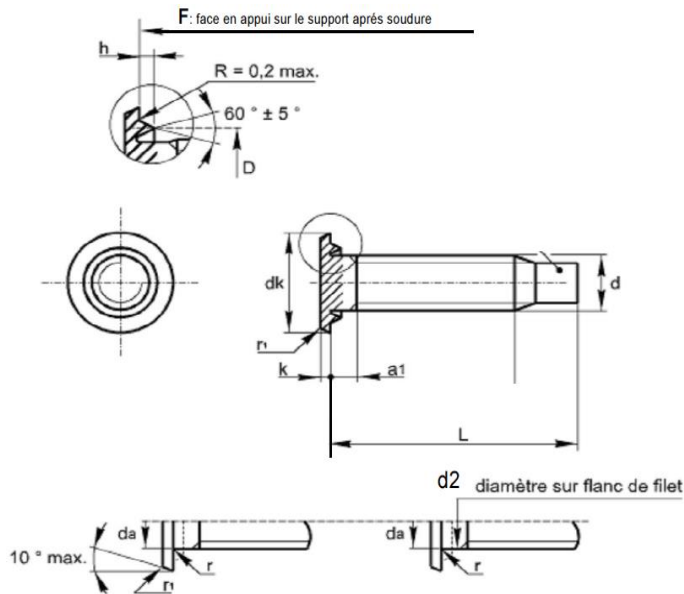
The following instructions, being used as reminder, are extracted from the STNG VIS 9691205699 for screws.

A request must be made to the PSA Peugeot Citroën qualified department for the values of the missing torques as well as the torques of special screws and bolts.

## 4.3.2.1. Case of the screws

**This chapter only deals with classic screws.**

Dimensional screws characteristics:

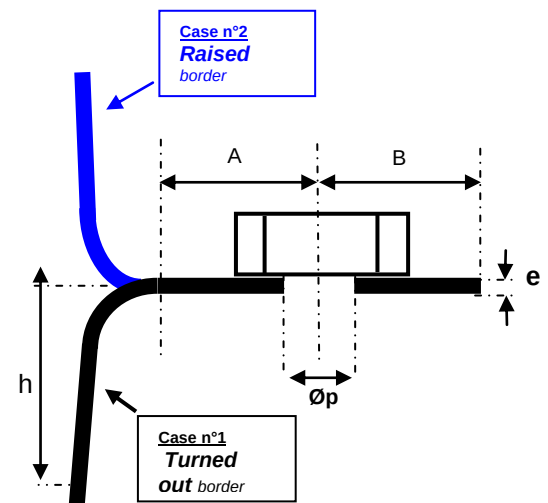
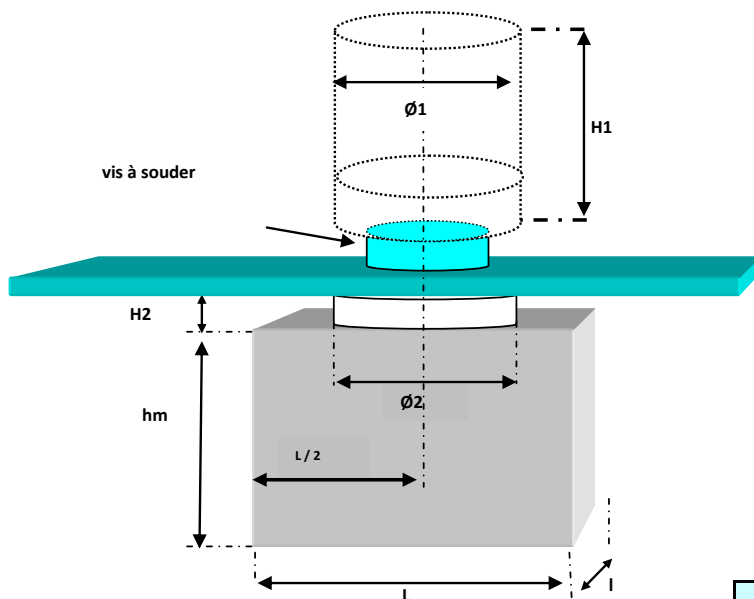


| Diamètre nominal d | dk   |       | k    |      | r1   | a1   | r    | da   | D     |      | h    |      | d2           | L            |
|--------------------|------|-------|------|------|------|------|------|------|-------|------|------|------|--------------|--------------|
|                    | max. | min.  | max. | min. | max. | max. | min. | max. | max.  | min. | max. | min. |              | js15         |
| M4                 | 11,5 | 11,23 | 2    | 1,75 | 0,7  | 1,95 | 0,2  | 4,7  | 8,75  | 8,5  | 1,25 | 0,9  | Voir § 8.3.2 | Voir § 8.1.2 |
| M5                 | 12,5 | 12,23 | 2,5  | 2,25 | 0,7  | 2,05 | 0,2  | 5,7  | 9,75  | 9,5  | 1,25 | 0,9  |              |              |
| M6                 | 14,5 | 14,23 | 2,5  | 2,25 | 0,7  | 2,25 | 0,3  | 6,8  | 10,75 | 10,5 | 1,25 | 0,9  |              |              |
| M8                 | 19   | 18,67 | 3,5  | 3,25 | 1    | 2,7  | 0,3  | 9,2  | 14,25 | 14   | 1,45 | 1,1  |              |              |
| M10                | 21   | 20,67 | 4    | 3,75 | 1    | 2,95 | 0,4  | 11,2 | 16,25 | 16   | 1,45 | 1,1  |              |              |
| M12                | 24   | 23,67 | 5    | 4,75 | 1    | 3,4  | 0,4  | 14,2 | 18,75 | 18,5 | 1,65 | 1,3  |              |              |

**Implantation screw characteristics :**

| Screw to weld | e<br>(mm) | B<br>(mm) | Øp<br>(mm) | A<br>(mm) |          | Mini in-between between 2 identical screws without any electrodes offset (air cylinder)<br>(mm) |                   |                                      |                                       |                                               |                                                |
|---------------|-----------|-----------|------------|-----------|----------|-------------------------------------------------------------------------------------------------|-------------------|--------------------------------------|---------------------------------------|-----------------------------------------------|------------------------------------------------|
|               |           |           |            | Case 1    |          | case 2                                                                                          | Welded on 2 tools | Welded rings on the same tool 6 bars | Welded rings on the same tool 10 bars | 3 embossment s welded on the same tool 6 bars | 3 embossment s welded on the same tool 10 bars |
|               |           |           |            | h ≤ 7 mm  | h > 7 mm |                                                                                                 |                   |                                      |                                       |                                               |                                                |
| M5            | 0,65      | 10        | 5,2        | 17,5      | 24       | 10,0                                                                                            | 30,0              | 79                                   | 71                                    | 79                                            | 71                                             |
| M6            | 0,65      | 10        | 6,2        | 17,5      | 24       | 11,0                                                                                            | 30,0              | 79                                   | 71                                    | 79                                            | 71                                             |
| M8            | 0,75      | 12        | 8,2        | 17,5      | 24       | 13,0                                                                                            | 35,5              | 90                                   | 71                                    | 79                                            | 71                                             |
| M10           | 0,95      | 13        | 10,5       | 19,5      | 24       | 14,0                                                                                            | 36,0              | 110                                  | 79                                    | 90                                            | 71                                             |
| M12           | 0,95      | 16        | 12,5       | 19,5      | 24       | 15,5                                                                                            | 37,0              | 110                                  | 79                                    | 90                                            | 71                                             |

- respect the minimal thickness of the steel sheet support : **e**
- guarantee a minimal board width: **B**
- define the diameter of steel sheet punching : **Øp**
- guarantee the **A** dimension for accessibility (control tool or welding tool)
- \* for the elastomer carrots see the PSA\_FER\_122 (01342 12 00210)



| Screw to weld | Superior volume |         | Lower volume |         |                        |    |    |
|---------------|-----------------|---------|--------------|---------|------------------------|----|----|
|               | Ø1 (mm)         | H1 (mm) | Ø2 (mm)      | H2 (mm) | Counter Electrode (mm) |    |    |
|               |                 |         |              |         | L                      | I  | hm |
| M5            | 20              | 80      | 30           | 7       | 48                     | 30 | 83 |
| M6            |                 |         |              |         |                        |    |    |
| M8            |                 |         |              |         |                        |    |    |
| M10           | 25              |         | 35           |         | 56                     | 56 | 83 |
| M12           |                 |         |              |         |                        |    |    |

**Particularities of the sealing functions:**

This function will be realizable only with annular embossment screws and respecting the following conditions:

- Level 0: no sealing exigency
- Level 1: sealing to runoff waters: only possible with the annular embossment screw.
- Level 2: sealing to pressurized water: no guaranteed.

The detail of the sealing characteristics required for the repetitive fixation pieces with the aim of assuring a sealing to water, air and acoustic are described in the reference norm [C00 1000](#) – “Etanchéité à l’eau, à l’air et acoustiques des pieces de fixation répétitives.”

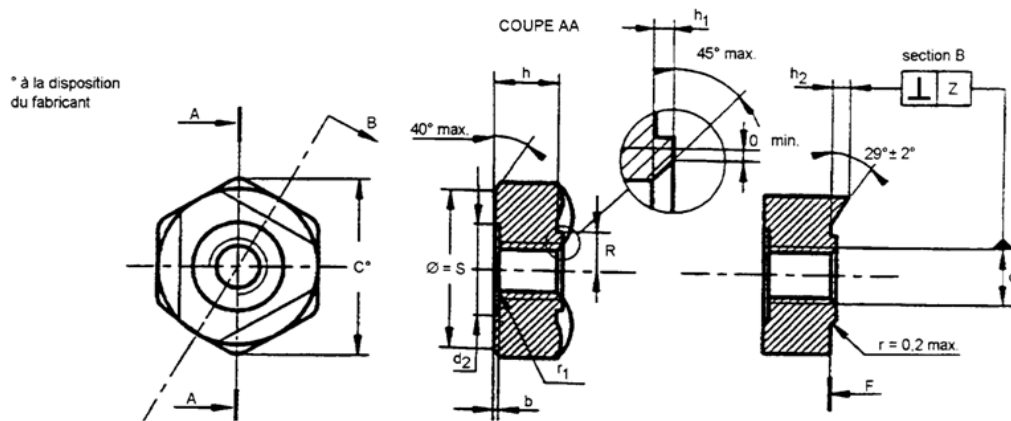
If the step is inferior to the mini, see the welding expert.

## 4.3.2.2.Case for nuts

**This chapter only deals with classic nuts**

Dimensional characteristics of the nuts:

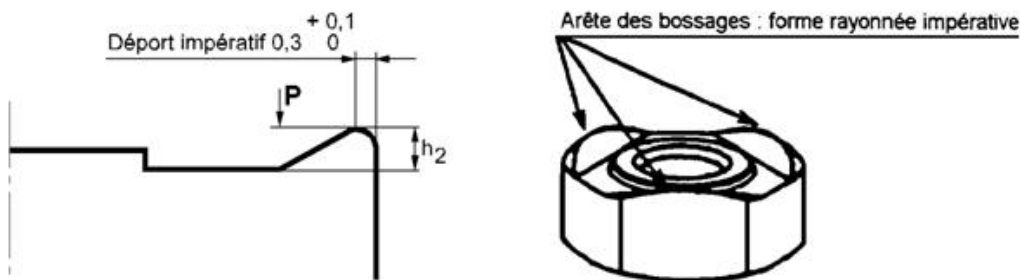
## Hexagonal resistance welding nuts



| D           | S<br>(0 / - 0,50) | h<br>± 0,25 | h <sub>1</sub><br>± 0,15 | R<br>± 0,15 | d <sub>2</sub><br>(0 / - 0,20) | b<br>(+0,15 / 0) | r <sub>1</sub> max. | h <sub>2</sub><br>± 0,10 | Z    |
|-------------|-------------------|-------------|--------------------------|-------------|--------------------------------|------------------|---------------------|--------------------------|------|
| M 6 x 1,00  | 13                | 5           | 0,6                      | 3,85        | 7,7                            | 0,3              | 0,5                 | 0,8                      | 0,25 |
| M 8 x 1,25  | 16                | 8           | 0,8                      | 5,25        | 9,6                            | 0,3              | 0,7                 | 1,0                      | 0,30 |
| M 10 x 1,50 | 19                | 9           | 0,8                      | 6,25        | 11,6                           | 0,3              | 1,0                 | 1,1                      | 0,35 |
| M 12 x 1,75 | 22                | 10          | 0,8                      | 7,25        | 14,6                           | 0,3              | 1,0                 | 1,2                      | 0,35 |

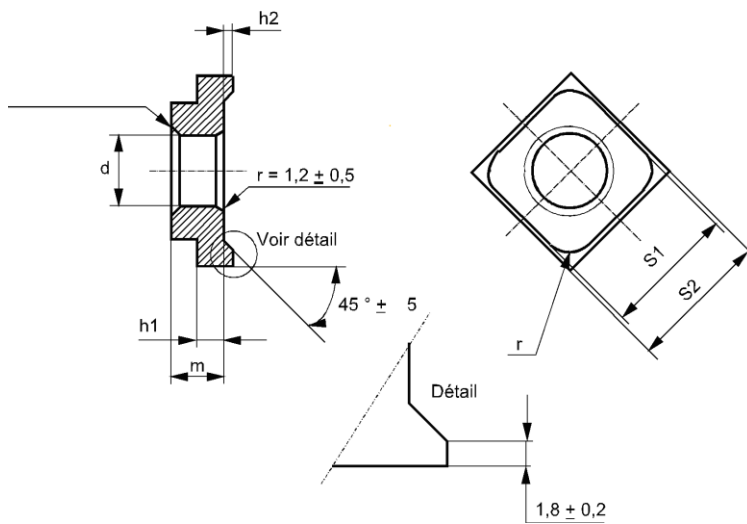
Special specifications:

P= plan going through the 3 embossment ridges.



## Squared nuts to weld by resistance

Special nuts (never screwed). These nuts must be used only for the gains of mass, in association with self-tapped screws specifics for this function.

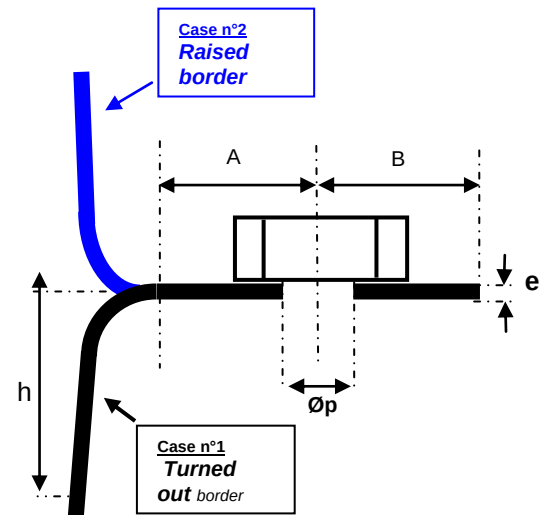
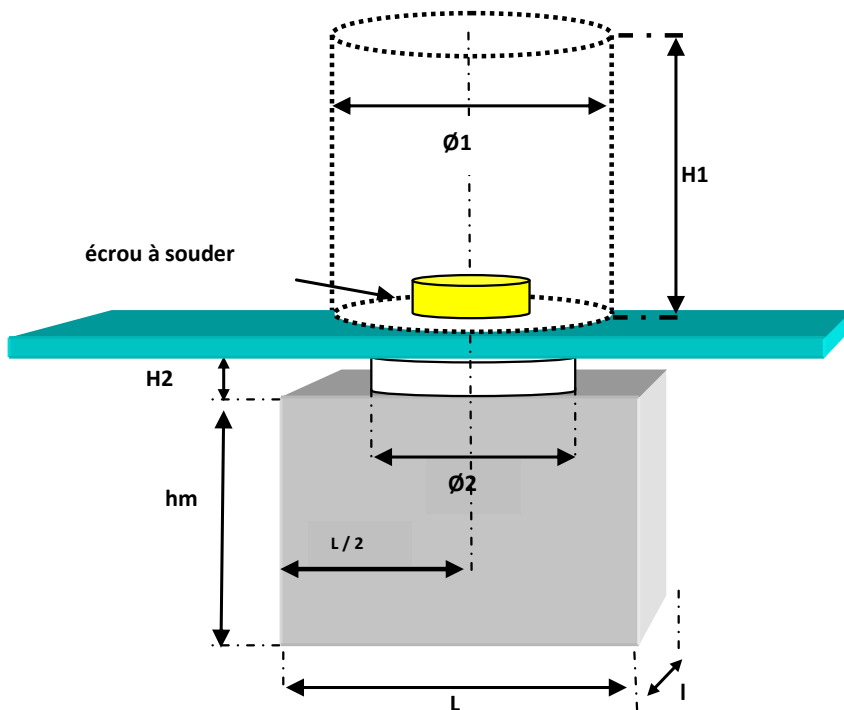


| Pour vis<br>D x pas | d<br>(0/+ 0,05) | S1<br>(0/- 0,2) | S2<br>$\pm 0,20$ | r<br>$\pm 0,20$ | m<br>$\pm 0,20$ | h1<br>$\pm 0,20$ | h2<br>$\pm 0,10$ |
|---------------------|-----------------|-----------------|------------------|-----------------|-----------------|------------------|------------------|
| M6 x 1,00           | 5,55            | 15,5            | 16               | 6               | 8               | 5,5              | 1,2              |
| M8 x 1,25           | 7,5             | 15,5            | 16               | 6               | 8               | 5,5              | 1,2              |

**Nuts implantation characteristics:**

| Hexagonal nuts to weld | e (mm) | B (mm) | Øp (mm) | A (mm)    |           | Mini in-between 2 nuts (air cylinder) (mm) |                   |                                |
|------------------------|--------|--------|---------|-----------|-----------|--------------------------------------------|-------------------|--------------------------------|
|                        |        |        |         | case 1    |           | case 2                                     | Welded on 2 tools | Welded on the same tool 6 bars |
|                        |        |        |         | h ≤ 21 mm | h > 21 mm |                                            |                   |                                |
| M5                     | 0,65   | 10     | 7,5     | 14,5      | 28        | 12,5                                       | 17,9              | 79                             |
| M6                     | 0,65   | 10,5   | 8,5     | 14,5      | 28        | 12,5                                       | 18,5              | 79                             |
| M8                     | 0,95   | 12     | 11      | 14,5      | 28        | 14,5                                       | 22,7              | 90                             |
| M10                    | 0,95   | 13,5   | 13      | 19,0      | 31        | 16,5                                       | 26,0              | 110                            |
| M12                    | 1,27   | 15     | 15,5    | 19,0      | 31        | 18,0                                       | 29,4              | 110                            |

- respect the minimal thickness of the steel sheet support : **e**
- guarantee a minimal board width: **B**
- define the diameter of steel sheet punching : **Øp**
- guarantee the **A** dimension for accessibility (control tool or welding tool)
- \* for the elastomer carrots see the PSA\_FER\_122 (01342 12 00210)



| Hexagonal<br>nuts to<br>weld | Higher volume |              | Lower volume |              |                                |    |    |
|------------------------------|---------------|--------------|--------------|--------------|--------------------------------|----|----|
|                              | Ø1 ( mm<br>)  | H1 ( mm<br>) | Ø2<br>( mm ) | H2<br>( mm ) | Counter<br>Electrode<br>( mm ) |    |    |
|                              |               |              |              |              | L                              | I  | hm |
| M5                           | 20            | 80           | 23           | 21           | 56                             | 56 | 67 |
| M6                           |               |              |              |              |                                |    |    |
| M8                           | 25            |              |              |              |                                |    |    |
| M10                          | 28            |              |              |              |                                |    |    |
| M12                          | 31,5          |              | 30           |              | 62                             | 56 |    |

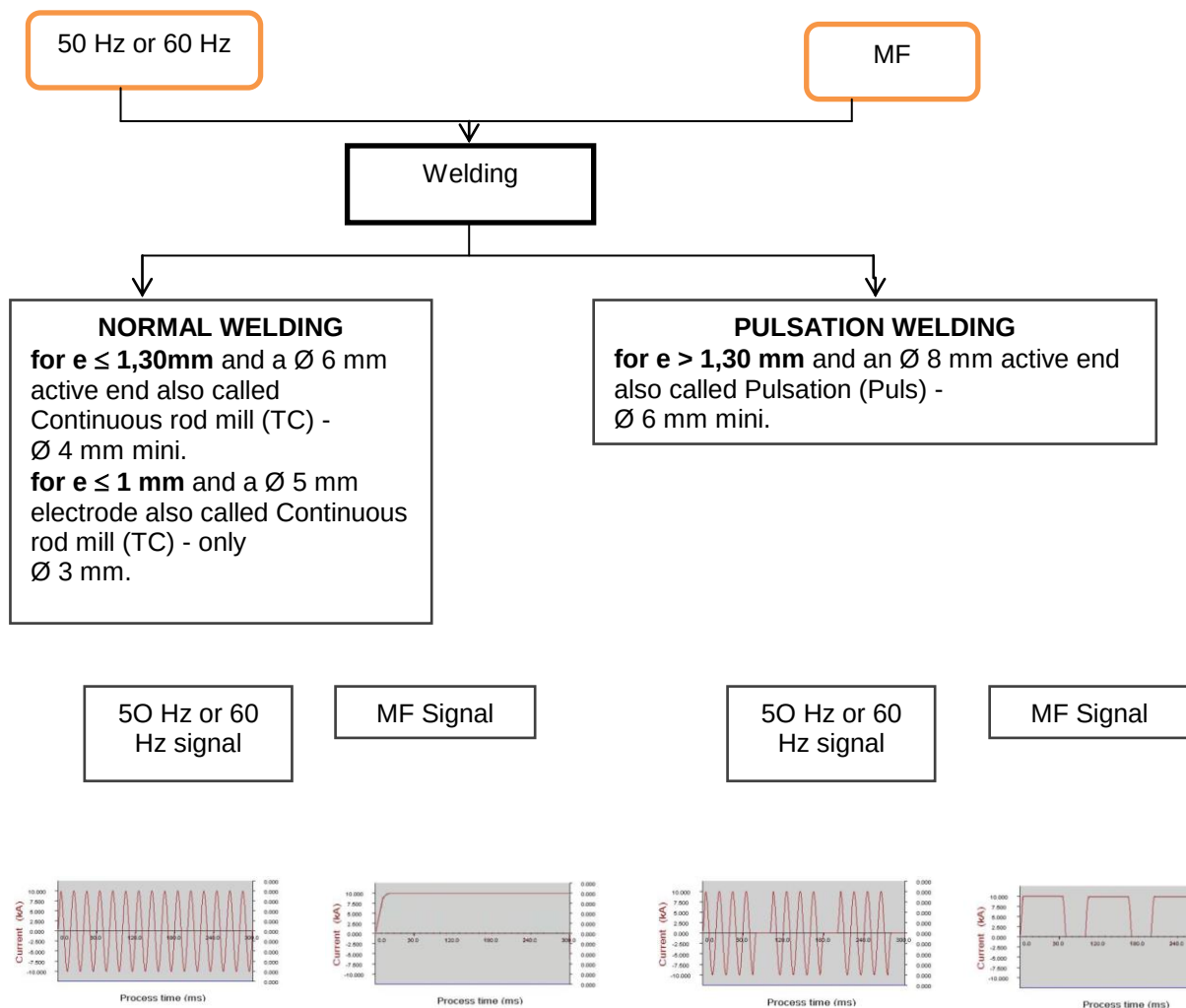
#### 4.4.TYPE OF WELDING

There are two types of welding:

There are two types of welding in 50Hz and in MF (normal welding and pulsation welding)

- Normal welding for  $e \leq 1,30$  mm and an  $\varnothing$  6 mm electrode also called Continuous rod mill (TC) - welding nugget :  $\varnothing$  4 mm mini.
- Normal welding for  $e \leq 1$  mm and a  $\varnothing$  5 mm active end also called Continuous rod mill (TC) – welding nugget :  $\varnothing$  3 mm.
- Pulsation welding for  $e > 1,30$  mm and an  $\varnothing$  8 mm electrode also called Pulsation (Puls) - welding nugget :  $\varnothing$  6 mm mini.

#### Normal and pulsation welding in 50Hz, 60Hz and in MF



**Starting parameters are the same in MF than in 50Hz or 60Hz**

## 4.5.WELDING TECHNIQUE

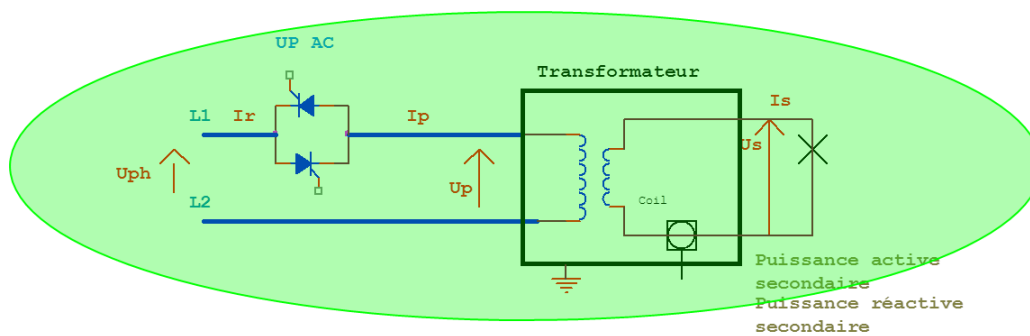
There are two kinds of welding current :

- Alternative (50Hz or 60 Hz (AC) depending on the countries).
- Direct, with the Frequency (**M.F.** (DC)) which can be used specifically after agreement of the Methods Door-fitting department.

(AC)\* : Alternatif Current

(DC)\* : Direct Current

### Welding principle 50Hz:

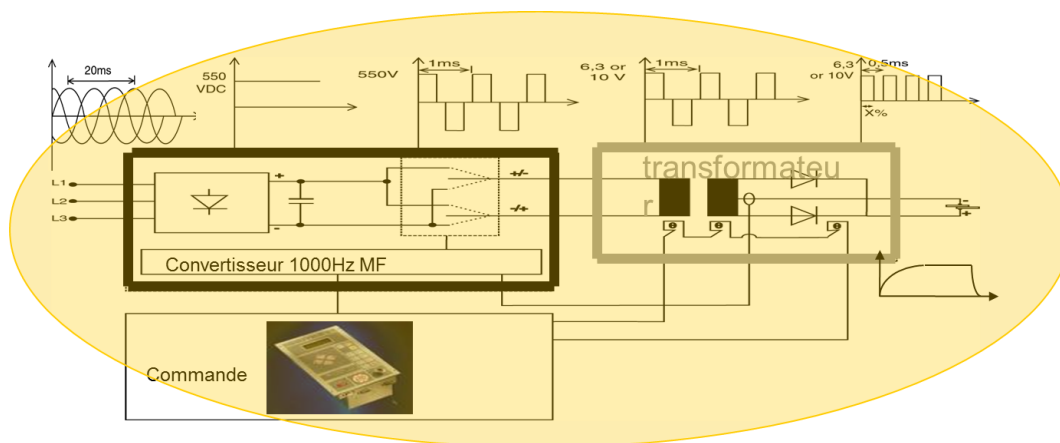


20ms regulation

**In 50Hz or 60Hz :**

- 1 Welding Programmable Command
- 1 thyristor panel
- 1 transformer

### Medium frequency welding principle:

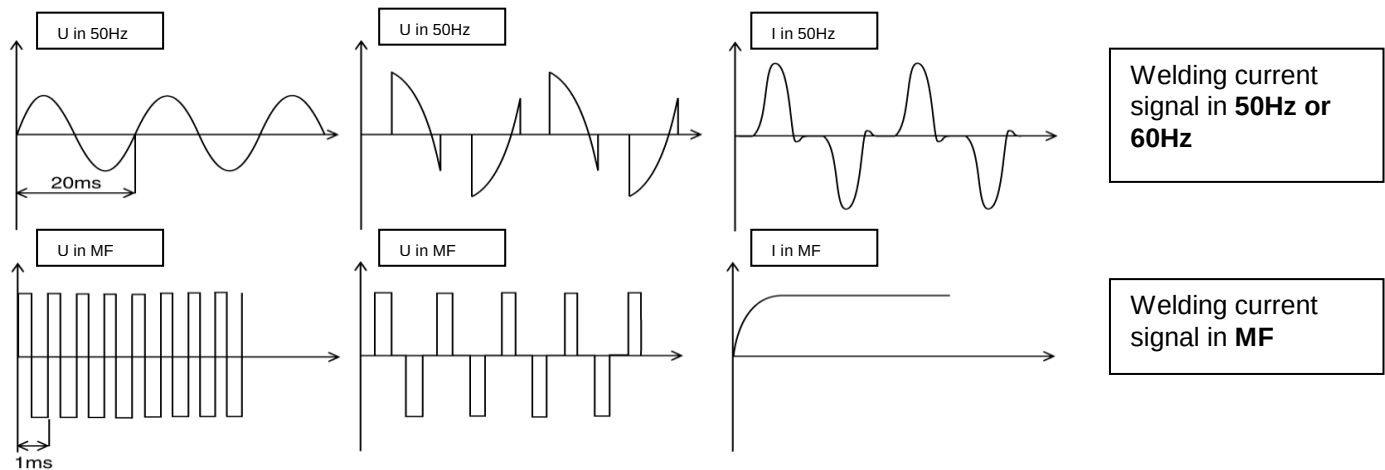


**In MF:**

- 1 Welding Programmable Command
- 1 Converter
- 1 transformer
- 1 diode bridge rectifier.

The three phase alimentation tension is straightened up, filtered and chopped at 1000Hz.

The regulation is possible from 1 ms.

**Shape of the welding current in 50Hz and in MF:****4.6.OUTSIDE THE NORMS OF PRODUCT & PROCESS WELDING**

For the design or the industrialization of the vehicle, any welding outside the norm requires a validation for:

- checking the guaranteed mini diameter, standard or lower,
- making sure that the expected service is guaranteed,
- defining the welding parameters,
- providing a report of test.

The results of this study will have to be recorded in a **PSE** file if there is a modification of the standard diameter.

**4.6.1.SPOT WELDING**

A Pt SE is considered outside the norm according to the below cases:

- Welding pulsation with active face electrode of  $\varnothing 6$  mm (see restrictive condition [§ 4.7](#)).
- Assembly 4 thicknesses (see restrictive condition [§ 4.7.1](#) et [§ 5.4](#)).
- No respect of norm PSA Peugeot Citroën [B13 1220](#).
- Weak force welding.
- Welding of new materials not validated by the qualified PSA Peugeot Citroën department (coatings, nature sheets); cf. over coding SAPHIR in docinfo (note [MXP\\_CEB00\\_0022](#)).

**4.6.2.WELDING SCREWS AND NUTS**

A screw or nut is considered outside the norm according to the following cases:

- Screws and nuts not in conformity with PSA Peugeot Citroën active norms: (STNG screw [01266\\_12\\_00682](#) and STNG nuts [01266\\_12\\_00912](#))
  - Assembly screw/sheet not in conformity (thickness definition of sheet min./max according to the diameters of the nuts).
  - Welding of screw or nut with a robot or a manual welding gun.

#### 4.6.3.EMBOSSMENT WELDING

(According to norm [B13 1230](#)).

An embossment is considered outside the norm according to the below cases:

- Assembly outside the norm (thickness of sheet, embossment except norm).
- Welding spot diameter not in conformity with norm PSA Peugeot Citroën [B13 1230](#).
- Embossment welding with a robot or a manual welding gun.

#### 4.7.MIXED RANGE

A mixed range is a welding range using the same working part (Ø 6 mm) to realize spots of normal welding (TC) (thick. ref. ≤ 1,3 mm) and spots of pulsation welding (thick. ref. > 1,30 mm).

For reasons of quality and Appearance, **the use of the mixed ranges is proscribed** at the time of the studies of new equipments within the framework of new vehicle projects.

Mixed range exclude **PtSE** in **ZF**

In case of product evolution or carry over, and after a validation of the calculation (**CCHE**), only the industrial processes specialist assisted by the person in charge of assembly fields can allow their use.

A derogation request must be addressed to them.

Mixte range involve the following restrictions:

- **The diminution of the welding spot diameter (from Ø6 mm to Ø5 mm) guaranteeing the provision, involving losses of mechanical strength of about 17% in pure traction and of 10% in traction shearing (the detail of these requests is in the norm [B13 1220](#)), => a validation CALCULATION REPORT is necessary for the PtSE Pulsation with a 5 mm diameter.**
- constraints of definition of welding equipment (electrodes, arm, motorizations of the robot and manual gun),
- increased general fees (from 20 to 40%) and penalized availabilities (frequency dressing constrain 330 even 290),
- a degradation of electric welding spots aspect.

#### 4.7.1.RESTRICTIONS OF USE

##### 4.7.1.1.Welding pulsation with Ø 6 mm electrodes of active face

Restrictions of industrialization to be taken into account :

| Electrodes D16/6 for force maxi 450 daN<br>(type P 911 310 112 and P 911 310 400)                                                           | Electrodes D20/6 for force maxi 600 daN<br>(type Z 000 484 257)                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• 30 % of pulsation spots maximum.</li> <li>• No more than 4 consecutive pulsation spots.</li> </ul> | <ul style="list-style-type: none"> <li>• 30 % of pulsation spots maximum.</li> <li>• No more than 4 consecutive pulsations spots .</li> </ul> |
| <ul style="list-style-type: none"> <li>• Water Flow of the manual welding gun: 4 l/min. minimum.</li> </ul>                                 | <ul style="list-style-type: none"> <li>• Water Flow of the manual welding gun: 6 l/min. minimum.</li> </ul>                                   |
| <ul style="list-style-type: none"> <li>• 6,5/100 per point of pulsation welding .</li> </ul>                                                | <ul style="list-style-type: none"> <li>• 6,5/100 per point of pulsation welding .</li> </ul>                                                  |

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 21/69 |
|----------------------------------------------------------|--------------|-------|

The parameters to be used are the welding parameters pulsation arranged for the electrode of active diameter of 6 mm according to the following rules:

**Rule N°1:** Force of unchanged welding.  
 $I = I_{\text{standard}}$   
 $T = \text{time of welding} - 1 \text{ train.}$

OR

**Rule N°2:** Force of unchanged welding.  
 $I = I_{\text{standard}} - 800 \text{ A.}$   
 $T = \text{time of standard welding.}$

OR

**Rule N°3:** Main conditions: matching of the steel sheets conform (included from family 2)  
 (the use of this rule is submitted to a derogation request)  
 $\text{Welding force} = \text{Force} - 20\%$   
 $\text{Intensity} = \text{Standard} - 1500 \text{ A}$   
 Welding time classic

4.7.1.2. Welding assembly thickness of reference  $\leq 1.30 \text{ mm}$  with tip electrodes of  $\varnothing 8 \text{ mm}$  of active face

**Strictly prohibited.**

4.7.1.3. Welding with tip active face  $\varnothing 5 \text{ mm}$  for  $\varnothing$  of PtSE other than 3 mm

The use of tip active face 5 mm is **strictly prohibited** to weld **PtSE** whose diameters of required minimal nugget are higher than 3 mm.

This makes **exclusive** the equipments of welding equipped with this kind of tip to the realization of **PtSE** of  $\varnothing 3 \text{ mm}$ .

#### 4.8. WELDING ON DEAD MECHANICAL WELDING GUN

A dead gun is a mechanical welding gun (use in MSA - to see [GE34-009G](#)) for which the current alimentation is made by a robot welding gun or a manual welding gun. The components used for the connection of the power supply of the dead welding gun as well as the rules of assemblies applied must respect the recommendations of the § "Secondary cables mounting" of the guide [GE34-009G](#).

The exploitation of this type of process is sensitive in mass production (management of the end of lifetime of electrodes, of the discs of supplies of current).

Their application will not be able to be made that in the last resorts after study of solutions of the types **M.S.A.** and a validation by the welding expert for the service Methods Door-fitting requires.

For their use, it is imperative to observe the following conditions:

- A specific program of welding.
- A management of the specific welding current incrementing (incrementing **MSA**).
- A specific management of the exchanges electrodes (**RAZ** partial of the timer).
- A minimum current ( $I_{\text{min.}}$ ) corresponding with a dephasing  $> 50 \%$ .

#### 4.8.1.IMPEDANCE

Valid only for welding in 50 Hz and 60 Hz (no valid in **MF**).

The impedance of the manual welding gun and of its secondary is to be added vectorially to the general calculation of the welding element transformer (robot welding gun or manual welding gun).

$$Z_{\text{total}} = \sqrt{(R_{\text{palim}} + R_{\text{pm}})^2 + (L\omega_{\text{palim}} + L\omega_{\text{pm}})^2}$$

With :

- $R_{\text{palim}}$  : Resistance of the robot welding gun or manual.
- $L\omega_{\text{palim}}$  : Inductance of the robot welding gun or manual.
- $R_{\text{pm}}$  : Resistance of the dead welding gun.
- $L\omega_{\text{pm}}$  : Inductance of the dead welding gun.

#### 4.8.2.TRANSFORMER OF THE ROBOT WELDING GUN OR MANUAL

To decrease the size of the main transformers, it is necessary to reduce to the maximum the impedance of the **dead** welding gun and of its secondary.

For that, it is necessary to use gun of small size and to privilege the silver shunts rather than of the braids.

Supply of current must be fixed on a fixed part independent of the welding gun in order to avoid important constraints on the operation of the latter (balancing, closing...).

## 4.8.3.SUPPLY OF CURRENT

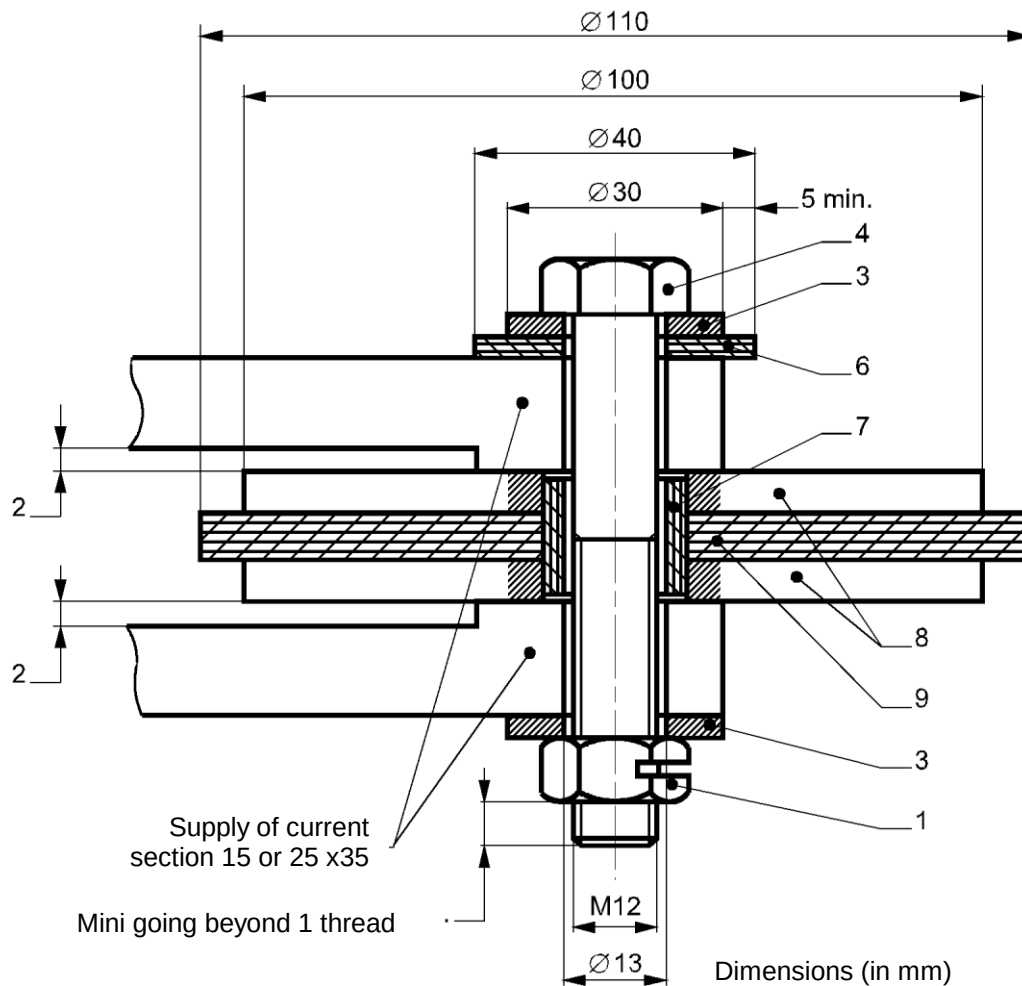


Figure 1

| Reference mark | Designation                                                                                                                                                                                              | MABEC Number           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1              | Self-locking nut                                                                                                                                                                                         | X 484 113 504          |
| 3              | Flat Disc                                                                                                                                                                                                | P 909 517 600          |
| 4              | Screw H12 classifies 8/8 - Length according to thickness of the assembly<br>(Note : for specific cases, it is possible to use screws H12 class 10,4 with a wrench torque same to the class 8,8 : 75 Nm). | -                      |
| 6              | Insulating disc                                                                                                                                                                                          | Z 000 165 800          |
| 7              | Insulating ring – cut to the length according to thickness of the element to be insulated                                                                                                                | Taken in X 213 811 706 |
| 8              | Plate of wear                                                                                                                                                                                            | Z 000 268 677          |
| 9              | Insulating slab                                                                                                                                                                                          | Z 000 291 479          |

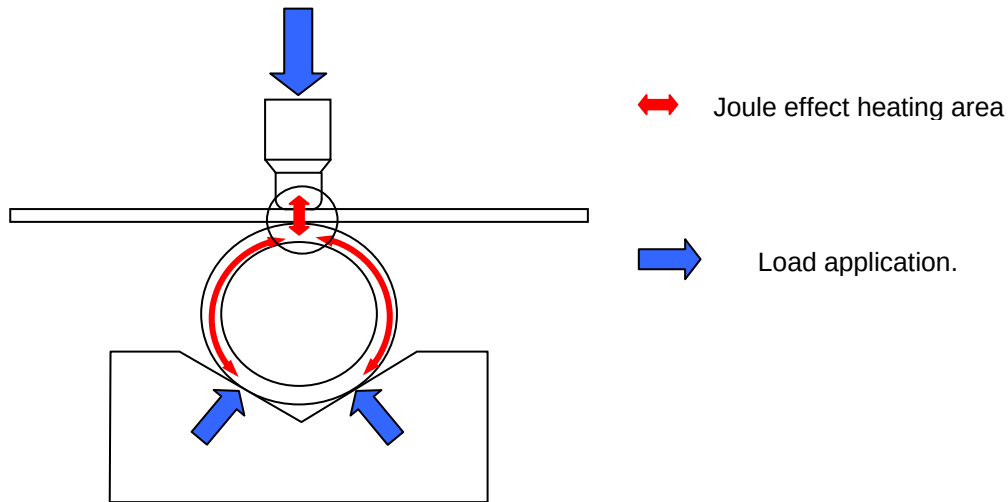
Table 3

## 4.9.SHEET WELDED ON TUBE

### 4.9.1.PRINCIPLE

This type of welding spot is also called "natural embossment".

Sheet assembly on tube.



4.9.2. THICKNESS RATIO → SEE NORM [B13 1220](#)

### 4.9.3.ELECTRODES

**Side sheet:** It is a standard tip of Ø 6 mm active face.

**Side tube:** It is in the shape of "V" which allows to:

- center the tube,
- limit the heating apart from the welding area and thus not to deteriorate the material with the contact tube/electrode.

## 4.10.CHARACTERISTICS FOR WELDING SHEETS

### 4.10.1.CLASSIFICATION OF THE SHEET SHADES

(See norm [B13 1220](#)).

The sheets shades referred in group PSA Peugeot Citroën are classified for weldability according to following families:

| Family 1 | Shades          | Family 2 | Shades      | Family 3 | Shades    | Family 2bis | Shades    |
|----------|-----------------|----------|-------------|----------|-----------|-------------|-----------|
|          | C, E and ES     |          | MS1500      |          | TRIP800 * |             | P220/P260 |
|          | E+BH            |          |             |          |           |             | DP450     |
|          | 1C, 3C          |          |             |          |           |             | DP600     |
|          | E275D / HD300LA |          | DP780       |          |           |             | DP590     |
|          | HE275           |          | DP980       |          |           |             | -         |
|          | E335D / HD340LA |          | DP1180      |          |           |             |           |
|          | E390D / HD380LA |          | HR45 / FB45 |          |           |             |           |
|          | HE335D          |          | HR60 / FB60 |          |           |             |           |
|          | HE390D          |          | 22MnB5      |          |           |             |           |
|          | HE445D          |          | CP800       |          |           |             |           |
|          | HD300LA         |          |             |          |           |             |           |
|          | HD340LA         |          |             |          |           |             |           |
|          | HD380LA         |          |             |          |           |             |           |

The coding of the steel shade is described in this document [MXP\\_CEB00\\_0022](#).

\*: This nuance is validated in welding only for thicknesses  $\leq 1.3$  mm with strong constraints on the parameters (long time and cycle of annealing) and is the object of a unit validation of all the configurations of assembly by the specialists in the entity door-fitting.

The various families impose a specific parameter setting on these assemblies, which requires regroupings and assignments of spots on specific programs.

The rules of assemblies and parameters will be defined progressively after homologations and validations. The indications specified above are used as orientation for the projects development.

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 26/69 |
|----------------------------------------------------------|--------------|-------|

| Families    | Welding parameters                                                                                                                                                                                                                                  |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Family 1    | <ul style="list-style-type: none"> <li>According to parameters of the § 5.</li> </ul>                                                                                                                                                               |
| Family 2    | <ul style="list-style-type: none"> <li><b>TS</b> = standard + 30 %.</li> <li>Force = standard + 25 %.</li> <li><b>TM*</b> = + 2 ~ in Continuous rod mill.</li> <li><b>TM</b> = + 3 ~ in Pulsation.</li> <li>Intensity = standard - 10 %.</li> </ul> |
| Family 2bis | <ul style="list-style-type: none"> <li>TS = standard + 30%</li> <li>Force = standard + 25%</li> <li><b>TM*</b> = + 2 ~ in Continuous rod mill</li> <li><b>TM</b> = +3 ~ in Pulsation</li> <li>Intensity = standard</li> </ul>                       |
| Family 3    | <ul style="list-style-type: none"> <li>The welding parameters are not to date defined.</li> <li>The use of time of preheats and of post heating in special cycles can be necessary. This involves a strong increase in the cycle times.</li> </ul>  |

**Note:** The round-offs of the new parameters will be done with the values immediately above (the unit for the welding times and about ten for the forces).

Forces:  $270 \text{ daN} + 25\% = 337,5 \text{ daN} \rightarrow$  value selected 340 daN.

Times of welding:  $12 \sim + 30\% = 15,6 \sim \rightarrow$  value selected 16 ~.

In the case of welding pulsation, the rule of calculation of the time of welding is as follows:

- Standard TS =  $4(5 \sim + 1)$ .
- Standard TS + 30 % =  $4(6,5 \sim + 1) \rightarrow$  value selected  $4(7 \sim + 1)$ .

\* TM: Holding time.

#### 4.10.2.DISSYMMETRICAL ASSEMBLIES

(See norm [B13 1220](#)).

**Symmetrical assembly:** Assembly answering to one of the following cases:

- Same family of sheets for three sheets: for example F1/F1/ F1.
- A sheet of family 2 (in the middle) for example F1/F2/ F1.
- Two sheets of the family 2 external: for example F2/F1/ F2.

**Dissymmetrical assembly:** Assembly answering to one of the same family:

- In two sheets: when the 2 sheets do not belong to the same family.
- In three sheets: an external sheet of a family different from both others: for examples F1 /F1/ F2 or F1/F2/ F2.

For these assemblies, the thickness ratio thickness between thin and strong sheet must be conforms to the norm [B13 1220](#).

A confirmation must be made individually by weldability tests carried out by the qualified PSA Peugeot Citroën department.

#### 4.10.3.WELDING OF ALUMINUM

General information to specify according to progressively applications projects.

You have to contact the industrial processes specialist in every cases for a parameters validation.

- **Materials**

Grade 5182, 5754, 6016.

Thickness from 1 to 1,5 mm.

- **Electrodes**

Use imperatively those definite with standard [GE34-007G](#) ( Z000 329 963, cylindrical tip Ø active of 12 mm, R=80mm).

- **Welding technique**

For the welding of aluminum sheet, the welding techniques 50 Hz or medium frequency (MF) can be used in mode simple point.

For this last Welding technique, when the Appearance of the point is specified with the drawing (tidy presentation), it is imperative that the electrode of polarity  $\ominus$  (Negative) is on the visible side of the welding spot.

- **Welding parameters (order of magnitude)**

Time of welding (TS):  $\approx$  4 periods ( $\sim$ ).

Force:  $\approx$  200daN.

Intensity from 23 to 25 kA for the grade 5000.

Intensity from 28 to 30 kA for the grade 6000.

- **Dressing (application known to date)**

Range 5000 in MF: the frequency of dressing is to be asked to the industrial processes specialist

Range 6000 in 50 Hz: this grade creates sticking of electrode on sheet which imposes an incrementing of the welding current between two dressings. The dressing frequency is to be asked to the industrial processes specialist.

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 28/69 |
|----------------------------------------------------------|--------------|-------|

## 5.WELDING PARAMETERS

The welding parameters (intensity, force, time) for the assemblies of two or three thicknesses are given for each thickness of reference.

These values are not given like absolute values, but like recommendations having to be used as a basis for the study of the welding equipments (determination of the secondary circuits of the transformers and estimate of the load of the welding cabinet or with the determination of the transformers of station with manual welding gun).

The welding parameters double point push-pull are identical to the parameters simple point.

### 5.1.WELDING PARAMETERS CONTINUOUS ROD MILL FOR THE 50HZ APPLICATIONS - SE PTS OF Ø 3 MM

| Thickness of reference mm | Ass 2 thicknesses                        |                                         | Ass 3 thicknesses                        |                                         | Holding time | Force daN (1) | Intensity KA with sheets |     |     |
|---------------------------|------------------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------|--------------|---------------|--------------------------|-----|-----|
|                           | < 4 rev and 2 <sup>nd</sup> T < 1,47 mm* | ≥ 4 rev or 2 <sup>nd</sup> T = 1,47 mm* | < 4 rev and 3 <sup>rd</sup> T < 1,47 mm* | ≥ 4 rev or 3 <sup>rd</sup> T = 1,47 mm* |              |               | Nb of coated faces 10µm  |     |     |
|                           | E maxi 1,47 mm*                          |                                         |                                          |                                         |              |               | 2                        | 4   | 6   |
| 0,6                       | 6                                        | 8                                       | 8                                        | 10                                      | 6            | 160           | 6,8                      | 7,7 | 8,4 |
| 0,7                       | 8                                        | 10                                      | 10                                       | 12                                      | 8            | 180           | 6,8                      | 7,8 | 8,5 |
| 0,8                       | 8                                        | 10                                      | 10                                       | 12                                      | 8            | 200           | 7                        | 8   | 8,6 |
| 0,9 **                    | 8                                        | 10                                      | 10                                       | 12                                      | 8            | 210           | 7,3                      | 8,2 | 8,8 |
| 1 **                      | 10                                       | 12                                      | 12                                       | 12                                      | 10           | 230           | 7,5                      | 8,4 | 9   |

**Table 4 : Class N (for electrode active face Ø 5 mm)**

(1) Force limits of behavior of the cone of the shanks 250daN.

\* Real thickness (it is the «true thickness» of the metal sheet which is in the table Chapter 4.2 and from which are defined the thicknesses for RESORE calculations.).

\*\* Not applicable for Family 2 and 2bis

**Note:** These parameters are valid only for thicknesses of coatings  $\leq 10 \mu\text{m}$  and a standard aspect. Any other configuration of outgoing assembly of these conditions must be the object of test of validation by service PSA Peugeot Citroën AssemblyTechnical entitled.

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 29/69 |
|----------------------------------------------------------|--------------|-------|

## 5.2.WELDING PARAMETERS CONTINUOUS ROD MILL FOR APPLICATIONS 50 HZ - PTS SE Ø 4 MM

### 5.2.1.COATINGS ≤ 10 µm

Parameters for sheet family 1 (F1):

| Thickness<br>of<br>reference<br>mm | Time of welding ( ~ )                             |                                                  |                                                     |                                                   | Holding time<br>~ | Force<br>daN<br>(1) | Intensity (kA) with sheets                     |      |      |      |      |      |      |
|------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|-------------------|---------------------|------------------------------------------------|------|------|------|------|------|------|
|                                    | Ass. 2 thick                                      |                                                  | Ass. 3 thick                                        |                                                   |                   |                     | Number of anti-corrosive coated faces<br>10 µm |      |      |      |      |      |      |
|                                    | < 4 rev and<br>2 <sup>nd</sup> sheet<br>≤ 1,5 mm* | = 4 rev or<br>2 <sup>nd</sup> sheet<br>> 1,5 mm* | < 4 rev and<br>3 <sup>rd</sup> sheet<br>≤ 1,95 mm * | ≥ 4 rev or<br>3 <sup>rd</sup> sheet<br>> 1,95 mm* |                   |                     |                                                |      |      |      |      |      |      |
|                                    | 0                                                 | 1                                                | 2                                                   | 3                                                 |                   |                     | 4                                              | 5    | 6    |      |      |      |      |
| 0,6                                | 8                                                 | 10                                               | 10                                                  | 12                                                | 8                 | 210                 | 8,5                                            | 9,0  | 9,7  | 10,5 | 11,0 | 11,5 | 12,0 |
| 0,7                                | 8                                                 | 10                                               | 10                                                  | 12                                                | 8                 | 210                 | 8,5                                            | 9,0  | 9,7  | 10,5 | 11,0 | 11,5 | 12,0 |
| 0,8                                | 8                                                 | 10                                               | 10                                                  | 12                                                | 8                 | 230                 | 8,6                                            | 9,3  | 10,0 | 10,7 | 11,2 | 11,8 | 12,3 |
| 0,9                                | 8                                                 | 10                                               | 10                                                  | 12                                                | 8                 | 250                 | 8,7                                            | 9,7  | 10,4 | 11,0 | 11,6 | 12,2 | 12,6 |
| 1,0                                | 10                                                | 12                                               | 12                                                  | 14                                                | 10                | 270                 | 8,8                                            | 9,8  | 10,5 | 11,0 | 11,6 | 12,2 | 12,7 |
| 1,2                                | 10                                                | 12                                               | 12                                                  | 14                                                | 10                | 300                 | 9,0                                            | 10,3 | 11,0 | 11,5 | 12,0 | 12,6 | 13,2 |

Table 5 : Class N (for end of active face Ø 6 mm)

Parameters for sheet family 2 (F2):

| Thickness of reference mm | Time of welding ( ~ )                      |                                           |                                              |                                            | Holding time ~ | Force daN (1) | Intensity (kA) with sheets                     |     |     |      |      |      |      |
|---------------------------|--------------------------------------------|-------------------------------------------|----------------------------------------------|--------------------------------------------|----------------|---------------|------------------------------------------------|-----|-----|------|------|------|------|
|                           | Ass. 2 thick                               |                                           | Ass. 3 thick                                 |                                            |                |               | Number of anti-corrosive coated faces<br>10 µm |     |     |      |      |      |      |
|                           | <4 rev and 2 <sup>nd</sup> sheet ≤ 1,5 mm* | =4 rev or 2 <sup>nd</sup> sheet > 1,5 mm* | <4 rev and 3 <sup>rd</sup> sheet ≤ 1,95 mm * | ≥4 rev or 3 <sup>rd</sup> sheet > 1,95 mm* |                |               |                                                |     |     |      |      |      |      |
|                           | 0                                          | 1                                         | 2                                            | 3                                          |                |               | 4                                              | 5   | 6   |      |      |      |      |
| 0,6                       | 10                                         | 13                                        | 13                                           | 16                                         | 10             | 260           | 7,6                                            | 8,1 | 8,7 | 9,4  | 9,9  | 10,3 | 10,8 |
| 0,7                       | 10                                         | 13                                        | 13                                           | 16                                         | 10             | 260           | 7,6                                            | 8,1 | 8,7 | 9,4  | 9,9  | 10,3 | 10,8 |
| 0,8                       | 10                                         | 13                                        | 13                                           | 16                                         | 10             | 290           | 7,7                                            | 8,4 | 9   | 9,6  | 10,1 | 10,6 | 11,1 |
| 0,9                       | 10                                         | 13                                        | 13                                           | 16                                         | 10             | 310           | 7,8                                            | 8,7 | 9,4 | 9,9  | 10,4 | 10,9 | 11,3 |
| 1,0                       | 13                                         | 16                                        | 16                                           | 18                                         | 12             | 340           | 7,9                                            | 8,8 | 9,5 | 9,9  | 10,4 | 10,9 | 11,4 |
| 1,2                       | 13                                         | 16                                        | 16                                           | 18                                         | 12             | 375           | 8,1                                            | 9,2 | 9,9 | 10,3 | 10,8 | 11,3 | 11,9 |

Table 6 : Class N (for end of active face Ø 6 mm)

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 30/69 |
|----------------------------------------------------------|--------------|-------|

## Parameters for sheets family 2bis (F2bis):

| Reference thickness<br>mm | Welding time (~)                                  |                                                  |                                         |                                       | Holding time<br>~ | Force<br>daN<br>(1) | Intensity (kA) with sheets      |      |      |      |      |      |      |
|---------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------------------|---------------------------------------|-------------------|---------------------|---------------------------------|------|------|------|------|------|------|
|                           | Ass. 2 thicknesses                                |                                                  | Ass. 3 thicknesses                      |                                       |                   |                     | Number of coated sides<br>10 µm |      |      |      |      |      |      |
|                           | <4 coat and<br>2 <sup>nd</sup> sheet<br>≤ 1,5 mm* | =4 coat or<br>2 <sup>nd</sup> sheet<br>> 1,5 mm* | <4 coat and<br>3rd sheet<br>≤ 1,95 mm * | ≥4 coat or<br>3rd sheet<br>> 1,95 mm* |                   |                     |                                 |      |      |      |      |      |      |
|                           | 0                                                 | 1                                                | 2                                       | 3                                     |                   |                     | 4                               | 5    | 6    |      |      |      |      |
| 0,6                       | 10                                                | 13                                               | 13                                      | 16                                    | 10                | 260                 | 8,5                             | 9,0  | 9,7  | 10,5 | 11,0 | 11,5 | 12   |
| 0,7                       | 10                                                | 13                                               | 13                                      | 16                                    | 10                | 260                 | 8,5                             | 9,0  | 9,7  | 10,5 | 11,0 | 11,5 | 12   |
| 0,8                       | 10                                                | 13                                               | 13                                      | 16                                    | 10                | 290                 | 8,6                             | 9,3  | 10,0 | 10,7 | 11,2 | 11,8 | 12,3 |
| 0,9                       | 10                                                | 13                                               | 13                                      | 16                                    | 10                | 310                 | 8,7                             | 9,7  | 10,4 | 11,0 | 11,6 | 12,2 | 12,6 |
| 1,0                       | 13                                                | 16                                               | 16                                      | 18                                    | 12                | 340                 | 8,8                             | 9,8  | 10,5 | 11,0 | 11,6 | 12,2 | 12 7 |
| 1,2                       | 13                                                | 16                                               | 16                                      | 18                                    | 12                | 375                 | 9,0                             | 10,3 | 11,0 | 11,5 | 12,0 | 12,6 | 13,2 |

Table 7

For welding on counter electrode (C.E.), add 1000 A following note DMEI/CEI/EVC N° 4277 of 12/89.

(1) Force limits of behavior of the cone of the shanks 450daN.

**Important:** The optimization of holding times is **PROHIBITED** for the assemblies comprising of sheets of F2 families.

**Note:** F1/F2: sheet of Family 1, sheet of Family 2 or 2bis.

\*: Real thickness.

## 5.2.2.COATINGS &gt; 10 µM

Starting from tables 4, apply the following conditions:

- Intensity with sheets: + 500 A.
- Time of welding: + 2 ~.

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 31/69 |
|----------------------------------------------------------|--------------|-------|

### 5.3.PULSATION WELDING PARAMETERS FOR APPLICATIONS 50 HZ

(For electrode Ø active face 8 mm).

Table of parameters for sheet family 1 (F1):

| Thickness<br>of<br>reference<br>mm | Time of welding<br>~ |                 | Holding<br>time ~ | Force<br>daN<br>(1) | Thickness<br>of<br>strongest<br>sheet<br>(except<br>thick. of<br>ref)<br>mm | Intensity (kA) with sheets<br>Thickness of the finest sheet (mm) |      |      |      |      |      |      |      |      |      |      |      |
|------------------------------------|----------------------|-----------------|-------------------|---------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|------|
|                                    | Ass.<br>3 thick      | Ass.<br>2 thick |                   |                     |                                                                             | 0                                                                | 0,6  | 0,7  | 0,8  | 0,9  | 1    | 1,2  | 1,5  | 2    | 2,5  | 3    |      |
| 1,5                                | 4(5+1)               | 4(4+1)          | 12                | 400                 | 1,5                                                                         | 12,5                                                             | 12,5 | 12,5 | 12,5 | 12,5 | 12,5 | 12,5 | 12,6 |      |      |      |      |
|                                    |                      | 4(5+1)          |                   |                     | 2                                                                           | 12,7                                                             | 12,7 | 12,7 | 12,7 | 12,7 | 12,8 | 12,8 | 13,0 |      |      |      |      |
|                                    |                      |                 |                   |                     | 2,5                                                                         | 12,8                                                             | 12,8 | 12,8 | 12,8 | 12,8 | 12,9 | 12,9 | 13,3 |      |      |      |      |
|                                    |                      |                 |                   |                     | 3                                                                           | 13,0                                                             | 13,0 | 13,0 | 13,0 | 13,0 | 13,2 | 13,2 | 13,5 |      |      |      |      |
| 2                                  | 5(5+1)               | 5(4+1)          | 15                | 450                 | 2                                                                           | 12,2                                                             | 12,2 | 12,3 | 12,3 | 12,3 | 12,3 | 12,4 | 12,6 |      |      |      |      |
|                                    |                      | 5(5+1)          |                   |                     | 2,5                                                                         | 12,6                                                             | 12,6 | 12,7 | 12,7 | 12,7 | 12,7 | 12,7 | 13,0 |      |      |      | 13,0 |
|                                    |                      |                 |                   |                     | 3                                                                           | 13,2                                                             | 13,2 | 13,4 | 13,4 | 13,4 | 13,4 | 13,4 | 13,5 |      |      |      | 13,5 |
| 2,5                                | 5(6+1)               | 5(5+1)          | 15                | 500                 | 2,5                                                                         | 13,5                                                             | 13,5 | 13,5 | 13,5 | 13,5 | 13,5 | 13,5 | 13,5 | 13,5 | 13,6 |      |      |
|                                    |                      |                 |                   |                     | 3                                                                           | 13,6                                                             | 13,6 | 13,6 | 13,6 | 13,6 | 13,6 | 13,6 | 13,7 | 13,8 | 14,0 |      |      |
| 3                                  | 6(6+1)               | 6(5+1)          | 20                | 550                 | 3                                                                           | 13,8                                                             | 13,8 | 13,8 | 13,8 | 13,8 | 13,8 | 13,8 | 13,8 | 13,9 | 14,0 | 14,2 |      |

Table 8

Table of parameters for sheet family 2 (F2):

| Thickness of reference mm | Time of welding ~ |              | Holding time ~ | Force daN (1) | Thicknes s of strongest sheet (except thick.. of ref) mm | Intensity (kA) with sheets<br>Thickness of the finest sheet (mm) |      |      |      |      |      |      |      |      |      |      |  |
|---------------------------|-------------------|--------------|----------------|---------------|----------------------------------------------------------|------------------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|--|
|                           | Ass. 3 thick      | Ass. 2 thick |                |               |                                                          | 0                                                                | 0,6  | 0,7  | 0,8  | 0,9  | 1    | 1,2  | 1,5  | 2    | 2,5  | 3    |  |
| 1,5                       | 4(7+1)            | 4(5+1)       | 15             | 500           | 1,5                                                      | 11,2                                                             | 11,2 | 11,2 | 11,2 | 11,2 | 11,2 | 11,2 | 11,3 |      |      |      |  |
|                           |                   | 2            |                |               | 11,4                                                     | 11,4                                                             | 11,4 | 11,4 | 11,4 | 11,5 | 11,5 | 11,7 |      |      |      |      |  |
|                           |                   | 2,5          |                |               | 11,5                                                     | 11,5                                                             | 11,5 | 11,5 | 11,5 | 11,6 | 11,6 | 11,9 |      |      |      |      |  |
|                           |                   | 3            |                |               | 11,7                                                     | 11,7                                                             | 11,7 | 11,7 | 11,7 | 11,9 | 11,9 | 12,1 |      |      |      |      |  |
| 2                         | 5(7+1)            | 5(5+1)       | 18             | 560           | 2                                                        | 10,9                                                             | 10,9 | 11   | 11   | 11,1 | 11,1 | 11,1 | 11,2 | 11,3 |      |      |  |
|                           |                   | 2,5          |                |               | 11,3                                                     | 11,3                                                             | 11,4 | 11,4 | 11,4 | 11,4 | 11,4 | 11,7 | 11,7 |      |      |      |  |
|                           |                   | 3            |                |               | 11,9                                                     | 11,9                                                             | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 |      |      |  |
| 2,5                       | 5(8+1)            | 5(7+1)       | 18             | 625* kept 600 | 2,5                                                      | 12,1                                                             | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 | 12,1 | 12,2 |      |  |
|                           |                   |              |                |               | 3                                                        | 12,2                                                             | 12,2 | 12,2 | 12,2 | 12,2 | 12,2 | 12,2 | 12,3 | 12,4 | 12,6 |      |  |
| 3                         | 6(8+1)            | 6(7+1)       | 23             | 690* kept 600 | 3                                                        | 12,4                                                             | 12,4 | 12,4 | 12,4 | 12,4 | 12,4 | 12,4 | 12,4 | 12,5 | 12,6 | 12,7 |  |

Table 9

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 32/69 |
|----------------------------------------------------------|--------------|-------|

## Tableau de paramètres pour tôle famille 2bis (F2bis):

Tableau 10

| Thickness of reference mm | Welding time ~ |              | Holding time ~ | Force daN (1)    | Thicknes s of strongest sheet (except thick.. of ref) mm | Intensity (kA) with sheets         |      |      |       |      |      |      |      |      |      |      |  |
|---------------------------|----------------|--------------|----------------|------------------|----------------------------------------------------------|------------------------------------|------|------|-------|------|------|------|------|------|------|------|--|
|                           | Ass. 3 thick   | Ass. 2 thick |                |                  |                                                          | Thickness of the finest sheet (mm) |      |      |       |      |      |      |      |      |      |      |  |
|                           |                |              |                |                  |                                                          | 0                                  | 0,6  | 0,7  | 0,8   | 0,9  | 1    | 1,2  | 1,5  | 2    | 2,5  | 3    |  |
| 1,5                       | 4(7+1)         | 4(5+1)       | 15             | 500              | 1,5                                                      | 12,5                               | 12,5 | 12,5 | 12,5  | 12,5 | 12,5 | 12,5 | 12,6 |      |      |      |  |
|                           |                | 2            |                |                  | 12,7                                                     | 12,7                               | 12,7 | 12,7 | 12,74 | 12,8 | 12,8 | 13,0 |      |      |      |      |  |
|                           |                | 2,5          |                |                  | 12,8                                                     | 12,8                               | 12,8 | 12,8 | 12,8  | 12,9 | 12,9 | 13,3 |      |      |      |      |  |
|                           |                | 4(7+1)       |                |                  | 3                                                        | 13,0                               | 13,0 | 13,0 | 13,0  | 13,0 | 13,2 | 13,2 | 13,5 |      |      |      |  |
| 2                         | 5(7+1)         | 5(5+1)       | 18             | 560              | 2                                                        | 12,2                               | 12,2 | 12,3 | 12,3  | 12,3 | 12,3 | 12,3 | 12,4 | 12,6 |      |      |  |
|                           |                | 5(7+1)       |                |                  | 2,5                                                      | 12,7                               | 12,7 | 12,7 | 12,7  | 12,7 | 12,7 | 12,7 | 13,0 | 13,0 |      |      |  |
|                           |                | 3            |                |                  | 13,2                                                     | 13,2                               | 13,4 | 13,4 | 13,4  | 13,4 | 13,4 | 13,4 | 13,5 | 13,5 |      |      |  |
| 2,5                       | 5(8+1)         | 5(7+1)       | 18             | 625*<br>kept 600 | 2,5                                                      | 13,5                               | 13,5 | 13,5 | 13,5  | 13,5 | 13,5 | 13,5 | 13,5 | 13,5 | 13,6 |      |  |
|                           |                |              |                |                  | 3                                                        | 13,6                               | 13,6 | 13,6 | 13,6  | 13,6 | 13,6 | 13,6 | 13,7 | 13,8 | 14,0 |      |  |
| 3                         | 6(8+1)         | 6(7+1)       | 23             | 690*<br>kept 600 | 3                                                        | 13,8                               | 13,8 | 13,8 | 13,8  | 13,8 | 13,8 | 13,8 | 13,8 | 13,9 | 14,0 | 14,2 |  |

4(5+1) = 4 continuous rod mills 5 hot periods followed by one period of cold time.

**Note :** (1) Force limits of behaviour of the cone of the shank 600 daN.

**Note :** \* : For these assemblies, RESORE calculates the force to 600 daN

625 daN\*: the shanks must be straight.

690 daN\*: there are not standard shanks and electrodes (cone) supporting this force.

**Important :** The optimization of holding timed is **PROHIBITED** for sheets of family 2 (see note [MXP CEB07 0644](#)).

**Note :** The limits of acceptance of the thickness ratio and position of sheets between them are defined in the standard [B13 1220](#).

**Note :** Significance of the code of color of the table (see in [B13 1220](#))

|                                                                 |                     |       |
|-----------------------------------------------------------------|---------------------|-------|
| <b>ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS</b> | <b>E34.03.180.G</b> | 33/69 |
|-----------------------------------------------------------------|---------------------|-------|

If the thickness of reference and 2<sup>nd</sup> sheet, the strongest, are of family 1:



Valid choices



3<sup>rd</sup> sheet, the thinnest, must being imperatively in inter sheet so that the assembly is realizable.



Outside the norm assembly

If the thickness of reference and/or 2<sup>nd</sup> sheet, the strongest, is of family 2 and/or if there is presence of glue at the interface opposed to the thin sheet (ex. E Soudo Col / E / e) :



Valid choice



3<sup>rd</sup> sheet, the thinnest, must being imperatively in inter sheet so that the assembly is realizable.



Outside the norm assembly

For exceptional cases of assemblies leaving these requirements, it is necessary to consult service PSA Peugeot Citroën entitled to validate the adopted solution.

For exceptional cases of assemblies leaving these requirements, it is necessary to consult service Assembly Technical Expertise to validate the adopted solution.

#### 5.4.WELDING OF THE ASSEMBLIES 4 THICKNESSES IN 50HZ

The perimeter of these assemblies is limited by the respect of the recommendations of the standard [B13 1220](#) Chap. 5.5.5 and the card [01353 09 00111](#).

Feasibility welding must be validated by Responsible welding methods door-fitting.

Determination thickness of reference: the thinnest of the 2 thickest sheets.

The interval of dressing must be limited: see § [8.12.7](#)

#### 5.5.WELDING TECHNIQUE 60 HZ

Times of welding are to be raised of 20 %. The intensities remain unchanged.

**5.6.LEVEL OF REGROUPING FOR WELDING PROGRAMS**

Welding in 50Hz:

Table 11

| Levels of regroup. | Applicable conditions only for sheets of family 1                                                 | Robot      | M.S.A.*<br>M. A. S.* | P. A. P.* | Attribu<br>te A/B | ZF     | Δ time of<br>welding<br>max. |
|--------------------|---------------------------------------------------------------------------------------------------|------------|----------------------|-----------|-------------------|--------|------------------------------|
| 0                  | Same codes sheets                                                                                 | OK         | OK                   | OK        | A/B               | ZF/HZF | 0                            |
| 1                  | Same thicknesses + same coatings                                                                  | OK         | OK                   | OK        | A/B               | ZF/HZF | 0                            |
| 2                  | Same thicknesses of reference + even Nb of coatings + same sheet quantity + respect of the E max. | OK         | OK                   | OK        | A/B               | ZF/HZF | 0                            |
| 3                  | Same thicknesses of reference + same quantity of coatings                                         | •          | •                    | OK        | A/B               | ZF/HZF | 2                            |
| 4                  | Same thicknesses of reference + delta of 1 on the coating                                         | •          | Prohibited           | •         | B                 | HZF    | 2                            |
| 5                  | same coating quantity+ delta of 1 on the sheet of reference                                       | •          | Prohibited           | •         | B                 | HZF    | 4                            |
| 6                  | Delta of 1 on the thickness of reference and on the coating                                       | Prohibited | Prohibited           | •         | B                 | HZF    | 4                            |

OK Applications authorized to the integrators.

• With agreement of the person in charge of welding.

**Auto adaptative Medium Frequency Welding:**

Table 12

| Levels of regroup. | Applicable conditions only for sheets of family 1                                                | Robot    | M.S.A.*<br>M. A. S.* | P. A. P.* | Attribu<br>te A/B | ZF     | Δ time of<br>welding<br>max. |
|--------------------|--------------------------------------------------------------------------------------------------|----------|----------------------|-----------|-------------------|--------|------------------------------|
| 0                  | Same codes sheets                                                                                | OK       | OK                   | OK        | A/B               | ZF/HZF | 0                            |
| 1                  | Same thicknesses + same coatings                                                                 | OK       | OK                   | OK        | A/B               | ZF/HZF | 0                            |
| 2                  | Same thicknesses of reference + even Nb of coatings + same sheet quantity + respect of the E max | OK       | OK                   | OK        | A/B               | ZF/HZF | 0                            |
| 3                  | Same thicknesses of reference + same quantity of coatings                                        | •        | •                    | OK        | A/B               | ZF/HZF | 2                            |
| 4                  | Same thicknesses of reference + delta of 1 on the coating                                        | •        | Interdit             | •         | B                 | HZF    | 2                            |
| 5                  | same coating quantity + delta of 1 on the sheet of reference                                     | •        | Interdit             | •         | B                 | HZF    | 4                            |
| 6                  | Delta of 1 on the thickness of reference and on the coating                                      | Interdit | Interdit             | •         | B                 | HZF    | 4                            |
| 7                  | Delta of 2 on the thickness of reference and on the coating                                      | Interdit | Interdit             | •         | B                 | HZF    | 4                            |

OK Applications authorized to the integrators.

• With agreement of the person in charge of welding.

For the levels of regrouping 3 to 6, take into account the time of maximum welding.

For the assemblies comprising at least 1 sheet of families 2, 2 bis, a specific program of welding must be allotted to him.

OK Applications authorized to the integrators.

- With agreement of the person in charge of welding.

**Note :** For the levels of regrouping 3 to 6, take into account the time of maximum welding.

For the assemblies comprising at least 1 sheet of families 2, 2 bis, a specific program of welding must be allotted to him.

## 5.7.WELDING PARAMETERS FOR SCREWS AND NUTS

### 5.7.1.SCREW OF DEFINITION CONFORM TO NORMS PSA PEUGEOT CITROËN STNG SCREW 9691205699 ([01266 12 00682](#))

Table 13

|            |                            | Screw with 3 embossments |         |                     |                | Screw with annular embossment |         |                    |                |
|------------|----------------------------|--------------------------|---------|---------------------|----------------|-------------------------------|---------|--------------------|----------------|
| Screw<br>Ø | Thick.<br>Sheet<br>real mm | Force<br>daN             | TS<br>~ | INT.<br>Sheet<br>kA | T of hold<br>~ | Force<br>daN                  | TS<br>~ | INT<br>Sheet<br>kA | T of hold<br>~ |
| 4          | 0,6 to 1,3                 | 350                      | 6       | 10                  | 8              | 400                           | 4       | 15                 | 8              |
|            | 1,31 to 3                  | 350                      | 6       | 10                  | 8              | 400                           | 6       | 15                 | 8              |
| 5          | 0,74 to 1,3                | 350                      | 6       | 10                  | 8              | 400                           | 6       | 17                 | 8              |
|            | 1,31 to 3                  | 350                      | 6       | 10                  | 8              | 400                           | 8       | 19                 | 8              |
| 6          | 0,74 to 1,3                | 350                      | 6       | 10                  | 8              | 400                           | 6       | 20                 | 8              |
|            | 1,31 to 3                  | 350                      | 6       | 10                  | 8              | 400                           | 8       | 22,5               | 8              |
| 8          | 0,74 to 1,3                | 400                      | 8       | 12,5                | 10             | 500                           | 8       | 27,5               | 10             |
|            | 1,31 to 3                  | 400                      | 8       | 12,5                | 10             | 500                           | 8       | 30,5               | 10             |
| 10         | 0,94 to 1,3                | 450                      | 8       | 13,5                | 10             | 600                           | 8       | 29,5               | 10             |
|            | 1,31 to 3                  | 450                      | 8       | 13,5                | 10             | 600                           | 10      | 34                 | 10             |
| 12         | 0,94 to 1,3                | 500                      | 8       | 15                  | 10             | 700                           | 10      | 31                 | 10             |
|            | 1,31 to 3                  | 500                      | 8       | 15                  | 10             | 700                           | 10      | 36                 | 10             |

The tests were carried out on sheets naked and galvanized LAF 10 µm like on sheets HLE (E335D) galvanized 10 µm.

**-IMPORTANT:** If welding on sheet 22MnB5, HR, DP≥780, consult the referential service.

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 36/69 |
|----------------------------------------------------------|--------------|-------|

### 5.7.2.NUT OF DEFINITION CONFORM TO NORM PSA PEUGEOT CITROËN STNG NUT 9691172999 (01266\_12\_00912)

| Nut<br>Ø     | Force<br>daN | Intensity<br>A | Time of welding<br>~ | Holding time<br>~ |
|--------------|--------------|----------------|----------------------|-------------------|
| M5           | 300          | 15 000         | 6                    | 8                 |
| M6           | 400          | 15 000         | 6                    | 8                 |
| M8           | 500          | 18 000         | 6                    | 8                 |
| M10          | 600          | 25 000         | 8                    | 10                |
| M12          | 800          | 30 000         | 8                    | 10                |
| 7/16" 20 UNF | 600          | 25 000         | 8                    | 10                |

Table 14

**Note:** If welding on sheet 22MnB5, HR, DP≥780, consult the referential service.

**IMPORTANT:** If welding on steel sheets shades 22MnB5, HR, DP≥780, consult the Technical Assembly department.

### 5.7.3.NUT OF MASS

| Nut<br>Ø | Force<br>daN | Intensity<br>A | Time of welding<br>~ | Holding time ~ | Number        |
|----------|--------------|----------------|----------------------|----------------|---------------|
| 5,5      | 400          | 15000          | 8                    | 10             | 96.176.717.80 |
| 5,5      | 400          | 23 000         | 8                    | 10             | 79.030.420.62 |

Table 15

**Note:** If welding on sheet 22MnB5, HR, DP≥780, consult the referential service.

**IMPORTANT:** If welding on steel sheets shades 22MnB5, HR, DP≥780, consult the Technical Assembly department.

### 5.7.4.WELDING OF EMBOSSEMENT (REF. NORM PSA PEUGEOT CITROËN [B13 1230](#))

#### 5.7.4.1.Welding parameters of embossment

**IMPORTANT:** If welding on steel sheets shades 22MnB5, HR, DP≥780, consult the Technical Assembly department.

| Number of<br>embossment | Force<br>daN | Intensity coated<br>sheet<br>A | Intensity naked<br>sheet<br>A | Time of welding<br>~ | Holding time<br>~ |
|-------------------------|--------------|--------------------------------|-------------------------------|----------------------|-------------------|
| 3                       | 180          | 12 000                         | 9 000                         | 5                    | 5                 |
| 4                       | 250          | 15 000                         | 10 000                        | 6                    | 6                 |
| 5                       | 400          | 16 500                         | 12 000                        | 7                    | 7                 |
| 6                       | 500          | 19 000                         | 15 000                        | 8                    | 8                 |
| 7                       | 600          | 20 000                         | 16 000                        | 10                   | 10                |

Table 16

For 2 embossments: x 1,5 force and intensity.

For 3 embossments: x 2 force and intensity.

For 4 embossments: x 3 force and intensity.

For 6 embossments: x 4 force and intensity.

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 37/69 |
|----------------------------------------------------------|--------------|-------|

#### 5.7.4.2. Embossments forms "D" for reduced flanges

This embossing makes it possible to carry out welding of strength equivalent to PtSE and possibly, in laser bead substitution (see standard [B13 1230](#)).

The specificity of the assemblies and the joint faces requires to consult systematically the specialists in the Reference frame Assemblies by Welding for feasibility and determination of the welding parameters.

#### 5.7.4.3. Embossments triangular for screw, nuts and special fixings (see standard [B13 1230](#))

Continuous or stopped annular triangular embossing (3 to 120 °).

#### Welding parameters :

The parameters can be estimated, at the stage studies, on the database per mm length of embossing, namely:

- FORCE: 20 daN per mm length of embossing.
- INTENSITY: 750 A per mm length of embossing. The intensity of welding is obtained with a reserve of power of 17,5% (initial dephasing with max. 85%).
- TIME: 8 to 10 ~ according to the relative geometries of the two parts, not having to compensate for a lack of power.

**IMPORTANT:** If welding on steel sheets shades 22MnB5, HR, DP $\geq$ 780, consult the Technical Assembly department.

The optimization of the parameters is to be realized according to the recommendations of note [PMXP\\_CEB09\\_0014](#).

## 6. MEDIUM FREQUENCY WELDING (ROBOTIZED PROCESS) AND AUTO ADAPTIVE MEDIUM FREQUENCY (MANUAL PROCESS)

### 6.1. PREAMBLE MF WELDING

#### 6.1.1. ROBOTIZED PROCESS

In a robotized process, only the medium frequency welding is authorized.  
The MF Auto Adaptive Welding is forbidden.

#### 6.1.2. MANUAL PROCESS

If it's necessary to have more than 1 welding program for each manual welding gun in 50Hz, then the Medium Frequency Auto Adaptive Welding is to be privileged.

- The Medium Frequency Auto Adaptive welding uses maximum only 1 welding program with manual manual welding gun.
- The Medium Frequency Auto Adaptive welding allows the suppression of mechanical changing programs ranges.

## 6.2.RULES TO RESPECT

### • Medium Frequency Welding

- According to the ratio  $E/e$  in the assembly, you have to refer to the abacus corresponding (Chap. 6.3) "rules of association with steel sheets", so as to assure the weldability of the assembly.

Lonely, the assemblies with a high thickness dissymmetry and with the thin sheet outside the assembly, obligate to respect the polarity (+) on the thin sheet side.

- If there's a lack of respect to these abacus, then this thickness ratio  $R=E/e$  in the assembly can be reduced, so more restrictive.
- Manual Auto Adaptive Welding**
  - In addition to these previous rules, the thickness ratio between the thickest assembly and the thinnest assembly of the range must not exceed  $R = 2$ .

## 6.3.WELDABILITY FIELD

Refer to norm [B13 1220](#).

**Assemblies E ref =<1,30mm (classic welding):**

If the thickness ratios are in the eligible limits of **the orange areas (see below)**, there is no management of the gun arms polarity compare with the assembly.

But, if the thickness ratios are **superior to the limits of the orange areas**, this time there is an imperative management of the  $\oplus$  polarity on the thin sheet side, respecting the eligible maxi.

**R maxi = E/e fonction type courant et famille de tôles**

| Assemblage | Famille 1       |              |           | Famille 2 ou 2bis |              |           |                 |              |           |
|------------|-----------------|--------------|-----------|-------------------|--------------|-----------|-----------------|--------------|-----------|
|            | Type de courant |              |           | Ass Symétrique    |              |           | Ass Dissym      |              |           |
|            | Type de courant |              |           | Type de courant   |              |           | Type de courant |              |           |
|            | AC              | DC<br>Coté e |           | AC                | DC<br>Coté e |           | AC              | DC<br>Coté e |           |
|            |                 | $\oplus$     | $\ominus$ |                   | $\oplus$     | $\ominus$ |                 | $\oplus$     | $\ominus$ |
| 2 Tôles    | 4 → 4           | 4            | 3         | 4 → 4             | 4            | 3         | 3 → 3           | 3            | 2         |
| 3 Tôles    | 3 → 3           | 3            | 2         | 3 → 3             | 3            | 2         | 2 → 2           | 2            | 2         |

**Symétrique**

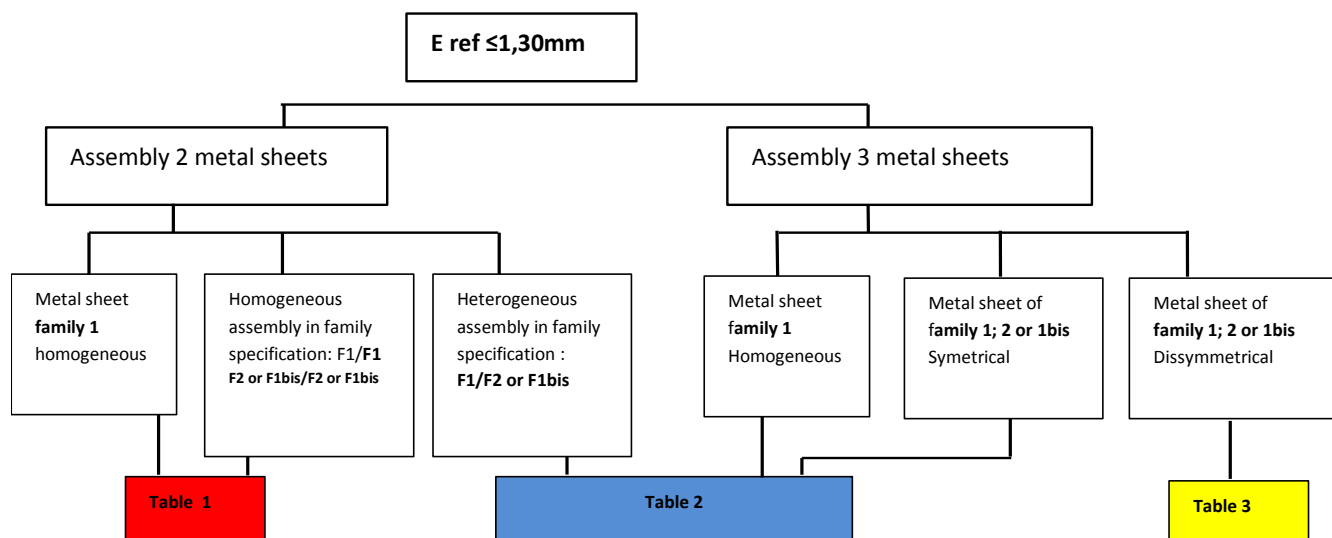
F1/F1/F1  
F2/F2/F2  
F1/F2/F1  
F2/F1/F2

**Dissymétrique**

F1/F1/F2  
F1/F2/F2

From this table, we define the association rules of the steel sheets in Medium Frequency:

*Minimum space between PtSE : 20mm*



**Abacus 1:**

| The thinnest sheet of the assembly (mm) | The thickest sheet of the assembly (mm) |             |             |             |             |             |             |       |       |       |      |       |       |
|-----------------------------------------|-----------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------|-------|-------|------|-------|-------|
|                                         | 0,65 à 0,74                             | 0,75 à 0,84 | 0,85 à 0,94 | 0,95 à 1,09 | 1,10 à 1,30 | 1,31 à 1,69 | 1,70 à 1,94 | ≥1,95 | ≥2,25 | ≥2,55 | ≥2,6 | ≥2,85 | >2,95 |
| 0,65 à 0,74                             |                                         |             |             |             |             |             |             |       |       |       |      |       |       |
| 0,75 à 0,84                             |                                         |             |             |             |             |             |             |       |       |       |      |       |       |
| 0,85 à 0,94                             |                                         |             |             |             |             |             |             |       |       |       |      |       |       |
| 0,95 à 1,09                             |                                         |             |             |             |             |             |             |       |       |       |      |       |       |
| 1,10 à 1,3                              |                                         |             |             |             |             |             |             |       |       |       |      |       |       |

**Abacus 2:**

| The thinnest sheet of the assembly (mm) | The thickest sheet of the assembly (mm) |             |             |             |             |       |      |      |      |       |       |      |       |       |
|-----------------------------------------|-----------------------------------------|-------------|-------------|-------------|-------------|-------|------|------|------|-------|-------|------|-------|-------|
|                                         | 0,65 à 0,74                             | 0,75 à 0,84 | 0,85 à 0,94 | 0,95 à 1,09 | 1,10 à 1,30 | ≥1,31 | ≥1,5 | ≥1,7 | ≥1,9 | ≥2,25 | ≥2,55 | ≥2,6 | ≥2,85 | >2,95 |
| 0,65 à 0,74                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 0,75 à 0,84                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 0,85 à 0,94                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 0,95 à 1,09                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 1,10 à 1,3                              |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |

**Abacus 3:**

| The thinnest sheet of the assembly (mm) | The thickest sheet of the assembly (mm) |             |             |             |             |       |      |      |      |       |       |      |       |       |
|-----------------------------------------|-----------------------------------------|-------------|-------------|-------------|-------------|-------|------|------|------|-------|-------|------|-------|-------|
|                                         | 0,65 à 0,74                             | 0,75 à 0,84 | 0,85 à 0,94 | 0,95 à 1,09 | 1,10 à 1,30 | ≥1,31 | ≥1,5 | ≥1,7 | ≥1,9 | ≥2,25 | ≥2,55 | ≥2,6 | ≥2,85 | >2,95 |
| 0,65 à 0,74                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 0,75 à 0,84                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 0,85 à 0,94                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 0,95 à 1,09                             |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |
| 1,10 à 1,3                              |                                         |             |             |             |             |       |      |      |      |       |       |      |       |       |

**Caption of the abacus 1-2-3 :**

|  |                                                                |
|--|----------------------------------------------------------------|
|  | Valid choice without polarity impact                           |
|  | Valid choice with respect of the polarity                      |
|  | Assembly out of norm                                           |
|  | Valid choice in MM without polarity impact (robot welding)     |
|  | Out of norm assembly in adaptative MF welding (manual welding) |

**Assemblies E ref > 1,3mm (pulsation welding) :****Table 17**

| Table 17                 |                                                                 |                                     |                |                |                |                |                |                |                |                |               |            |
|--------------------------|-----------------------------------------------------------------|-------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|------------|
| Reference thickness (mm) | Strongest steel sheet thickness (mm)<br>(except ref. thickness) | Thinnest steel sheet thickness (mm) |                |                |                |                |                |                |                |                |               |            |
|                          |                                                                 | 0                                   | 0,55<br>à 0,64 | 0,65<br>à 0,74 | 0,75<br>à 0,84 | 0,85<br>à 0,94 | 0,95<br>à 1,09 | 1,10<br>à 1,30 | 1,31<br>à 1,69 | 1,70<br>à 2,29 | 2,3<br>à 2,79 | 2,8<br>à 3 |
| 1,31 à 1,69              | 1,31<br>à 1,69                                                  |                                     |                |                |                |                |                |                |                |                |               |            |
|                          | 1,7<br>à 2,29                                                   |                                     |                |                |                |                |                |                |                |                |               |            |
|                          | 2,3<br>à 2,79                                                   |                                     |                |                |                |                |                |                |                |                |               |            |
|                          | 2,8<br>à 3                                                      |                                     |                |                |                |                |                |                |                |                |               |            |
| 1,70 à 2,29              | 1,7<br>à 2,29                                                   |                                     |                |                |                |                |                |                |                |                |               |            |
|                          | 2,3<br>à 2,79                                                   |                                     |                |                |                |                |                |                |                |                |               |            |
|                          | 2,8<br>à 3                                                      |                                     |                |                |                |                |                |                |                |                |               |            |
| 2,3 à 2,79               | 2,3<br>à 2,79                                                   |                                     |                |                |                |                |                |                |                |                |               |            |
|                          | 2,8<br>à 3                                                      |                                     |                |                |                |                |                |                |                |                |               |            |
| 2,8 à 3                  | 2,8<br>à 3                                                      |                                     |                |                |                |                |                |                |                |                |               |            |

Additive limit restriction of e eligible in comparison with the 50 Hz.

If the thickness of the 3rd steel sheet is equal to 0, this matches with the case of a **2 steel sheets assembly**. The reference thickness is still > 1,3, there's no effect on the polarity for the MF DC welding (pulsation).

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 41/69 |
|----------------------------------------------------------|--------------|-------|

**Meaning of the table color code:**

**Case 1:** If the reference thickness **and** the 2<sup>nd</sup> strongest steel sheet are from the **family 1 or family 2**:

|  |                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------|
|  | Valid choice in MF (DC) welding without polarity impact                                                          |
|  | Valid choice in MF (DC) welding with respect of the polarity + on the side thin sheet                            |
|  | Valid choice in MF(DC) with avec thin sheet in the middle (essential). In this case, there is no polarity impact |
|  | Valid choice in MF (DC) welding with respect of the polarity + on the side thin sheet                            |
|  | Assemblies out of norm                                                                                           |

**Case 2:** If the reference thickness **and/or** the 2<sup>nd</sup> strongest steel sheet are from the **family 2 or 2bis and** the thin steel sheet from the family 1 **and/or** there is glue on thin sheet opposite interface (ex E Soudo Col / E / e):

|  |                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------|
|  | Valid choice in MF (DC) welding without polarity impact                                                          |
|  | Valid choice in MF (DC) welding with respect of the polarity + on the side thin sheet                            |
|  | Valid choice in MF(DC) with avec thin sheet in the middle (essential). In this case, there is no polarity impact |
|  | Valid choice in MF(DC) with avec thin sheet in the middle (essential). In this case, there is no polarity impact |
|  | Assemblies out of norm                                                                                           |

In exceptional cases of assembly out of these imperatives, it's necessary to see the assembly specialists so as to validate the elected solution.

## 6.4.WELDING SETTING

### 6.4.1. MEDIUM FREQUENCY (ROBOT OR MANUAL)

Apply the RESORE parameter (identical 50Hz/60Hz)

### 6.4.2. ADAPTIVE MEDIUM FREQUENCY (MANUAL)

- The starting of a manual welding gun in MF Auto Adaptive is specific to each type of welding cabinet (supplier)
  - The starting operating mode is to be requested to the technical leader of the referential position with manual welding gun.
  - In the MF Auto Adaptive manual welding, there is no classic welding settings table as in 50Hz.
  - The starting setting is used to define the reference curve (called MASTER\*) of the range. The settings are defined according to the following indications.
  - This setting must satisfy the quality of every assembly of the range in Auto Adaptive Medium Frequency.
  - The making of the reference curve (called MASTER) may need a modification of this setting.
- \*MASTER: the Master is dynamic resistance curve of the welding spot, registered from various welding spots made on the thinnest assembly of the range.*
- This curve is then used by the auto adaptive system to define, following an algorithm, the welding settings (time and current) according to the dynamic resistance of the welding spots on the various assemblies of the range*

**WELDING FORCE**

**Steel sheets family 1, 2 & 2bis => F = medium F** of the various assemblies of the range  
**and** medium F must be at least = to 80% of the maxi range force F.

**WELDING INTENSITY**

**I = I** of the thinnest assembly of the range  
(Intensity to optimize on a case to case basis => diminution of the I welding)

**\*WELDING TIME – HOLDING TIME***Table 18*

| Thickness reference of the thinnest assembly | Welding time                   | Holding time    |                                |
|----------------------------------------------|--------------------------------|-----------------|--------------------------------|
|                                              | Steel sheets F1, F2 and F2bis  | Steel sheets F1 | Steel sheets F1 +F2 (or F2bis) |
| e ref (mm)                                   | Ts (ms)                        | Tm (ms)         |                                |
| 0,6                                          | 320                            | 200             | 240                            |
| 0,7                                          | 320                            | 200             | 240                            |
| 0,8                                          | 340                            | 200             | 240                            |
| 0,9                                          | 340                            | 200             | 240                            |
| 1                                            | 360                            | 200             | 240                            |
| 1,2                                          | 360                            | 200             | 240                            |
| 1,5                                          | T = T of the thinnest assembly | 240             | 300                            |
| 2                                            |                                | 300             | 360                            |
| 2,5                                          |                                | 300             | 360                            |

**\*DRESSING FREQUENCY:**

If the abacus of the Chapter [6.3](#) are respected => dressing frequency = 500 PtSE

## 7.ASSEMBLIES RANGES

The tool retained by PSA Peugeot Citroën for the realization of the ranges of assemblies is software RESORE (Referential Welding Resistance)

It must contain the totality of the ranges of the whole of the equipments of assemblies of a workshop (equipments of spot welding, studs, arc, glue, riveting, clinching, laser...).

### 7.1.RESORE

The software RESORE is used:

- create a standard file of the equipments,
- with the design of the ranges of assemblies of the equipments (according to the norms and CdC of design specific to PSA Peugeot Citroën),
- calculate the welding parameters of the assemblies of the ranges,
- with equipments dimensionings,
- with the calculation of energies connection,
- with the exploitation in mass production.

#### 7.1.1.INSTALLATION AND MAINTENANCE

The request for installation is carried out by the computer system and validated by the Method Door-fitting expert.

In the same way, for any problem of software error, ask them to solve it.

#### 7.1.2.RESORE FILE

It must contain the whole of information necessary to the good exploitation of the equipments in mass production.

Following the brand of the welding cabinet => a configuration sheet associated can be created.

#### 7.1.3.MAKING OF RANGE

The milestone for the procurement of RESORE are described in the "Guide et jalons des activités des assembleurs en projet [01342\\_11\\_00036](#)"

## 8.DEFINITION OF THE PROCESS

Welding quality depends on the equipment, the environment and their management (conformity produced, preventive, management quality, quality of the fluid networks and energy, material correctly chosen, used and parameterized, qualification operator, etc.).

The application of a surveillance plan, process established from the basics trades, make it possible to ensure obtaining and the maintenance of the specified performances quality (norm PSA Peugeot Citroën [B13 1226](#)).

According to the Functional and Nonfunctional Areas as well as Pt SE with tidy presentation defined by the services vehicle studies, the Methods services work out the ranges and invest to limit the angle of attack of the environment and to reduce the failure rate.

#### Preliminary conditions:

- The definition of the assemblies is in conformity with norm PSA Peugeot Citroën [B13 1220](#). Their characteristics must be transmitted to the Manufacturing Department before the RO (Tools Achievements).
- File **PSE** must being filled for the **RO** then regularly updated by the calculation service.
- Information which describes the welding spot, in particular the Functional Areas, is contained in file **PSE**. This information is extracted from file **PSE** by the Methods services.

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 44/69 |
|----------------------------------------------------------|--------------|-------|

## 8.1.EQUIPMENTS

They are in conformity with the norms and CdC of design specific to PSA Peugeot Citroën.

| Type                                                       | Functions                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manual welding gun to weld robot or mechanized welding gun | <ul style="list-style-type: none"> <li>Regulation of the secondary intensity.</li> <li>Manual welding gun 50 Hz, 60 Hz, MF.</li> <li>Automatic dressing and if necessary, automatic changer for electrodes.</li> </ul>                                                                                       |
| P. A. P.<br>(Station With Manual welding gun)              | <ul style="list-style-type: none"> <li>Regulation of the secondary intensity + gun guide.</li> <li>Manual welding gun 50 Hz, 60 Hz, MF.</li> <li>The selection of program by the operator (trigger) is <b>prohibited</b> except for the equipments concerned with the reference frame small rate.</li> </ul> |
| M.S.A.<br>(Automatic Machine of Welding)                   | Compensation of tension and control of the intensity to the primary or primary regulation of intensity.<br>Maximum 1 spot by cycle and thyristor panel.<br>In the case of the DPPP, 2 spots max. per cycle and thyristor panel .                                                                             |
| M. A. S.<br>(Pedestal Machine)                             | Compensation of tension and control of intensity to the primary.<br>Other alternative: Welding MF.                                                                                                                                                                                                           |
| Recoated material (*)                                      | According to recovery.                                                                                                                                                                                                                                                                                       |
| Equipments of recovery (**)                                | Functions identical to mass production equipments.                                                                                                                                                                                                                                                           |

(\*) A material known as "is recoated" if, **at least, one** chain link of welding is recoated (sequence, manual welding gun, cupboard, etc). The degree of handing-over on level, decided by the structure project, will be a function :

- state of the material,
- functions included,
- level of the investments of produced launching and level of the costs of exploitation.

(\*\*) The equipments of recovery are different from the equipments of mass production but they are capacitors because they will allow the use of the standard parameters. The time of recovery will be different from the cycle time.

**Note :** For the embossment applications, the speed of collapse of embossments and the fast evolution of the resistance of contact can disturb the operation of the regulation of the secondary or primary current. It is thus recommended to use the function "Compensation of tension primary and control of the primary intensity".

### 8.1.1.INITIAL ADJUSTMENTS AND CONDITIONS OF IMPLEMENTATION

Methodology entitled "Collection of the procedures of bodyshop tests" n° PSA.VEH.182, (01342\_09\_00020 list in an exhaustive way adjustments and conditions of implementation of the equipments installed.

### 8.1.2.USE IN MASS PRODUCTION

To control and correct the drifts, a process surveillance plan must be set up and adapted according to the type of equipment and its level of failure (norm PSA Peugeot Citroën [B13 1226](#)).

## 8.2.PRIMARY CONNECTION OF THE EQUIPMENTS

Increasing energy needs dependent:

- with the increase in the thicknesses to be welded,
- with the arrival of the new grades of sheets,
- with the optimization of the cycle times of the installations,

involve an important rise in the primary thermal intensities.

For stage with parametric adaptations necessary to obtaining quality welding, a reserve, known as of safety, equalizes to 5% on the permitted primary thermal intensities values according to Appendix 2 must being applied to the new vehicles projects.

These reasons oblige us to return our capacitors installations by a rigorous dimensioning of the sections of the cables and primary catches.

In the event of installation of welding on **Medium Frequency**, the rules hereafter are also applicable. Fuel supply IG2 towards welding cabinet is done in three-phase current + covers over with soil.

When the welding gun is declared in medium frequency in RESORE software, the sizing of the primary fuses box is calculated by RESORE in three phases

### 8.2.1.CONNECTION IG2 / WELDING CABINET

The dimensioning of the electrical fuses boxes for the supply of the welding cabinet is given by software RESORE.

Commonly, these boxes of the types "CANALIS" are connected to the IG2 of the workshops. If an electric service of distribution of the power would be carried out differently, the same rules are to be respected, namely :

- Choose the fuses of the box to nominal box – 160A for a box 160, 250A for a 250, for applications MSA and MAS. For the robotized and manual applications, refer to the § 8.2.1.1.
- The fuses are of type aM.
- The adjustment of the thermals of the circuit breaker of the Welding cabinet must being adjusted with more close to real consumption of the range.
- The section of the electric cables (box/welding cabinet), must :
  - guarantee a maximum voltage drop of 5 % of the nominal voltage for the maximum current of the range at the end of the lifetime electrode,
  - allow to carry the effective thermal current corresponding to the fuses gauge for the applications MSA and MAS (except the box KH) according to Appendix 2.
- The quality of the cables must being conform to the H07 range - RNF.

## 8.2.1.1. Specific case of the applications robotized and manual for primary cables lower than 30 meters

We fixed a limit of length at 30 meters between Canalis boxes and welding cabinet.

For lengths of cables higher than 30 meters, it is necessary, in addition to the checking of the compatibility the length and of the section of the cables, to validate the generated voltage drops.

In order to simplify the management of the sections of cables for the robotized and manual applications, for lengths of primary cables lower or equal to 30 meters, the elements of electric connection to take into account are described here after.

**For robotized applications:**

| Primary Ith<br>calculated range<br>by RESORE (A) | Box Canalis (A) |                                    |                           | Gauge<br>fusible<br>(A) | Section of cable (mm <sup>2</sup> )                       |                               |                              |
|--------------------------------------------------|-----------------|------------------------------------|---------------------------|-------------------------|-----------------------------------------------------------|-------------------------------|------------------------------|
|                                                  | MF<br>50 Hz     | Box<br>calculat<br>ed by<br>RESORE | Box to<br>be<br>installed |                         | Canalis /<br>Welding<br>cabinet<br>(Length max =<br>30 m) | Cabinet/manual<br>welding gun |                              |
|                                                  |                 |                                    |                           |                         |                                                           | Robot<br>gun                  | Fixed<br>gun<br>for<br>robot |
| 0< Ith <=125 A                                   | MF              | 160                                | 160                       | 125                     | 25                                                        | 25                            | 25                           |
| 0< Ith <= 125 A                                  | 50 Hz           | 160                                | 160                       | 160                     | 35                                                        | 35                            | 35                           |
| 126 A < Ith ≤ 162 A                              | 50 Hz/MF        | 250                                | 250                       | 250                     | 50                                                        | 35                            | 35                           |
| 162 A < Ith ≤ 200 A                              | 50 Hz/MF        | 250                                | 250                       | 250                     | 50                                                        | prohibited                    | 50                           |
| 200 A < Ith ≤ 210 A                              | 50 Hz/MF        | 400                                | 250                       | 250                     | 50                                                        | prohibited                    | 50                           |

**For manual applications:**

| Primary Ith<br>calculated range<br>by RESORE (A) | Box Canalis (A) |                                   |                        | Gauge<br>fusible<br>(A) | Section of cable (mm <sup>2</sup> )                       |                               |
|--------------------------------------------------|-----------------|-----------------------------------|------------------------|-------------------------|-----------------------------------------------------------|-------------------------------|
|                                                  | MF<br>50 Hz     | Box<br>calculated<br>by<br>RESORE | Box to be<br>installed |                         | Canalis /<br>Welding<br>cabinet<br>(Length max =<br>30 m) | Cabinet/manual<br>welding gun |
|                                                  |                 |                                   |                        |                         |                                                           |                               |
| 0< Ith ≤ 125 A                                   | MF              | 160                               | 160                    | 125                     | 25                                                        | 25                            |
| 0< Ith ≤ 125 A                                   | 50 Hz           | 160                               | 160                    | 160                     | 25                                                        | 25                            |
| 126 A < Ith ≤ 140 A                              | 50 Hz/MF        | 160                               | 160                    | 160                     | 25                                                        | 25                            |
| 141 A < Ith ≤ 162 A                              | 50 Hz/MF        | 160                               | 160                    | 160                     | 35                                                        | 35                            |

**Important:** The sections of cables (Canalis/Welding cabinet) stipulated in the table above correspond to the minimal sections to use for the connection of each type of box.

## 8.2.2.CONNECTION WELDING CABINET / WELDING EQUIPMENTS

By plug connectors and electric cables, it is understood :

- For the robots with embarked manual welding gun (applications R.S.E.) :
  - Harness, Welding cabinet / axis 1 – axis 1 / axis 3 – axis 3 / axis 6.
  - Plug connectors Welding cabinet / axis 1 / axis 3 / axis 6.
- For the robots with fixed welding gun on the ground (applications R.P.F.) :
  - Harness Welding cabinet / welding gun.
  - Plug connectors Welding cabinet / welding gun.
- For the P.A.P. (applications P. A. P.):
  - Harness Welding cabinet / welding gun.
  - Plug connectors Welding cabinet / welding gun.
- For the M.S.A. (applications MSA) :
  - Harness Welding cabinet / transformers.

## 8.3.THERMAL INTENSITY

The thermal intensity (also called permanent intensity) makes it possible to check the compatibility of the constituent components of the welding chain (cables, plugs, transformer, circuits secondary – shunts, electrodes,...) It gives "the thermal equivalent" of the heating of the components.

### 8.3.1.PRIMARY THERMAL INTENSITY

Its simplified formula is as follows:

**50Hz single-phase:**

$$I_{th} = I_s \times \sqrt{\frac{\sum \text{Time of welding}}{\text{Time of cycle} \times 50}} \times KR \times \frac{U_{20}}{U_1}$$

**MF Three-phases:**

$$I_{th} = I_s \times \sqrt{\frac{\sum \text{Time of welding}}{\text{Time of cycle} \times 50}} \times KR \times \frac{U_{20}}{U_1 \sqrt{3}}$$

$I_s$  Intensity of standard welding in Amp.

$U_{20}$  Vacuum secondary tension of the transformer (given manufacturer) in Volt.

$U_1$  Primary supply voltage welding (variable value according to the sites of productions) in Volt.

$KR$  Coefficient of reserve which is a function of the type of equipments :

Robot → 1,20 / PaP. → 1,25 / M.S.A. → 1,3 / Welding of screws and nuts → 1,17

Welding Time in period.

Cycle time of the line in second.

If the welding currents or welding times vary with the various cycles, to replace the current of  $I_s$  welding by the current of medium welding called "quadratic Intensity" ( $I_q$ ) given by the following formula:

$$I_q = \sqrt{\frac{\sum I_s^2 t}{\sum t}}$$

$t$  Number of periods for each point.

$I_s$  Intensity of standard welding in Amp.

### 8.3.2.SECONDARY THERMAL INTENSITY

The secondary thermal intensity is commonly called permanent secondary intensity (I<sub>2p</sub>).

The following approximate formula makes it possible to obtain a sufficient precision compared to the exact formula which can be consulted in the norm ISO 5826.

It is usable when the  $\delta/T$  report/ratio is lower or equal to 5. It is the case for the welding applications used in the bodyshops of group PSA Peugeot Citroën.

It is this formula which is used in calculations RESORE.

$$I_{2p} = I_q \times \sqrt{\frac{\sum \text{Time of welding}}{\text{Time of cycle} \times 50}} \times KR$$

*I<sub>q</sub>* Same as § 8.3.1.

*KR* Coefficient of reserve which is a function of the type of equipments :

*Robot* → 1,20 / *PAP.* → 1,25 / *M.S.A.* → 1,3 / *Welding of screws and nuts* → 1,17

*Cycle time of the line in second.*

### 8.4.WORKING FACTOR

The duty ratio of an equipment is expressed in % and represents the operating time in welding compared to the cycle time.

It allows, like the thermal intensity, to check the compatibility of the constituent components of the welding chain.

$$FDM = \frac{\sum T_s}{T_{cycle} \times 50} \times 100$$

(In this formula, T cycle is multiplied by par 50)

*T<sub>s</sub>* Time of welding in periods.

*T cycle* Cycle time of the line in second.

*50* Coefficient of conversion second/periods.

**Note :** The cycle time to take into account is the cycle time of the line.

### 8.5.COOLING WATER OF THE WELDING EQUIPMENTS

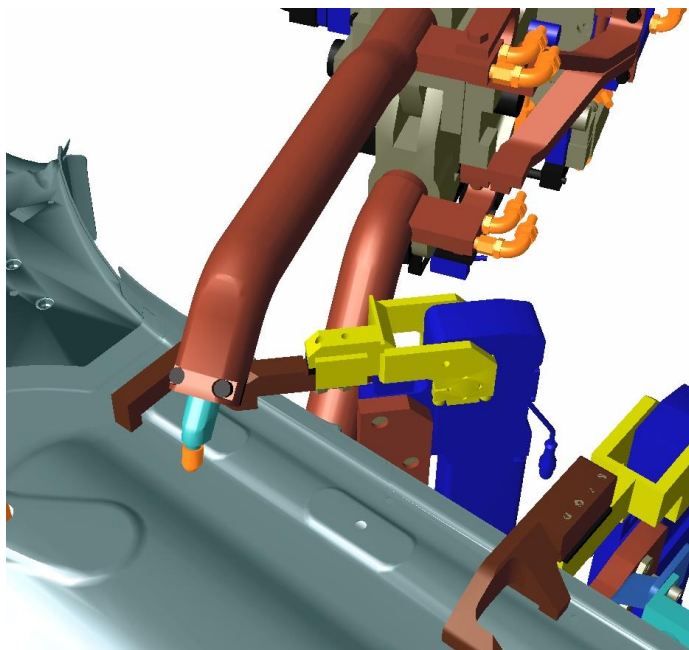
The welding equipments need a cooling water tarp filtered at 200 µm.

### 8.6.MAGNETIC AND ELECTRICAL INSULATION

In a 50 Hz welding or a Medium Frequency welding.

- *In the MSA or tool for MAS (with a PtSE like roman welding, nut welding, screw welding)*  
When the welding is under a distance of 30mm from geometry elements, the latter have to be isolated and in non-magnetic stainless (material to use: X6CrNi18-10)
- *In the MSM, MSR or MSA (for fixed gun)*  
All the geometric elements in the manual welding gun field during the welding must be in non-magnetic stainless (see example below)  
Their insulation is not necessary.

For any other case, there won't be neither insulation nor non-magnetic material.



*Example of a geometry element swallowed by a welding gun during the making of a welding spot.*

## 8.7.USE OF CONNECTION BY BARS

The principles describes after are valid for all welding equipments of assembly used (M.S.A, M.A.S., application dead gun...). They are taken from the guide [GE34-009G](#).

An adaptation of the ends of the bars will be made to be compatible with the need.

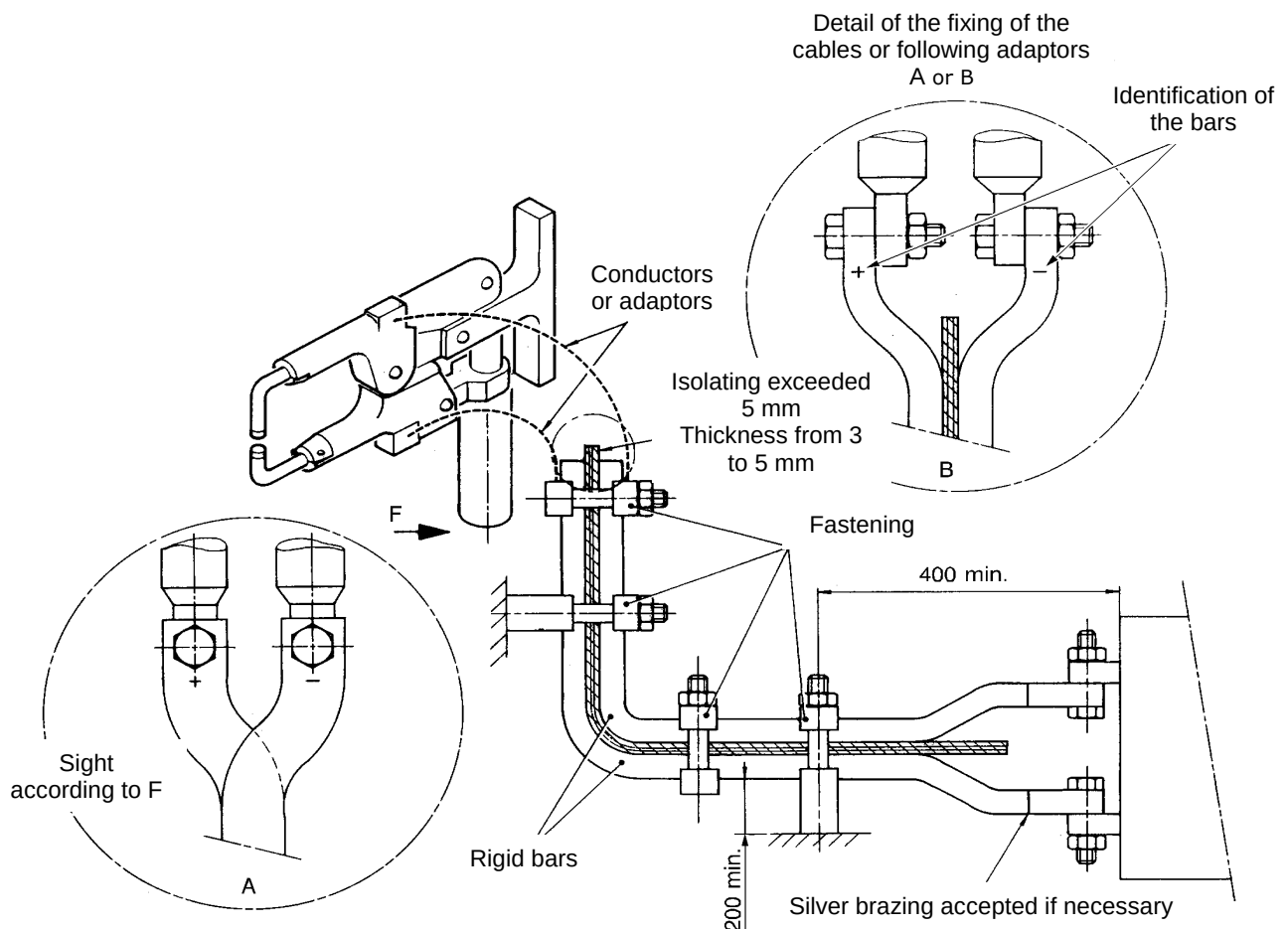
Connection adapter and bar can be brazed with silver. In the same way, the bars can be joined up in bevel, by brazing with the silver. Surfaces to be brazed are striated for diffusion by capillarity with the filler metal.

The first fastening with the frame must being to the maximum to 400 mm of connection. One at least needs a fastening for the frame at each end of the bar. Fixings of the bars must imperatively to be done by fastening and not by drilling through bars.

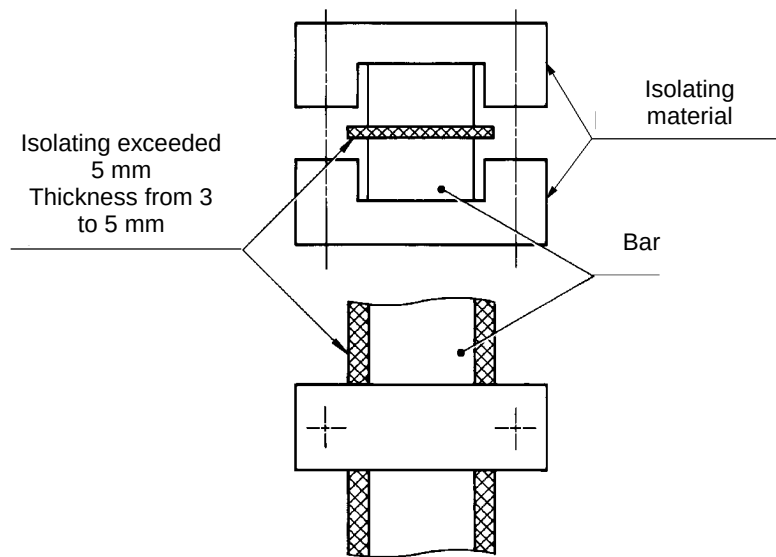
To locate the polarity + and - on the bars at the time of the BI/bar connection - bar/cable and bar/bar.

In the objective to quickly repair the installation in the event of contacts arcing of a bar, it is necessary to envisage an additional drilling by point of fixing (of the braids, BI or thimbles) near the clamp holes envisaged being studied.

The use of round bars is **prohibited** for reasons of maintainability.

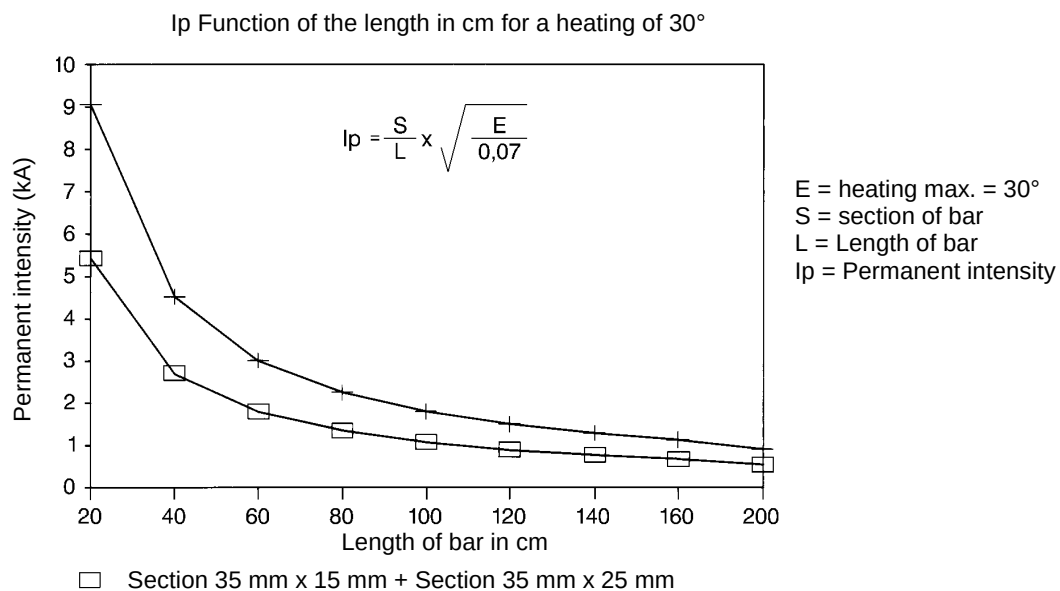


## 8.7.1.U FASTENING FOR THE BAR HOLD UP



## 8.7.2.DETERMINATION OF THE BAR SECTION

The sections of the bars are given according to the thermal intensity and the duty ratio of the installation.  
See with the § 8.3 and 8.4 applicable formulas of computations.



## 8.8.ATTRIBUTE OF SPOTS WELD - FUNCTIONAL AREAS

The attribute of **Pt SE** is the whole of the characteristics of the assemblies defined in the PSE file which is the produced reference frame of the assemblies.

The concept of attribute of **Pt SE** is not any more used on the new projects since 2006 and is replaced by the Functional Areas. (See 01342\_11\_00009 et [B13 1220](#) et [B13 1226](#)).

### 8.8.1.REGULATION OF WELDING SPOTS

The regulation of Pt SE (A or B) is defined by the services Studies vehicle group PSA Peugeot Citroën.

**The restrictions to be only taken into account for PtSE of attribute A are:**

- To use the modes of simple welding spot or double spot push-pull.
- To limit the regroupings of welding parameters (see § 5.6).
- To prohibit the mixed ranges.

### 8.8.2.WELDING SPOT IN FUNCTIONAL AREA

The rule is defined by the services Vehicle studies group PSA Peugeot Citroën.

**The restrictions to be taken into account for Pt SE in functional area are:**

- Use the modes of simple welding spot or double spot push-pull.
- Prohibit the mixed ranges for thicknesses of reference > 1.3 mm.
- Prohibit Pt SE very tidy aspect.
- Pts SE of a Functional Area must be realized in only one sector of manufacturing and controlled before the sub-assembly or the part leaves this sector.

## 8.9.DIAMETERS OF SPOTS WELD

(See norm [B13 1220](#)).

For thicknesses of references lower or equal to 1,3 mm, the minimum diameter of the nugget to guarantee is 4 mm.

It is obtained by classic welding with electrodes of diameter of active face Ø 6 mm.

For thicknesses of references higher than 1,3 mm, the minimum diameter to guarantee of the nugget is 6 mm and only welding pulsation, with electrodes of diameter of active face Ø 8 mm, makes it possible to control the development of the nugget.

## 8.10.WELDING SPOTS ASPECT

(see [B13 1220](#) chap. [5.7](#))

The 3 required characteristics of aspect are:

### Standard presentation (Pst) :

- No specific imposition related to the process.

### Tidy presentation (Pso) :

The following restrictions are imperative:

- Flatness, steel sheet matching conform.
- Restriction of the steel sheet thickness ratio.
- No thickness of coating higher than 10 µm
- The zinc coated allows a better aspect than galvanized sheets.
- The coated steel sheet mono-gal G10/01 is the ideal solution (if we can apply it)

No thickness of coating higher than 10 µm

If we can't, we have to apply a galva brushing range in the aspect PtSE areas before welding.

- Lifetime according to § [8.6](#).
- PtSE perfectly circular
- PtSE without any indentation in excess.
- PtSE without any deformation or pin
- Respect of the electrode/welding plan normality
- Use a Ce robot or manual welding gun with assembly MSM or MSR (set as geometry)
- Use of an electric embarked robot welding gun, with a calibration function by soft robot.
- The robot fixed welding guns are forbidden
- No gathering of welding parameters (choice of level 0, see Chap. [5.6](#)).
- (the mastering of welding settings)
- Dressing every 190 spots
- Mixed range is forbidden
- Use of straight centered electrodes (P911 310 112 in robot or manual)

### Additional conditions for M.S.A. :

- An tip with a dressing shape can be placed on the side of the appearance to limit the indentation.
- 1 only PtSE by cycle.
- Change of the electrodes every 475 or 725 spots depending on the coating (§ 8.12)

### Additional conditions for the P.A.P.:

- The manual welding gun has a complete fastening in position (arm guide + gun guide)

### Very tidy presentation (PtS) :

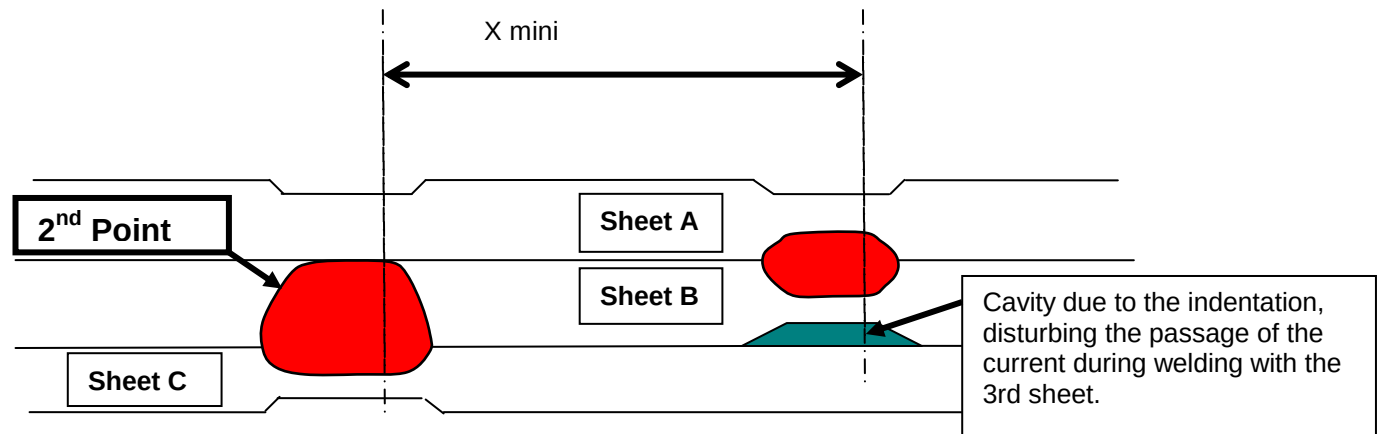
Ongoing V.I. of the process => waiting for the results before setting it up in this norm.

## 8.11.SPOTS MADE IN 2 THEN 3 THICKNESSES

(Constraint process - see norm [B13 1220](#)).

In this case, at the time of the assembly with 3 sheets it is imperatively necessary to shift the position of the spot compared to that made in 2 sheets of:

- X mini = 12 mm for sheet assemblies  $\leq 1.3$  mm
- X mini = 15 mm for sheet assemblies  $> 1.3$  mm



At the time of the realization of the first spot in 2 thicknesses, it creates an indentation which will prevent then any development of spot with a 3<sup>rd</sup> sheet, even if the electrodes are well positioned.

The shift of the 2<sup>nd</sup> Pt SE will allow to develop a spot between B and C in spite of a shunting between A and B which will not make it possible to develop a spot with this interface

In short, the junction is broken up into two:

1. PtSE for interface A/B.
2. PtSE for interface B/C, interface A/B not having any mechanical resistance.

## 8.12.INCREMENTAL LAW FOR WELDING CURRENTS

The laws of incrementing given in this chapter are laws by stages.

Programming of the timers (C.P.S.) can be realized with laws by stages or linear segments respecting the forms of the incrementings below.

### REMINDER:

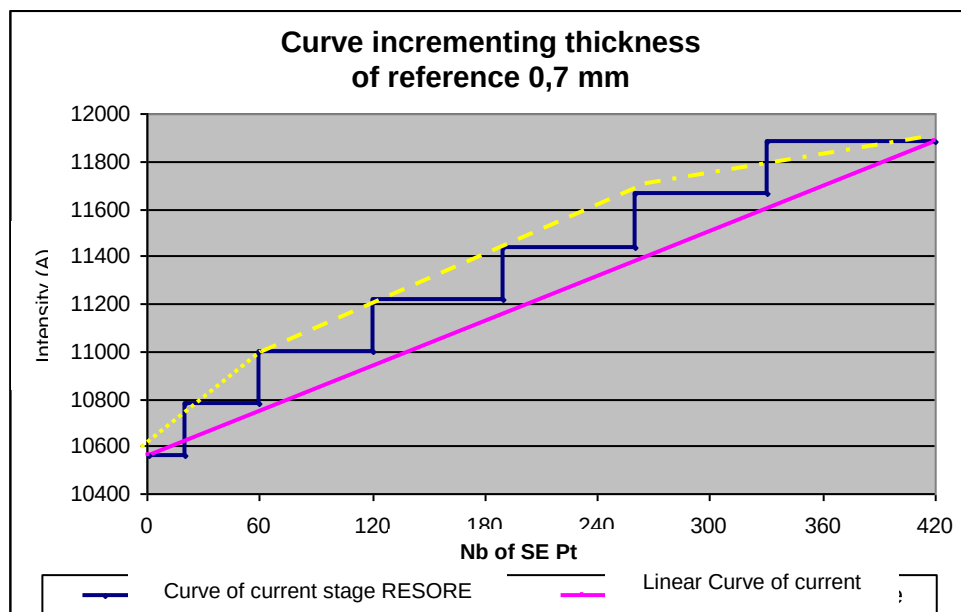
The curves of incrementings for robots were defined in order to guarantee a constant density of current to the contact electrode/sheet.

The form of the incrementing by stage is of logarithmic curve type.

The use of a segment mode, by keeping only the two extreme values of the law by stage, connects the two spots in a line passing below the theoretical values (-300 A to -400A)

To follow the theoretical form of the law of incrementing, several segments should be created connecting the tops of the steps (3 segments in dotted lines in the example below)

The programmed incrementings must correspond to the shape logarithmic curve of the standard curves.



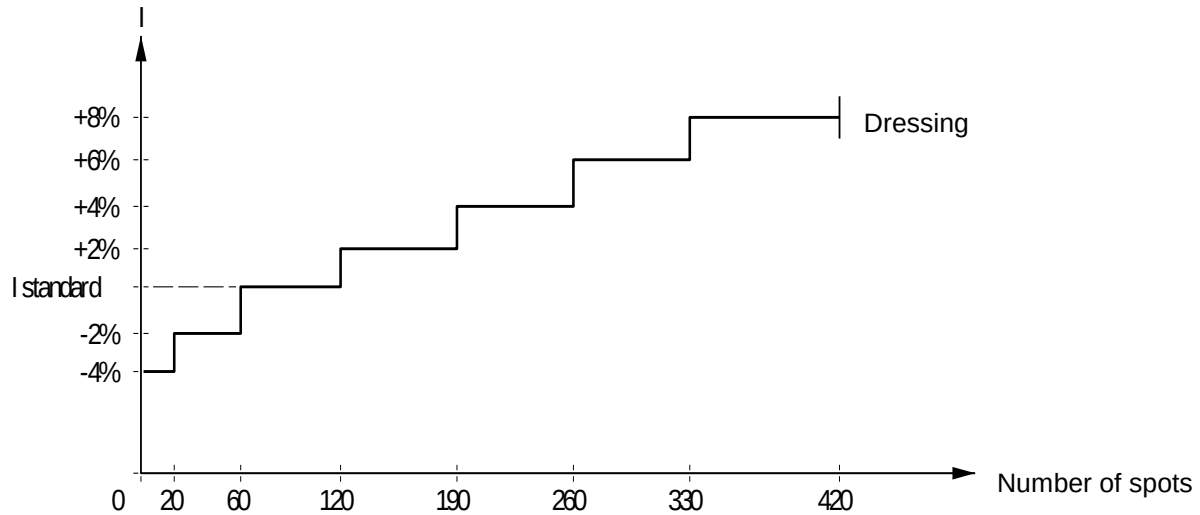
### REMINDER:

The PtSE declared out of norms produce and/or process will have their curves of incrementings limited by the removal of the last stage.

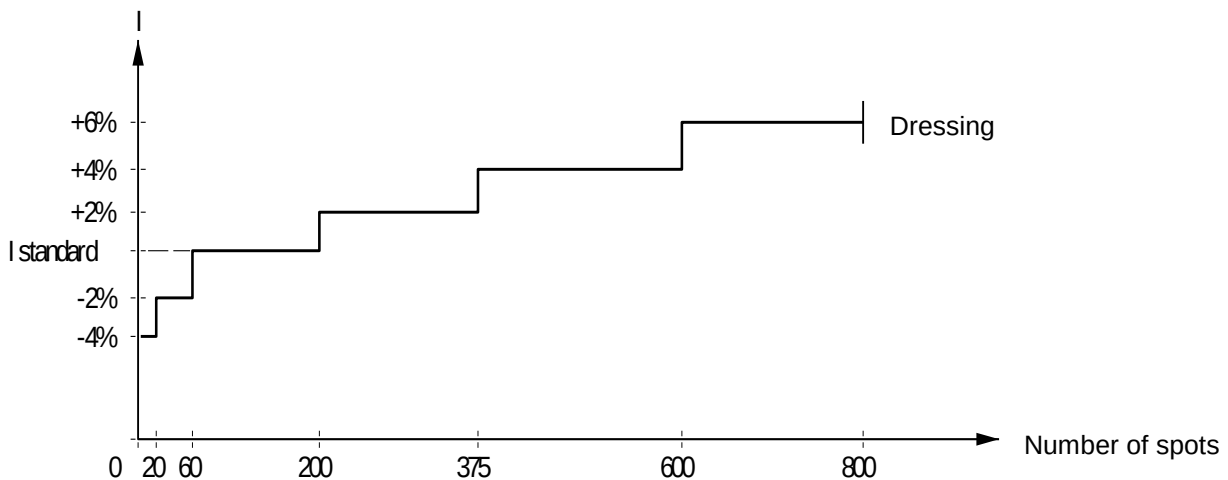
8.12.1.FOR ROBOT AND COATING THICKNESSES  $\leq 10 \mu\text{M}$ 

Being given the diversity of the percentage of coating in the ranges of welding on robot, it was validated a type of law making it possible to take into account cross flows and the versatility vehicle on lines PSA Peugeot Citroën.

## 8.12.1.1.Robot – Standard presentation



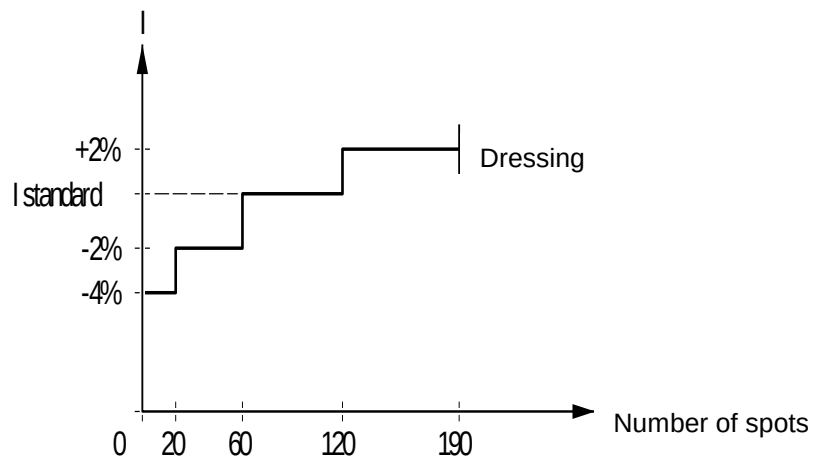
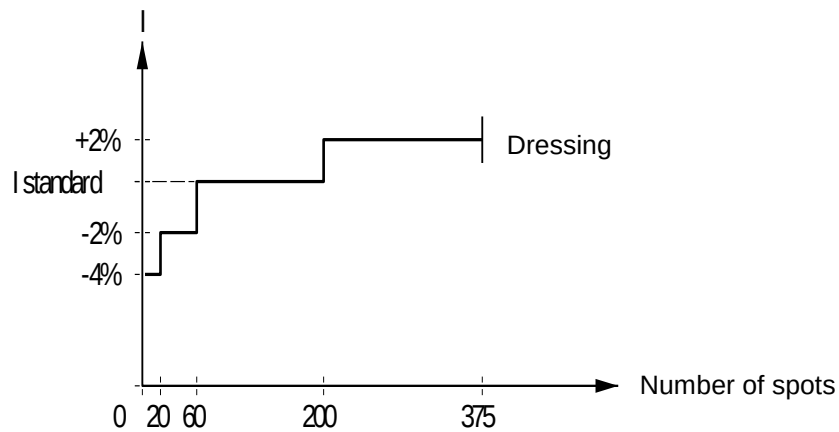
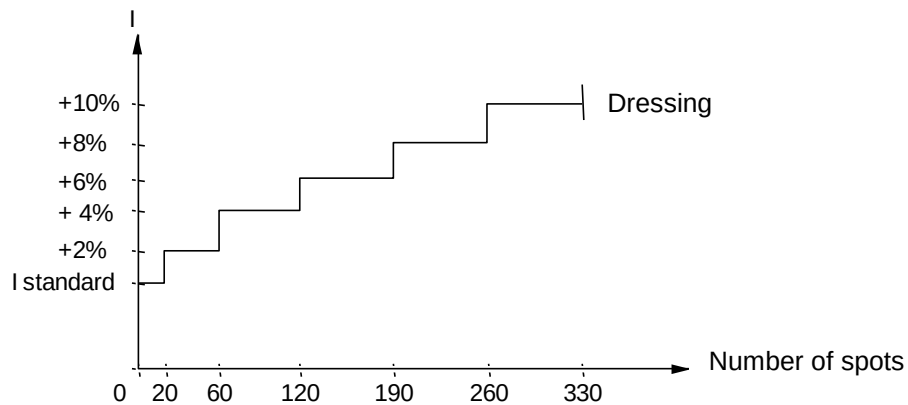
*Curve 1: Range comprising of the coated assemblies  $\geq 49\%$  of coated faces*



*Curve 2 : Range comprising of the coated assemblies  $< 49\%$  of coated faces*

**Note:** In the case of a range of welding pulsation, to divide by 2 the percentages of incrementing (1 % instead of 2 %).

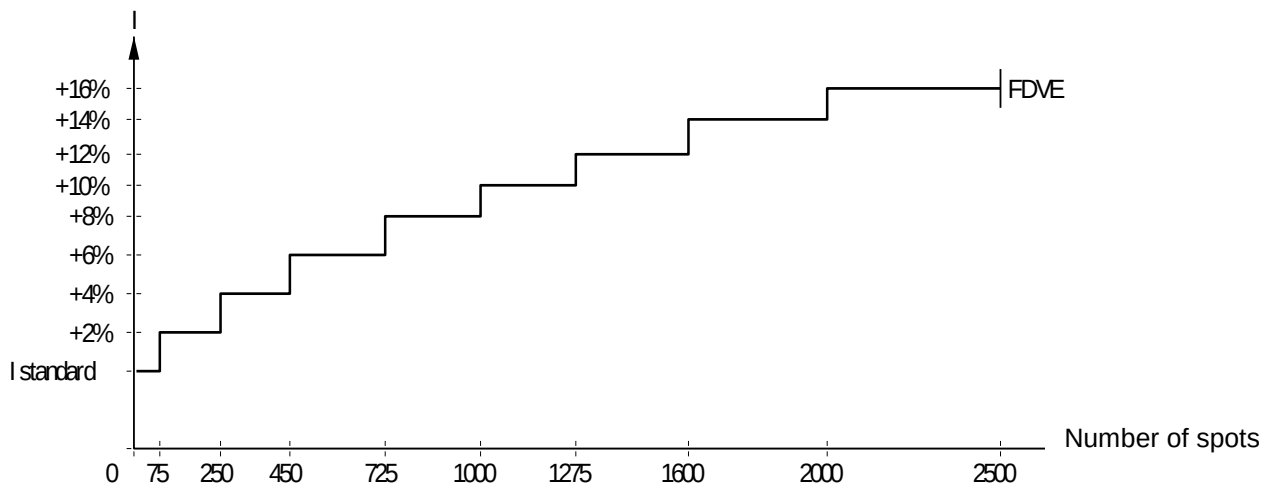
## 8.12.1.2.Robot – Tidy presentation

Curve 3: Range comprising of the coated assemblies  $\geq 49\%$  of coated facesCurve 4: Range comprising of the coated assemblies  $< 49\%$  of coated faces8.12.1.3.Robot – Case of PtSE  $\varnothing 3$  mm – Standard presentation

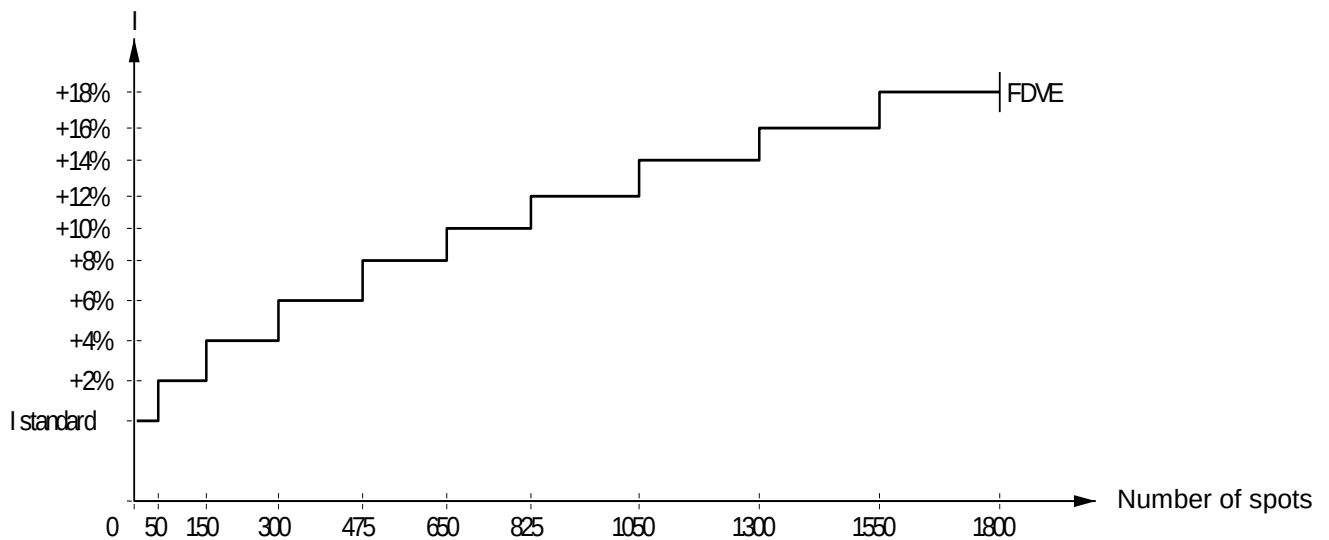
8.12.2.FOR M.S.A AND COATING THICKNESSES  $\leq 10 \mu\text{M}$ 

Three curves are defined corresponding to the electrozinged assemblies, galvanized and without coating.

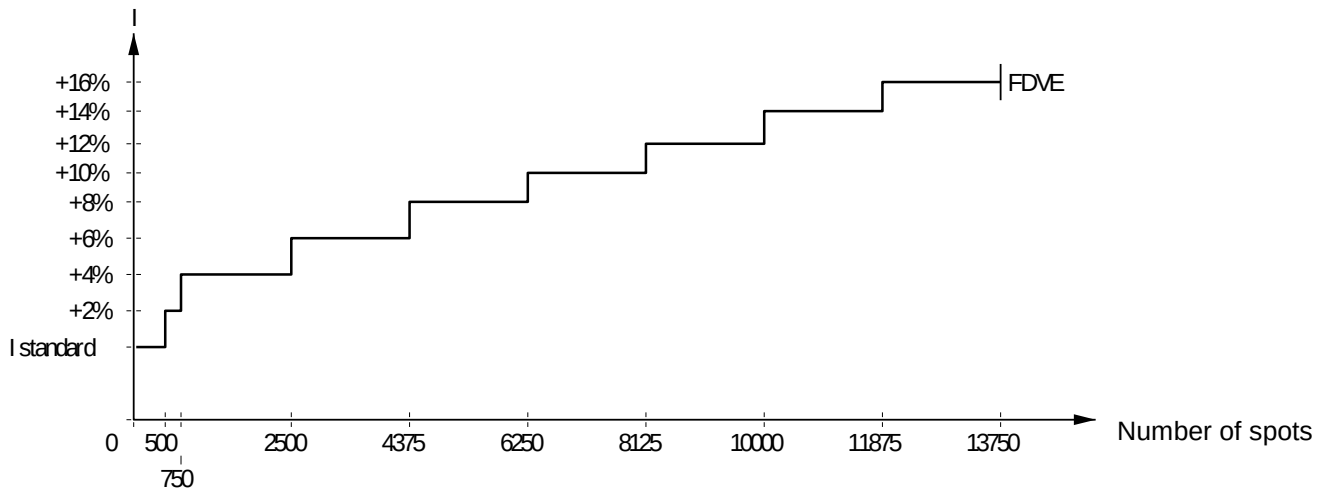
## 8.12.2.1.M.S.A. – Standard presentation



*Curve 5: Incrementing for spots having 1 or 2 coated faces electrozinged in contact with the electrodes*



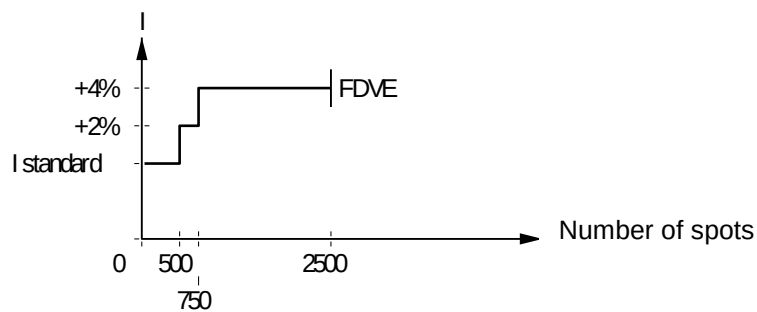
*Curve 6: Incrementing for spots having 1 or 2 coated faces galvanized in contact with the electrodes*



**Curve 7: Incrementing for spots having no face coated in contact with the electrodes**

**Note:** In the case of a range of welding pulsation, to divide by 2 the percentages of incrementing (1 % instead of 2 %).

#### 8.12.2.2.M.S.A. – Tidy presentation

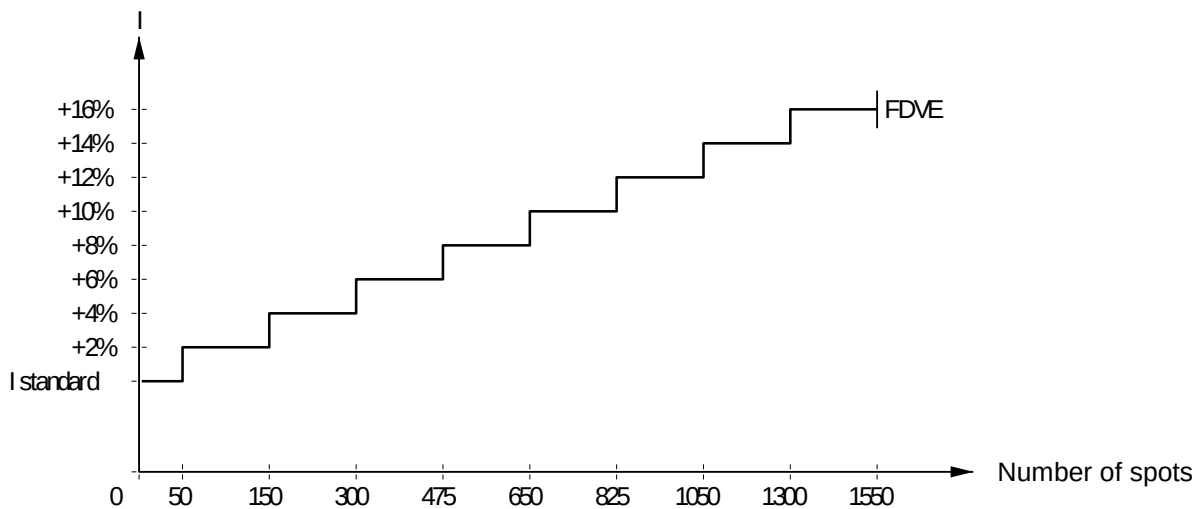


**Curve 8: Incrementing for spots not having any face coated in contact with the electrodes**

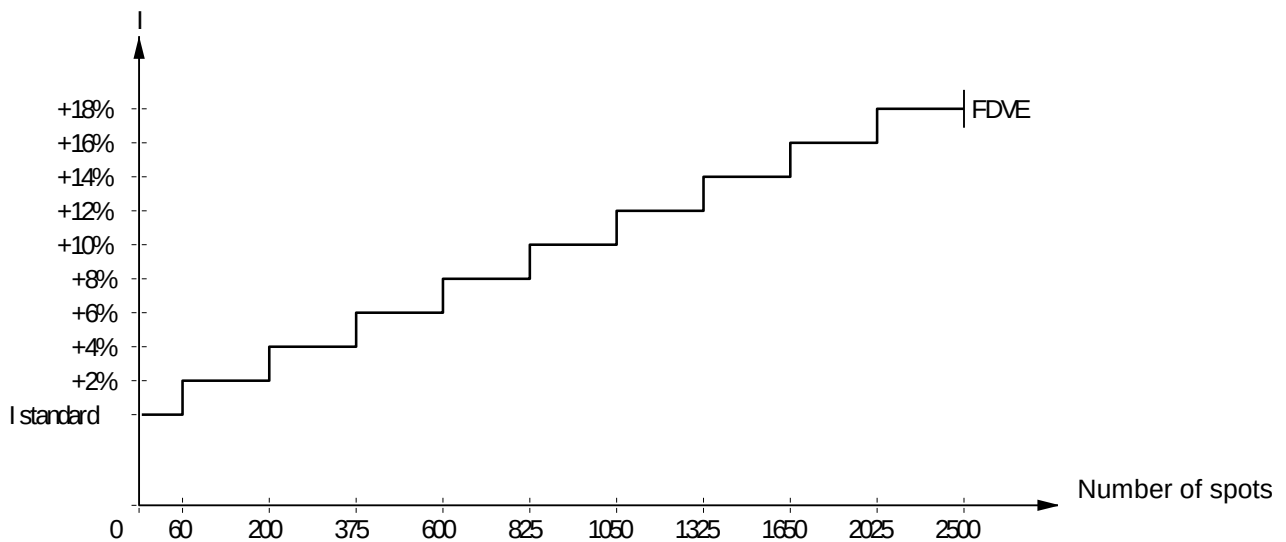
8.12.3.FOR P.A.P. AND COATING THICKNESSES  $\leq 10 \mu\text{M}$ 

Three curves are defined corresponding to the galvanized assemblies and without coating.

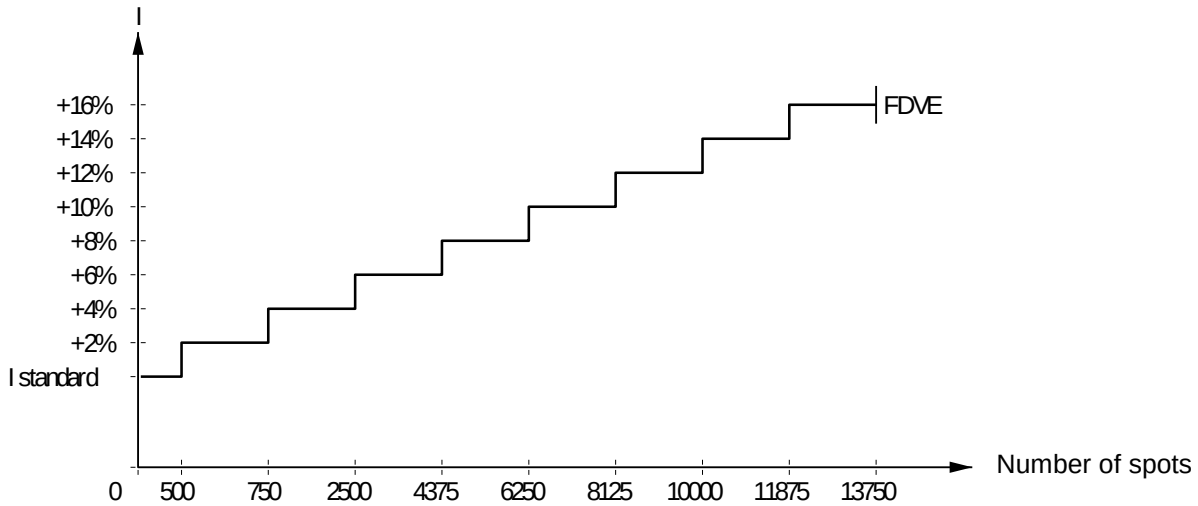
## 8.12.3.1.P.A.P. – Standard presentation



*Curve 9: Incrementing for coated ranges > 49 % of coated face*



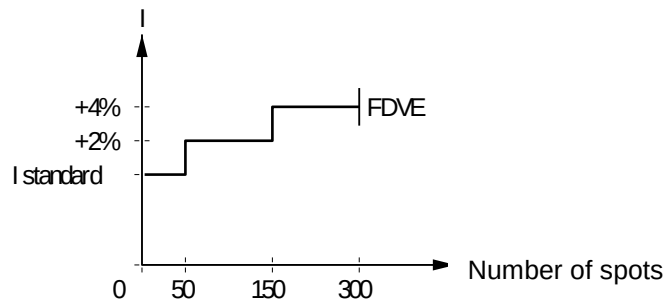
*Curve 10: Incrementing for coated ranges ≤ 49 % of coated face*



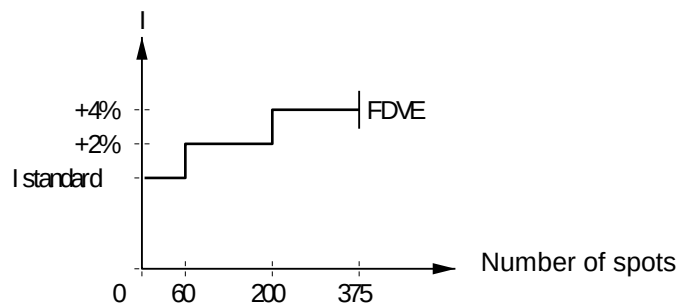
**Curve 11: Incrementing for naked ranges**

**Note:** In the case of a range of welding pulsation, to divide by 2 the percentages of incrementing (1 % instead of 2 %).

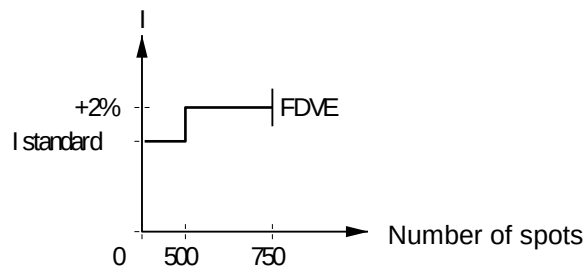
#### 8.12.3.2.P.A.P. – Tidy presentation



**Curve 12: Incrementing for coated ranges > 49 % of coated face**

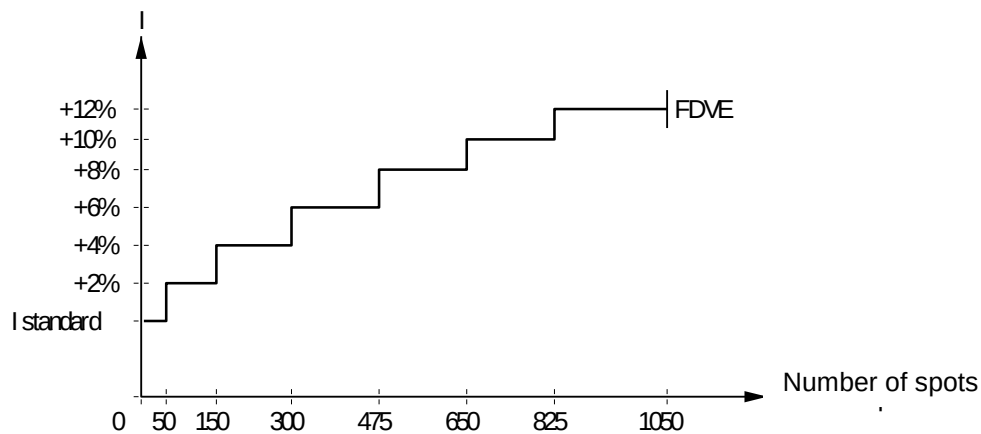


**Curve 13: Incrementing for coated ranges ≤ 49 % of coated face**



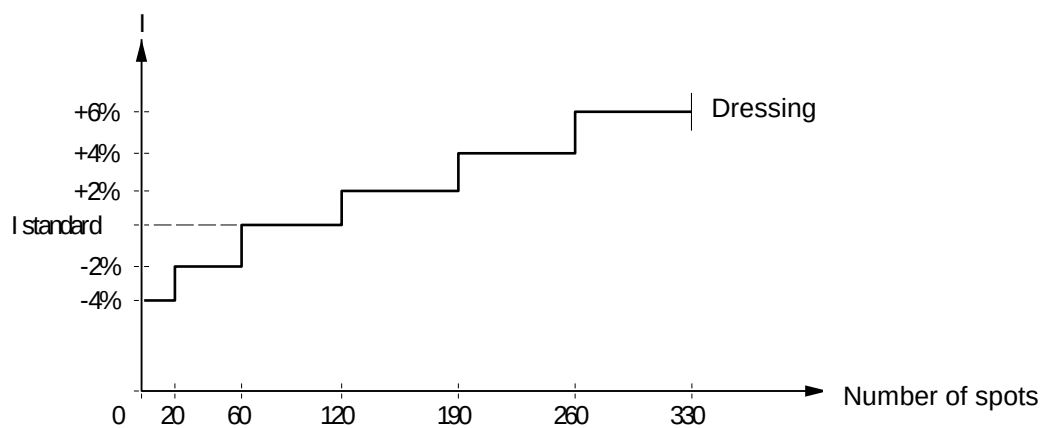
Curve 14: Incrementing for naked ranges

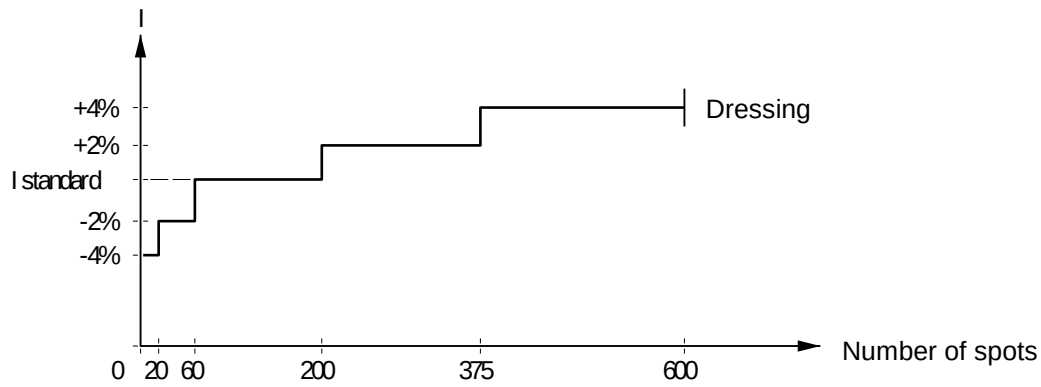
8.12.3.3.For P.A.P. - Case of PtSE of  $\varnothing$  3mm - Standard presentation



8.12.4.FOR ROBOT AND COATING THICKNESSES  $> 10 \mu\text{m}$

8.12.4.1.Robot - Standard presentation

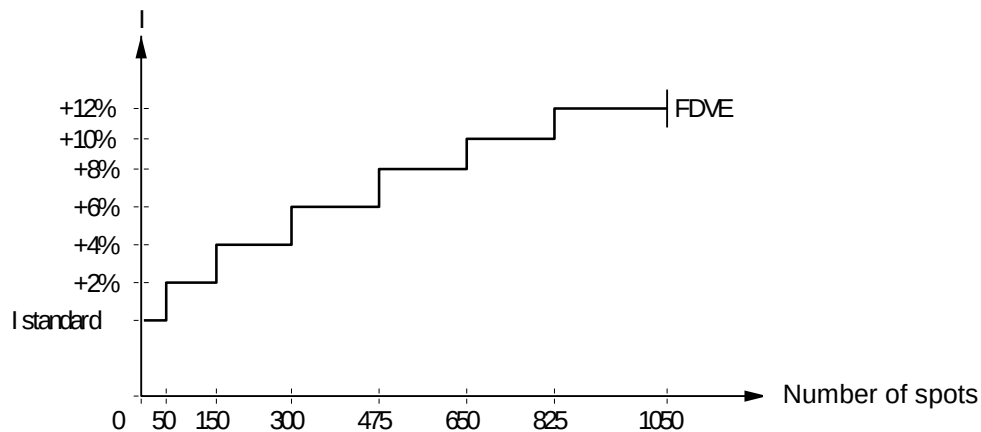
Curve 15: Range comprising of the coated assemblies  $\geq 49 \%$  of coated faces



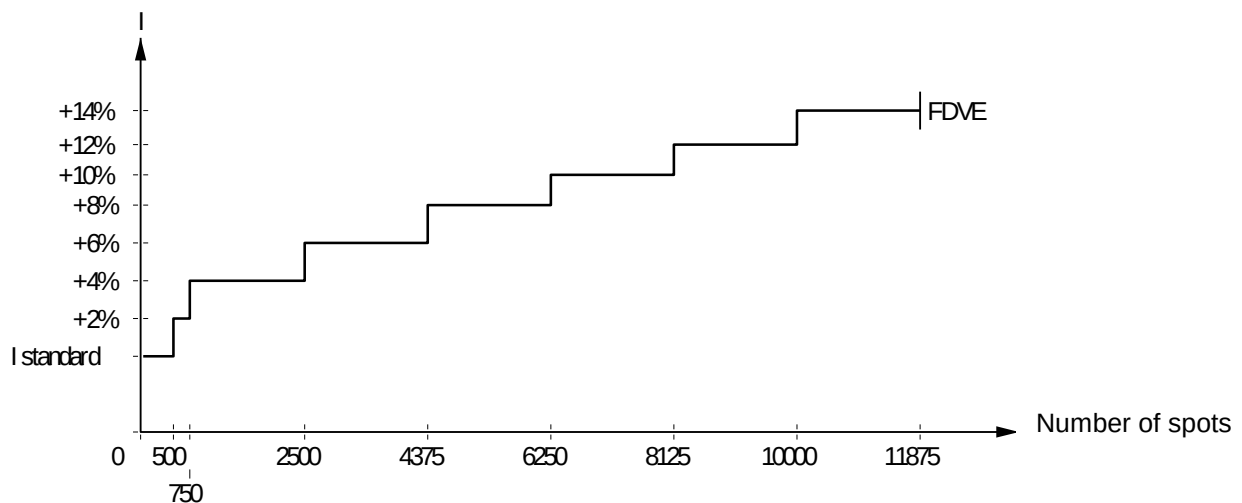
*Curve 16: Range comprising of the coated assemblies < 49 % of coated faces*

#### 8.12.5.FOR M.S.A. AND FOR COATING THICKNESSES > 10 $\mu$ M

##### 8.12.5.1.M.S.A. – Standard presentation



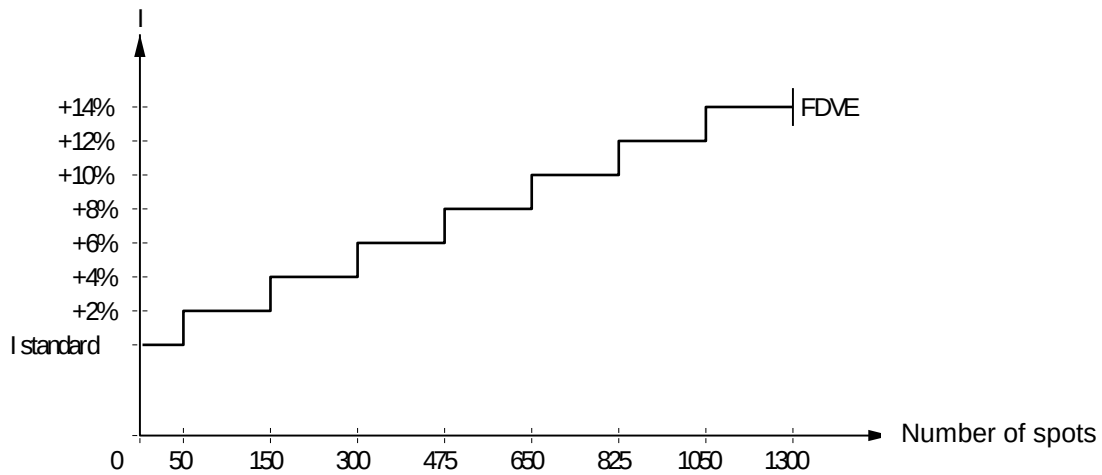
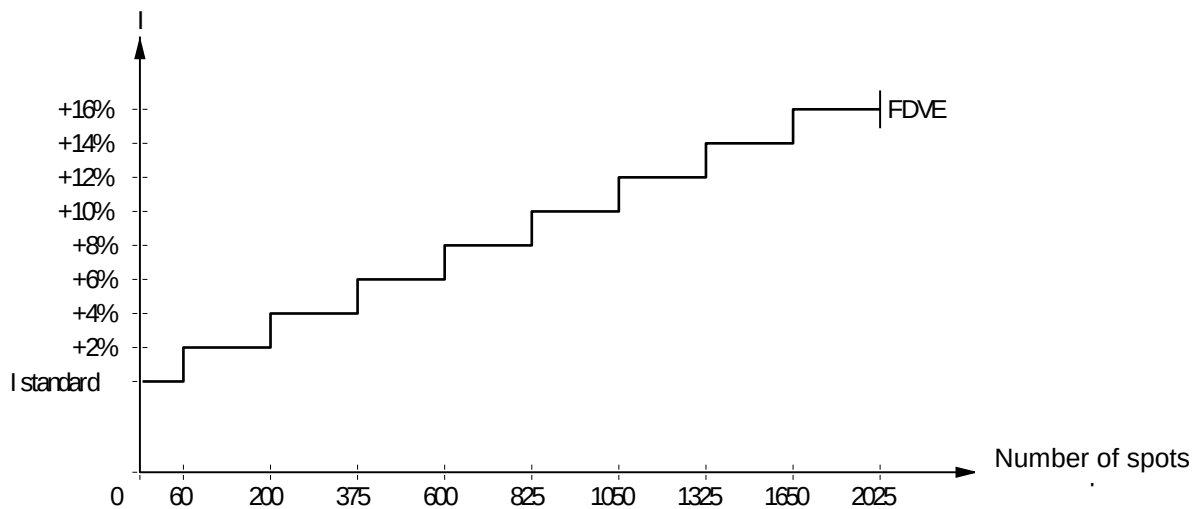
*Curve 17: Incrementing for spots having 1 or 2 coated faces galvanized in contact with the electrodes*



*Curve 18: Incrementing for spots having any face coated in contact with the electrodes*

8.12.6.FOR P A P. AND COATING THICKNESSES  $> 10 \mu\text{M}$ 

## 8.12.6.1.P. A P. - Standard presentation

Curve 19: Incrementing for coated ranges  $> 49 \%$  of coated faceCurve 20: Incrementing for coated ranges  $\leq 49 \%$  of coated face

### 8.12.7.INTERVALS OF DRESSING FOR ROBOT ACCORDING TO THICKNESSES OF REFERENCE AND TYPICAL CASES (50HZ AND MF)

**Note:** In RESORE, these parameters must be optimized by the assembler.

| Standard tip | Exterior thickness | Standard grinding interval | Reduced grinding interval                       |                |                   |
|--------------|--------------------|----------------------------|-------------------------------------------------|----------------|-------------------|
|              |                    |                            | - 4 thicknesses assembling<br>- quality problem | reduced flange | aspect imperative |
| P911 310 112 | 0,67               | 420                        |                                                 | 190            | 190               |
| P911 310 112 | 0,65               | 330                        | 260                                             | 190            | 190               |
| P911 310 112 | 0,62               |                            | 190                                             |                | 190               |
| P911 310 112 | 0,57               |                            | 190                                             |                | 190               |
| Z000 484 257 | (mixte range)      | 330                        | 260                                             |                |                   |

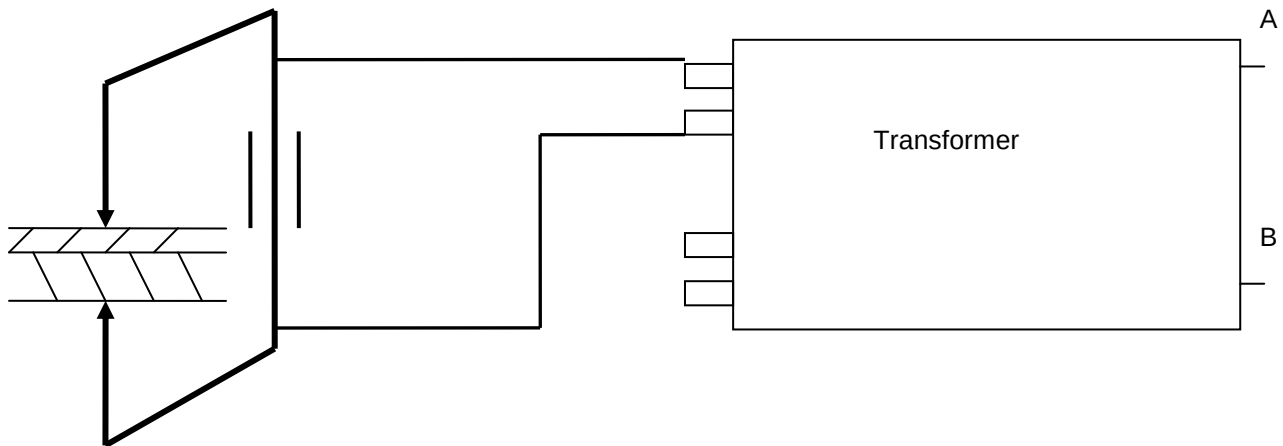
**Table 18: Range including coated ranges  $\geq 49$  % of coated face**

| Standard tip | Exterior thickness | Standard grinding interval | Reduced grinding interval                       |                |                   |
|--------------|--------------------|----------------------------|-------------------------------------------------|----------------|-------------------|
|              |                    |                            | - 4 thicknesses assembling<br>- quality problem | reduced flange | aspect imperative |
| P911 310 112 | 0,67               | 800                        | 600                                             | 375            | 375               |
| P911 310 112 | 0,65               | 600                        | 480                                             | 375            | 375               |
| P911 310 112 | 0,62               |                            | 375                                             |                | 190               |
| P911 310 112 | 0,57               |                            | 375                                             |                | 190               |
| Z000 484 257 | (mixte range)      | 600                        | 480                                             |                |                   |

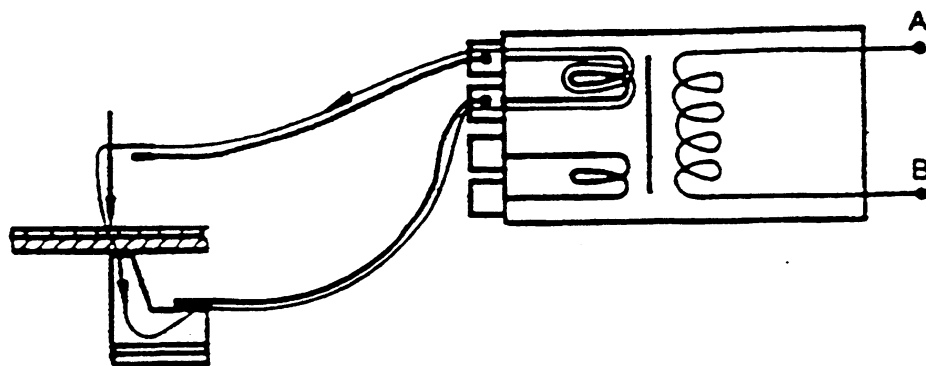
**Table 19: Range including coated ranges  $< 49$  % of coated face**

## Appendix 1 Methods of welding

*Simple spot (1 welding spot per secondary)*

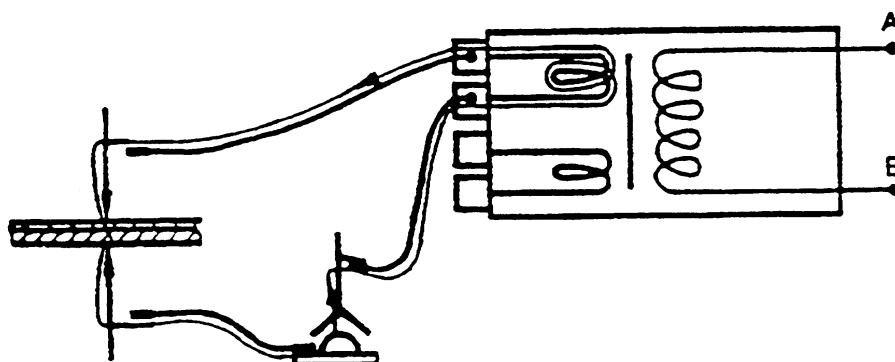


*Figure 2 : With welding gun (simple direct spot)*

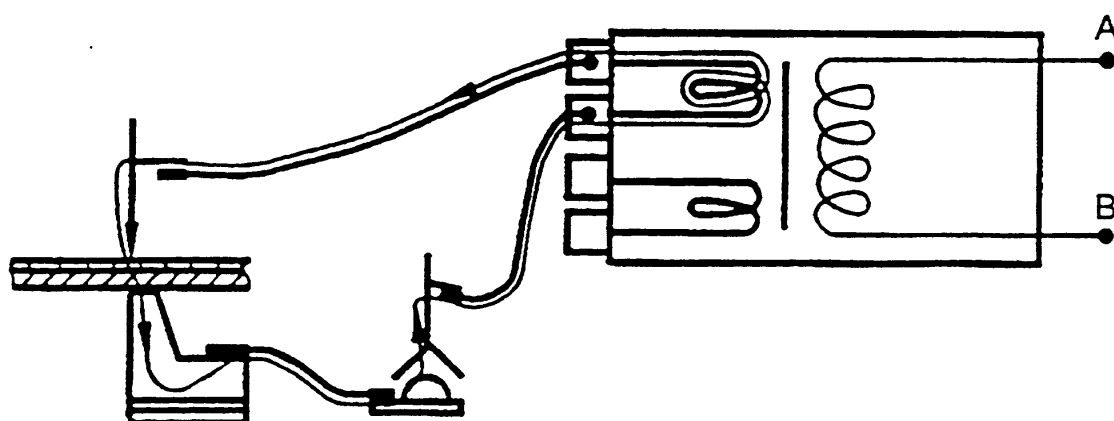


*Figure 3 : Simple direct spot on opposite electrode*

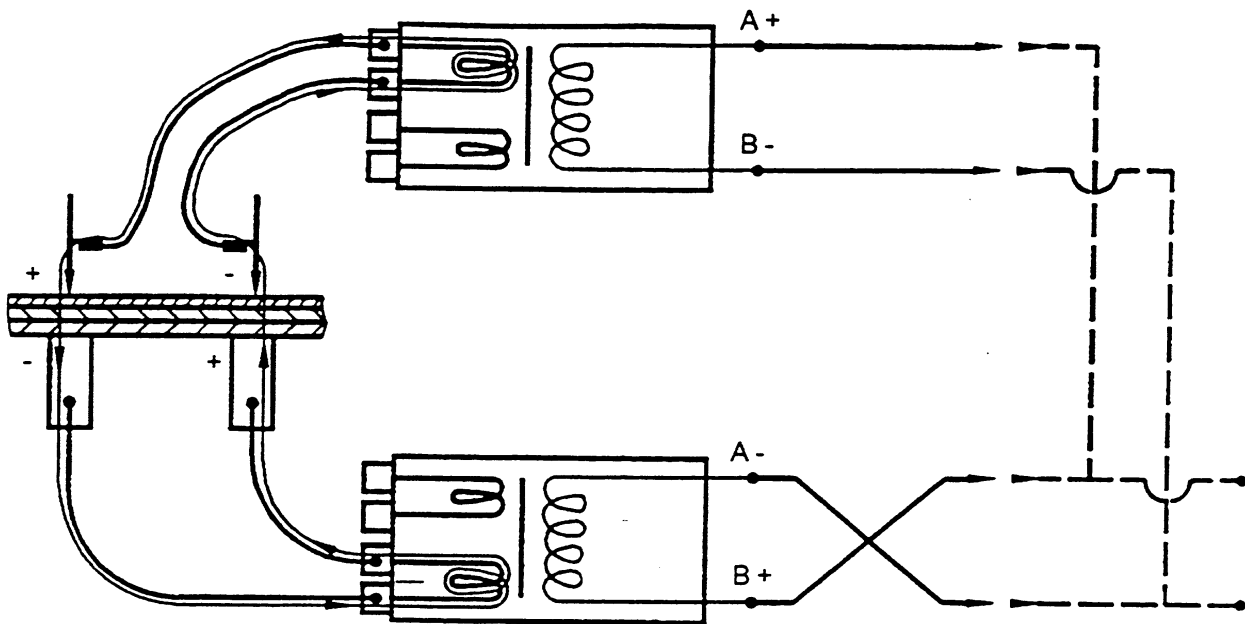
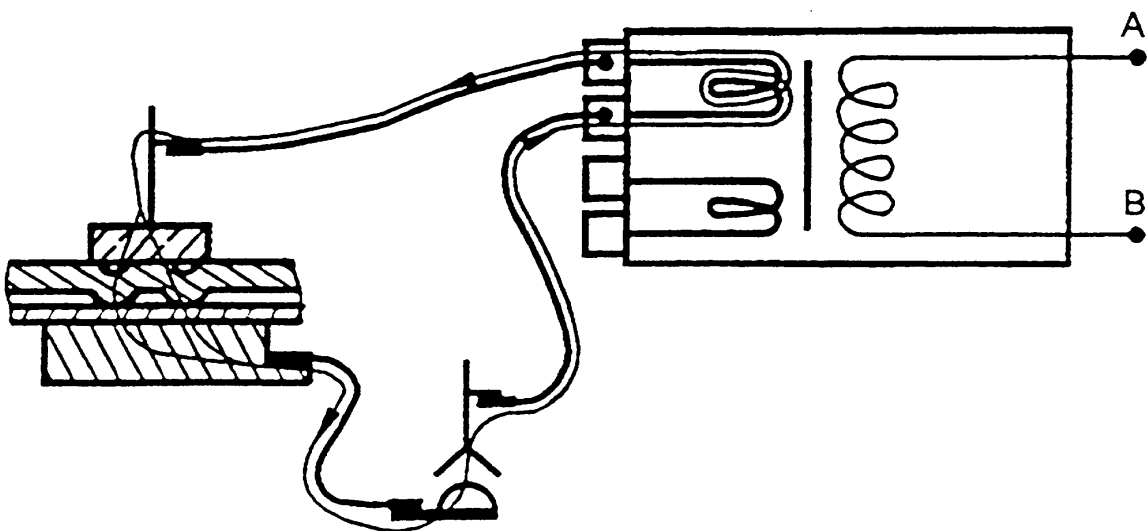
*Simple spot (1 welding spot per secondary)*



*Figure 4 : Simple indirect spot*



*Figure 5: Simple indirect spot on opposite electrode*

**Double push-pull spot (2 welding spots with 2 secondaries)***Figure 6 : On opposite electrode***Welding by embossment in simple spot (indirect)***Figure 7*

|                                                          |              |       |
|----------------------------------------------------------|--------------|-------|
| ELECTRICAL RESISTANCE WELDING - TECHNIQUES - GEN.SPECIFS | E34.03.180.G | 69/69 |
|----------------------------------------------------------|--------------|-------|

## Appendix 2

### Table of definition of the sections of cable (50Hz) Valid for cables type H07RNF and A07RN-F

|                             |                                                                                                                                      |                    |                    |                   |                    |                    |                     |                     |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|-------------------|--------------------|--------------------|---------------------|---------------------|
| ADEE/ISES<br>JL PUREN       | WELDING RESISTANCE NETWORKS 380V.<br>ACCEPTABLE INTENSITY IN STEADY OPERATION<br>CABLES INSTALLATIONS WITH THE FREE AIR (fall U: 5%) |                    |                    |                   |                    |                    |                     |                     |
|                             | SECTIONS of the CABLES                                                                                                               |                    |                    |                   |                    |                    |                     |                     |
| LENGTH of the<br>CONNEXIONS | 16 mm <sup>2</sup>                                                                                                                   | 25 mm <sup>2</sup> | 35 mm <sup>2</sup> | 50mm <sup>2</sup> | 70 mm <sup>2</sup> | 95 mm <sup>2</sup> | 120 mm <sup>2</sup> | 150 mm <sup>2</sup> |
| 145 meters                  | 59                                                                                                                                   | 95                 | 129                | 169               | 232                | 303                | 360                 | 421                 |
| 110 meters                  | 78                                                                                                                                   | 126                | 170                | 223               | 232                | 303                | 360                 | 421                 |
| 90 meters                   | 90                                                                                                                                   | 146                | 170                | 223               | 232                | 303                | 360                 | 421                 |
| From 1 to 80 meters         | 106                                                                                                                                  | 146                | 170                | 223               | 232                | 303                | 360                 | 421                 |

|                           |                                                  |      |      |       |      |      |      |      |
|---------------------------|--------------------------------------------------|------|------|-------|------|------|------|------|
| INSTANTANEOUS<br>CURRENTS | FALL OF SINGLE-PHASE VOLTAGE for 10m (cos = 0.8) |      |      |       |      |      |      |      |
|                           | Volt                                             | Volt | Volt | Volt  | Volt | Volt | Volt | Volt |
| 250 A                     | 5,5                                              | 3,75 | 2,75 | 1,92  | 1,4  | 1,15 | 0,95 | 0,8  |
| 500 A                     | 11                                               | 7,5  | 5,5  | 3,84  | 2,8  | 2,3  | 1,9  | 1,6  |
| 1000 A                    | 22                                               | 15   | 11   | 7,68  | 5,6  | 4,6  | 3,8  | 3,2  |
| 1500 A                    | 33                                               | 22,5 | 16,5 | 11,55 | 8,4  | 6,9  | 5,7  | 4,8  |
| 2000 A                    | 44                                               | 30   | 22   | 15,36 | 11,2 | 9,2  | 7,6  | 6,4  |
| 2500 A                    | 55                                               | 37,5 | 27,5 | 19,25 | 14   | 11,5 | 9,5  | 7,5  |

**Note :** The table "falls of voltage" above, is indicated starting from calculations carried out with data from supplier SOCABEL 5 (or TENAFLEX SR) for single-core cables H07RNF and A07RN-F :

|                                     |     |     |     |      |      |      |      |      |
|-------------------------------------|-----|-----|-----|------|------|------|------|------|
| Section cable<br>(mm <sup>2</sup> ) | 16  | 25  | 35  | 50   | 70   | 95   | 120  | 150  |
| Fall U /A/km<br>(volts)             | 2,2 | 1,5 | 1,1 | 0,77 | 0,56 | 0,46 | 0,38 | 0,32 |

Example of calculations:

Checking of the calculation of voltage drop to 10m for 2500 A for a cable of 35 mm<sup>2</sup>:

Case of a cable 35 mm<sup>2</sup>: 1.1 V of voltage drop per A and km;

So, voltage drop = 0,01 x 2500 x 1,1 = 27,5 V.